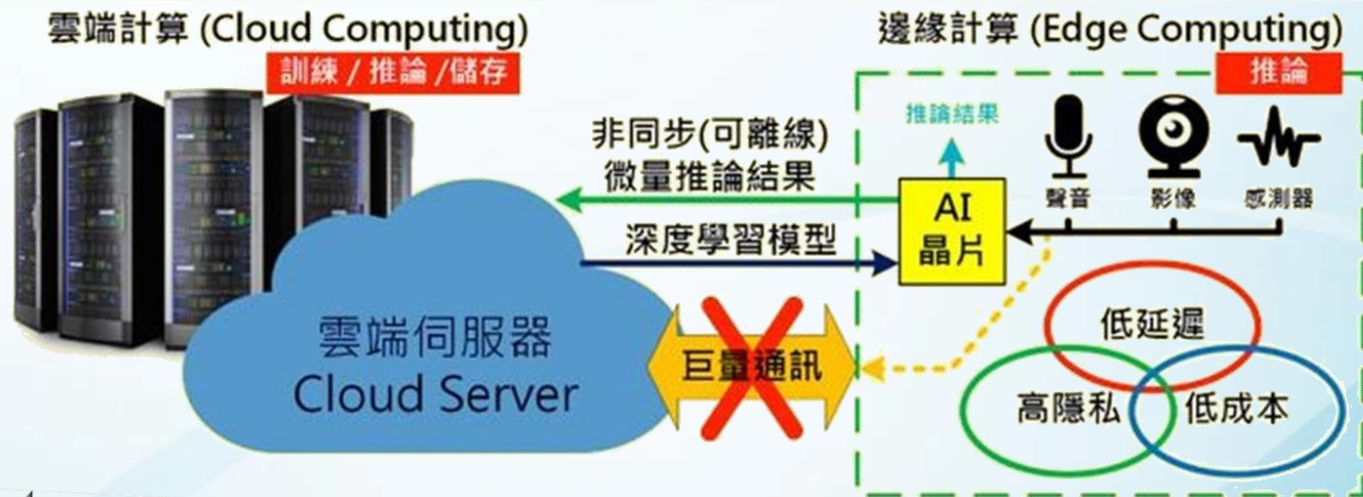
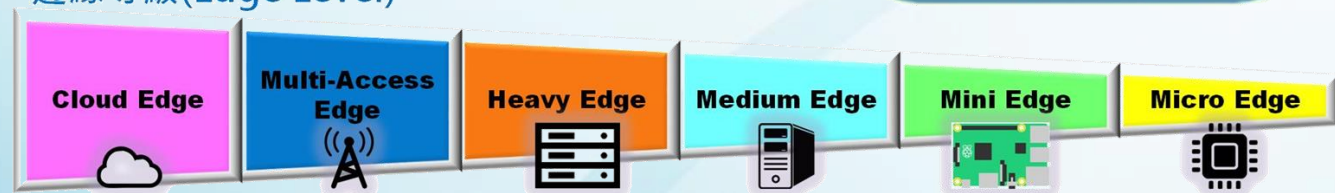


邊緣人工智慧實務 (EE5354701)



Edge AI

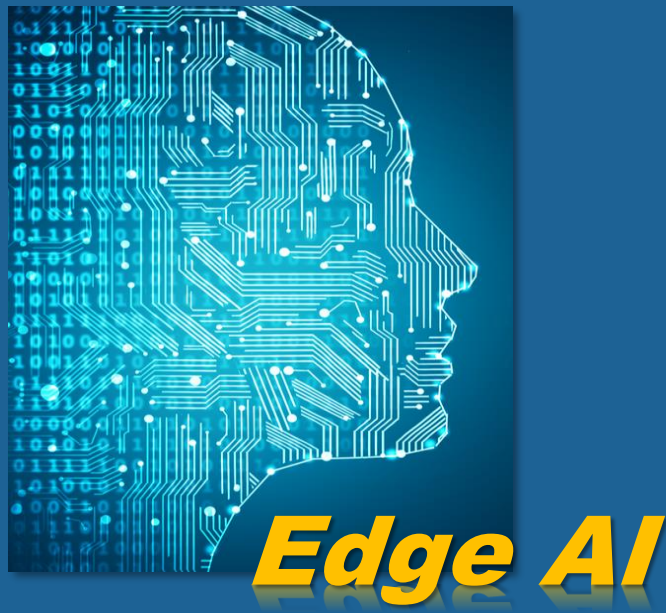
邊緣等級(Edge Level)



12 生成式邊緣人工智慧

電機工程系 人工智慧應用產碩專班 許哲豪 兼任助理教授

簡報大綱

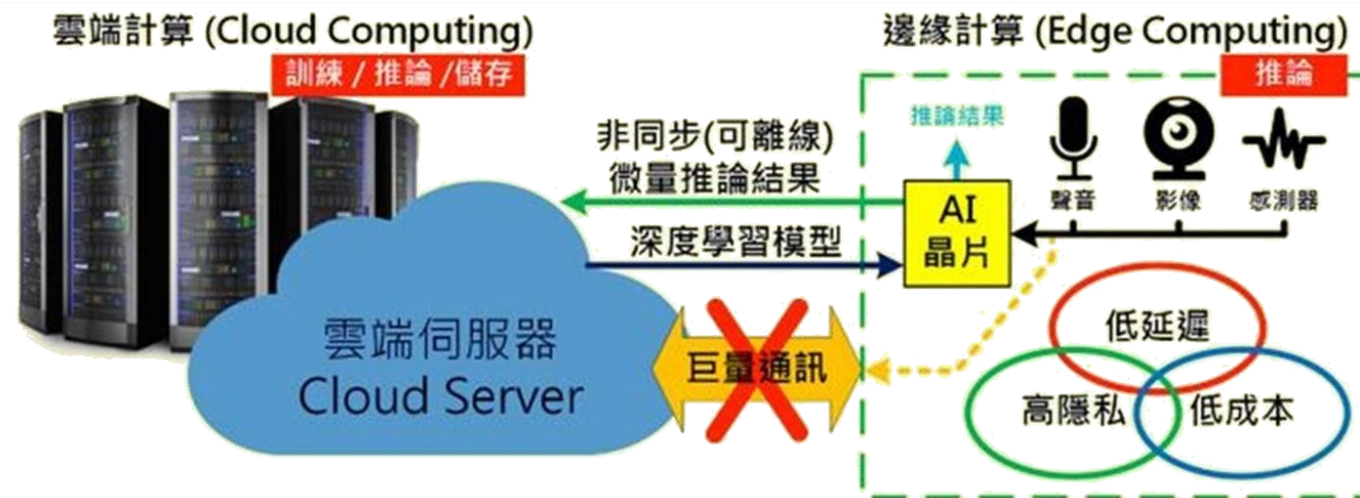


- **12.1. 邊緣與生成智慧**
- **12.2. 產業應用與消費者解決方案**
- **12.3. 代理式人工智慧與分散式系統**
- **12.4. 硬體優化與平台資源**
- **12.5. 學術與研究基礎**

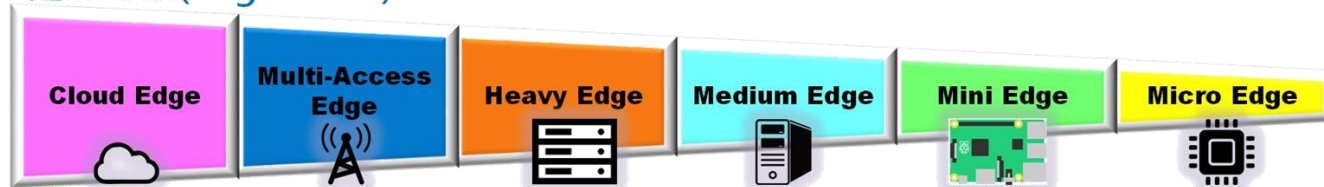
https://hackmd.io/@OmniXRI-Jack/NTUST_EdgeAI_2026



Edge AI



邊緣等級(Edge Level)

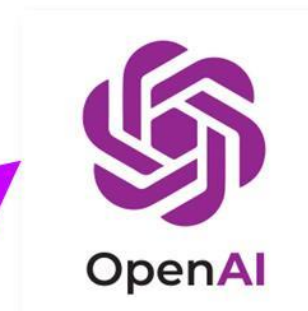
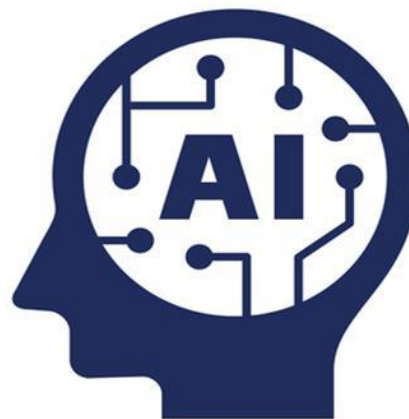
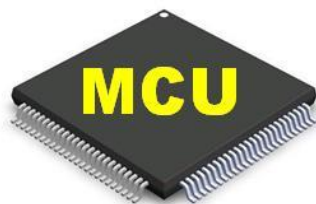


12.1. 邊緣與生成智慧

邊緣智慧 vs. 生成智慧

- 邊緣端極少資源
儲存(Flash, RAM)、計算
- 智慧感測
聲音、影像、運動...
- 超小模型、超低功耗

極小化

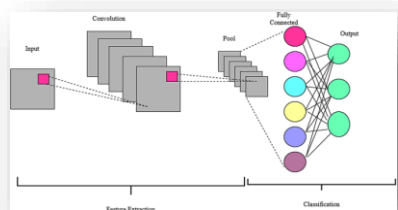


極大化

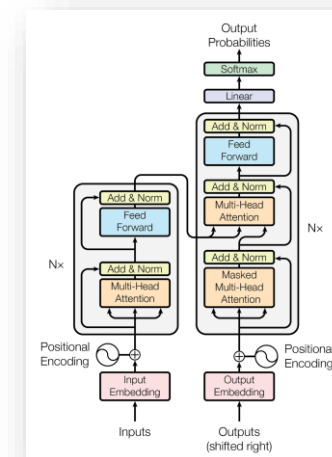
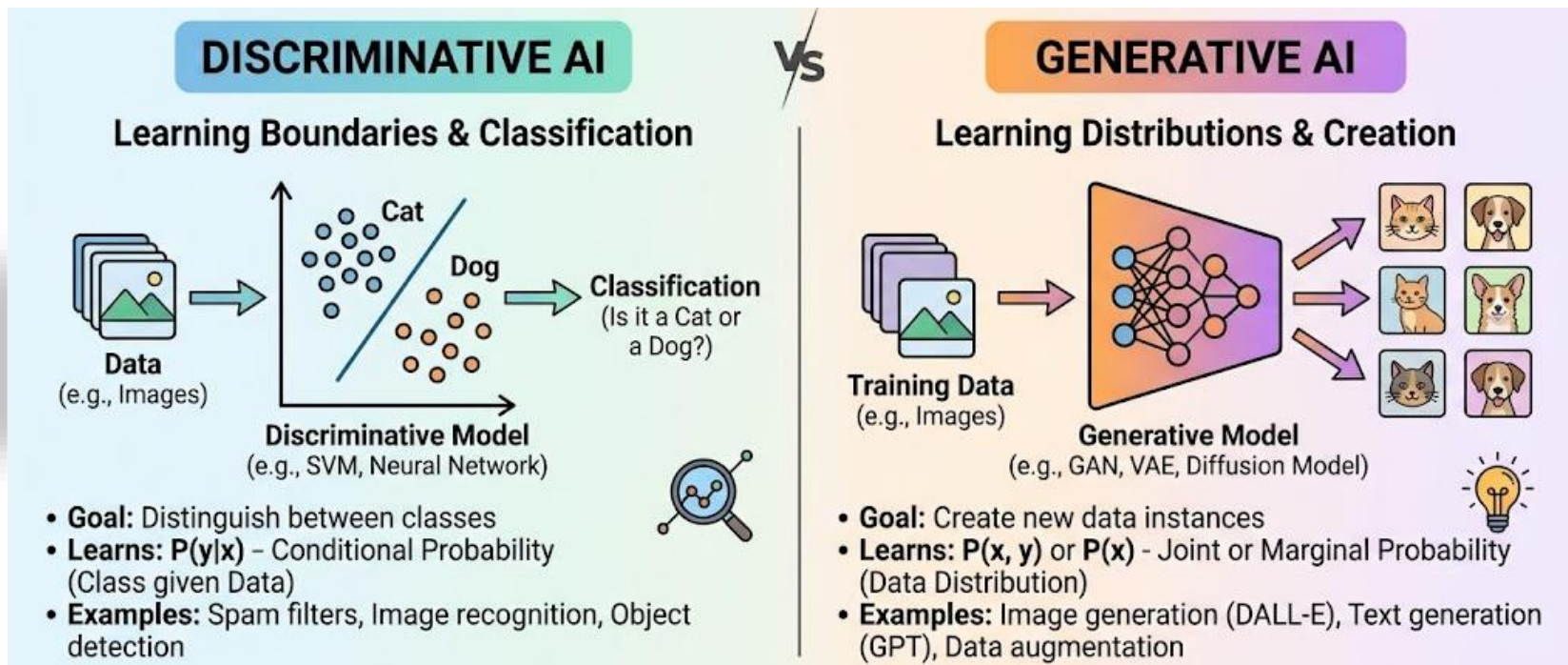
- 雲端無限資源
儲存、計算、頻寬、功耗
- AI生成
對話、影像、程式...
- 超巨大資料集、模型

資料來源：<https://omnixri.blogspot.com/2023/04/20230420aiexpo-aiedge-ai.html>

鑑別式(Discriminative) vs. 生成式(Generative)



CNN Based

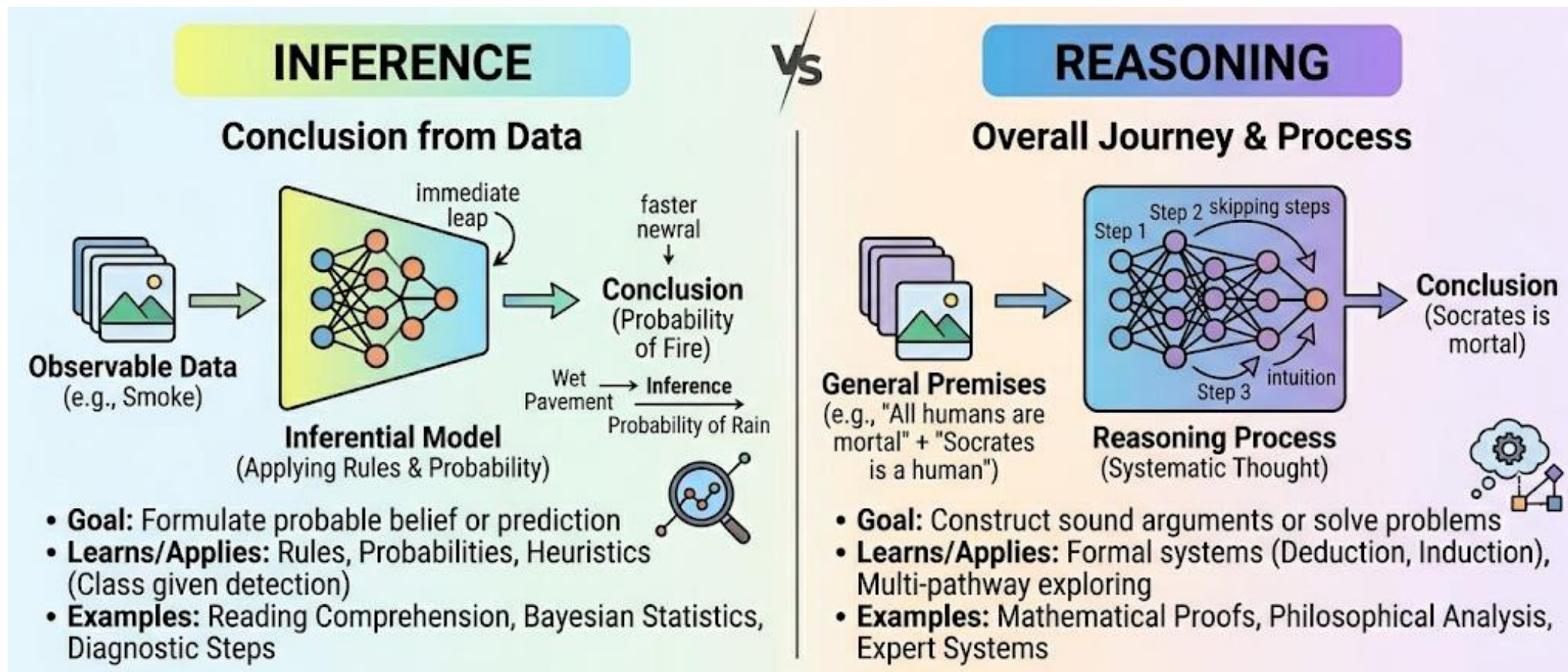


Transformer Based

辨識、分類、預測
找出不同類別邊界

數值、文字、聲音、影像輸入
產生類人輸出的全新內容

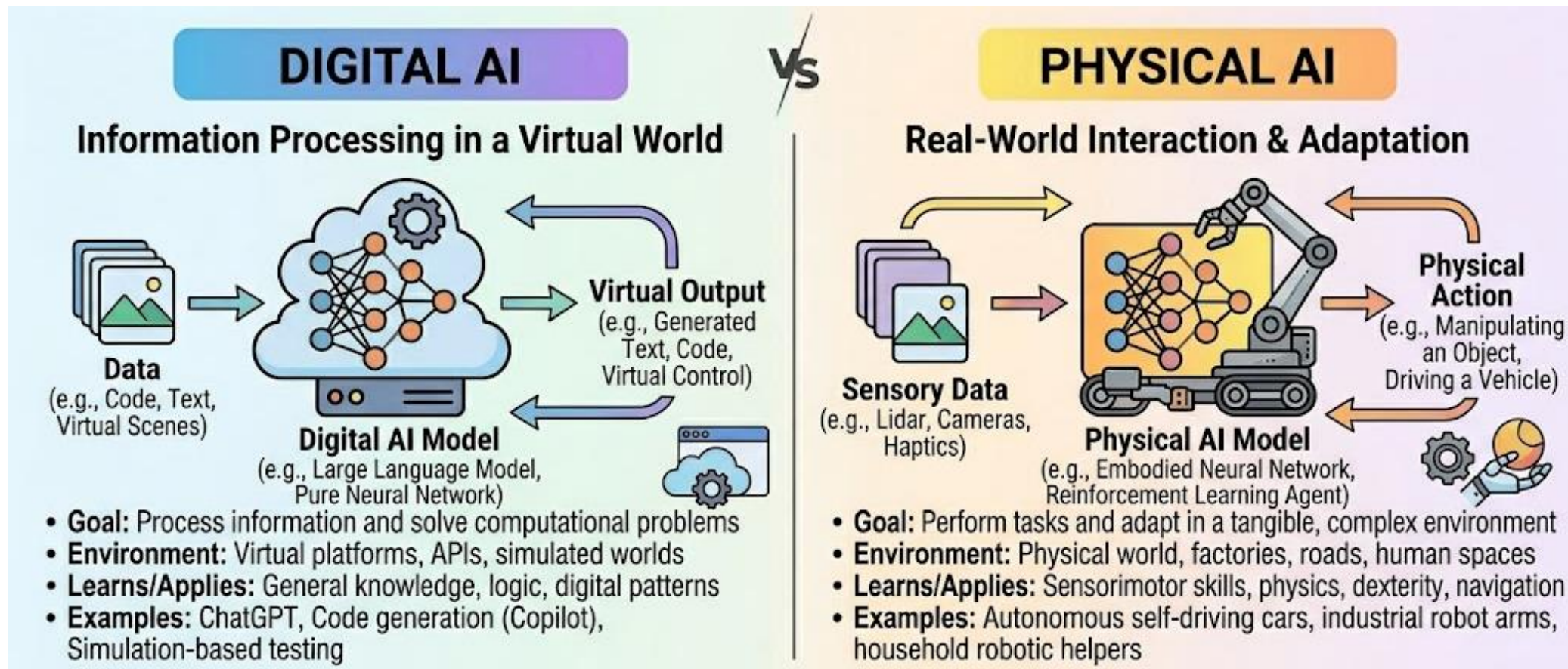
推論(Inference) vs. 推理(Reasoning)



依模型計算出機率最高答案

找出所需工作的步驟並逐一執行

數位 (Digital AI) vs. 實體(Physical AI)



在數位環境進行計算及輸出
包括AI晶片、雲端API、虛擬實境

結合真實世界影音、感測資訊
進行分析並推動仿生機電元件

Edge AI (tinyML) Foundation

2018 -
2024/11



2024/11
- now



<https://www.edgeaifoundation.org/>

Generative Edge AI Livestreams v5

推理、代理與實體智慧



Day 1:

<https://www.edgeaifoundation.org/livestreams/generative-edge-ai-reasoning-agentic-physical-intelligence-3>

Day 2 :

<https://www.edgeaifoundation.org/livestreams/generative-edge-ai-reasoning-agentic-physical-intelligence-2>

Day 1 (04:13:58)

Session 1:

產業應用與消費者解決方案

Session 2:

代理式人工智慧與分散式系統

Day 2 (02:56:54)

Session 3:

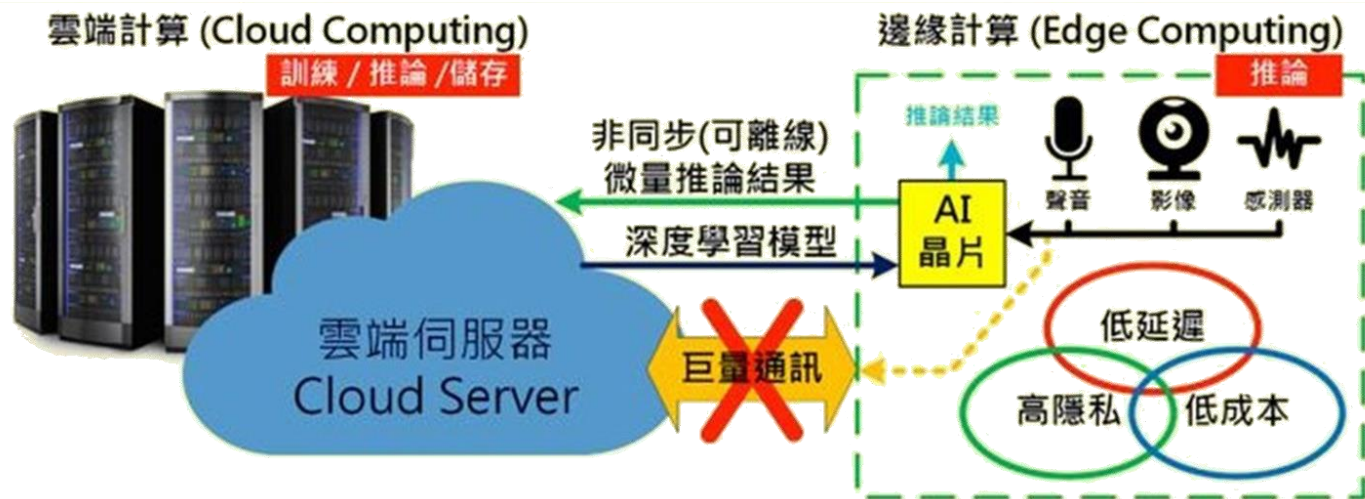
硬體優化與平台資源

Session 4:

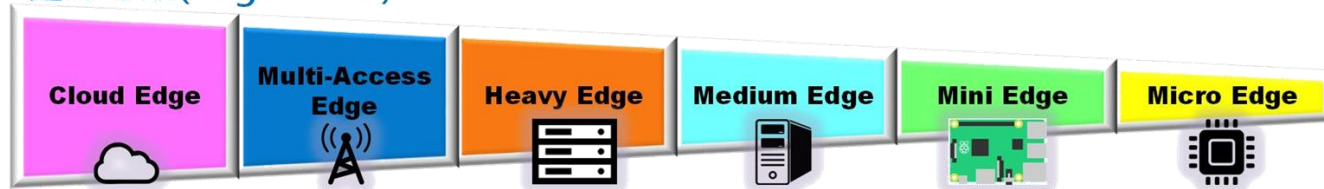
學術與研究基礎



Edge AI



邊緣等級(Edge Level)



12.2. 產業應用與消費者解決方案

Fireside Chat – Generative Edge AI

Generative Edge AI Perception + intelligence **感知+智慧**

Definition:
AI capable of sensing, reasoning, multimodal understanding, and generative behavior deployed directly at the edge – the point of data acquisition and action

The edge AI playing field:

The diagram shows a horizontal axis representing the deployment environment from Cloud to On-device. The vertical axis represents the parameter count on a logarithmic scale from 1M to >27T. Four models are shown as horizontal bars:

- Frontier** (Cloud): >27T parameters
- Edge LLM** (Datacenter): 7B - 400B parameters
- VLM / VLA** (Operations): 500M - 70B parameters
- SLM** (Operations): 100M - 8B parameters
- TinyML** (On-device): <300M parameters

Below the diagram, two arrows indicate the scope of context: "Business context and actions" (covering Cloud to Operations) and "Operational context and events" (covering Operations to On-device).

MOOR INSIGHTS & STRATEGY

主要領域：

- 雲端
- 資料中心
- 操作端
- 裝置端

模型規模：

- Edge LLM (7B ~ 400B)
- VLM/VLA (500M~700B)
- SLM(100M~8B)
- TinyML(<300M)

<https://youtu.be/kPJgEWDRotM?t=446>

Fireside Chat – Generative Edge AI

EDGE AI FOUNDATION

The Generative Shift

From observation to participation
"Moving intelligence as close to data and operations as possible is the smartest thing enterprises can do." Patrick Moorhead

Value shifts operational ROI drivers

Understand Observations → Situations	Decide Dashboards → Agentic interaction	Act Alerts → Actions	Optimize Reports → Interventions
--	---	--------------------------------	--

Tailwinds converging trends

Enterprise motivation AI ROI follows data to the edge	Edge models Scale from servers to battery powered	Turnkey platforms Build products, not system software	Edge AI silicon Constraint-optimized AI
---	---	---	---

Headwinds AI won't fix this stuff

Governance and safety High stakes. No framework	IoT readiness AI will not fix your broken IoT stacks	Platform fragmentation Fragmentation kills scale	Toolchain maturity AI is cloud-to-edge. Tools are not
---	--	--	---

MOOR INSIGHTS & STRATEGY

價值移動：

- 理解
- 決策
- 行動
- 優化

順境：

- 商業動機
- 邊緣模型
- 完整方案
- 半導體

逆境：

- 治理和安全
- 物聯網狀態
- 平台破碎化
- 工具鏈成熟度

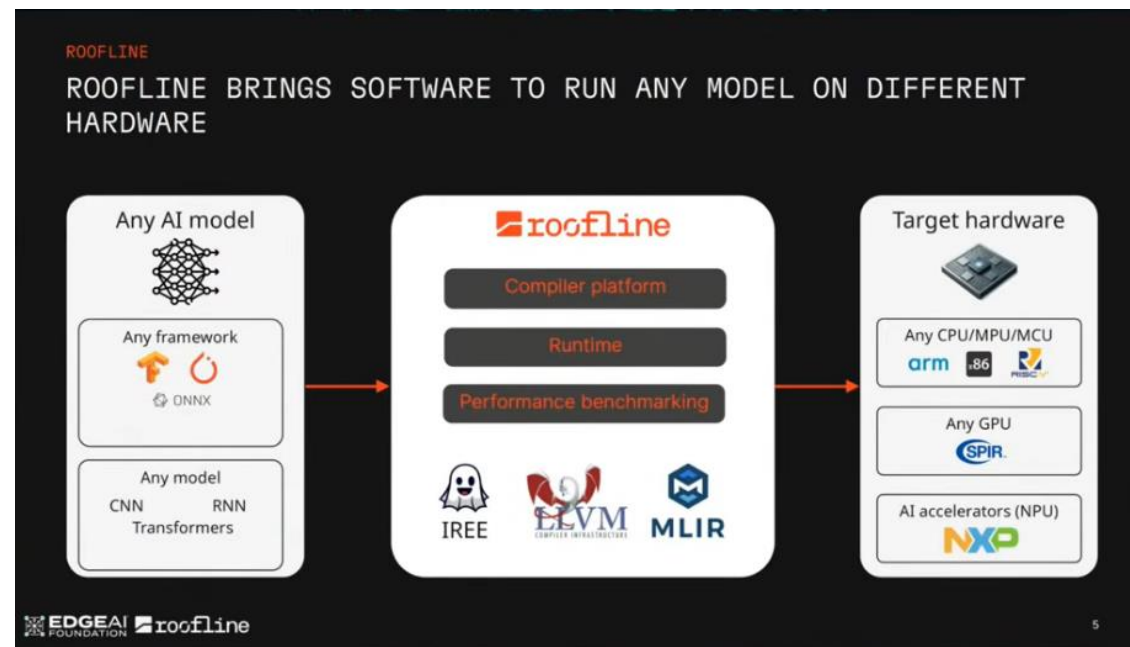
<https://youtu.be/kPJgEWDRotM?t=903>

Roofline – 加速邊緣代理系統創新速度



<https://youtu.be/kPJgEWDRotM?t=1838>

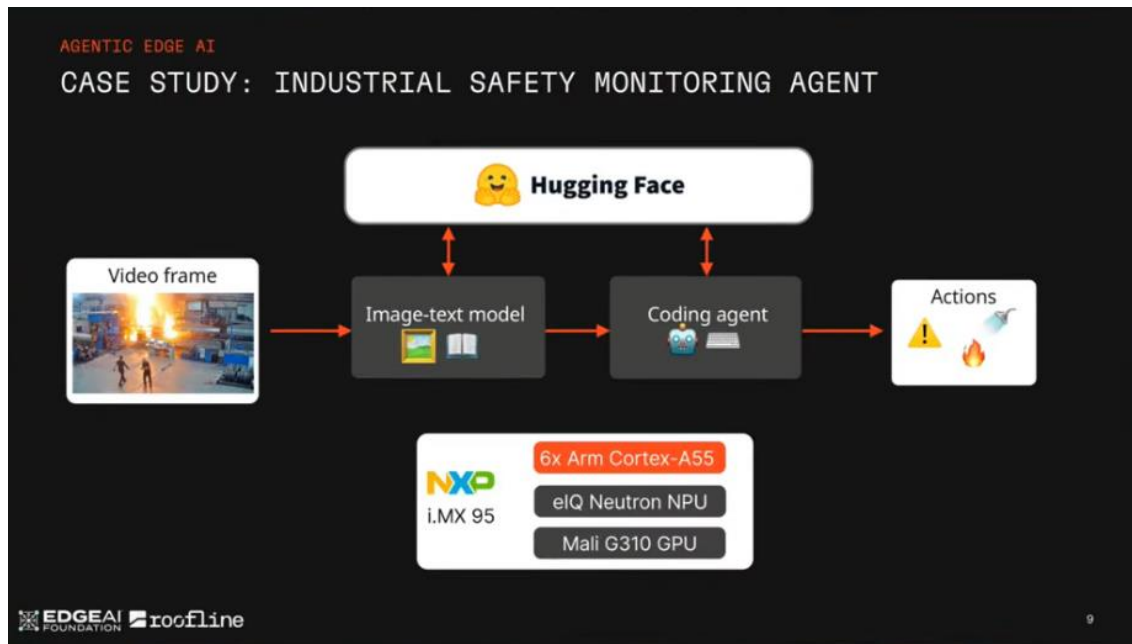
Roofline – 加速邊緣代理系統創新速度



傳統開發從模型選定測試到部署完成約需3~7個月，而 Roofline 提供一鍵部署，僅需 1 小時就能完成。

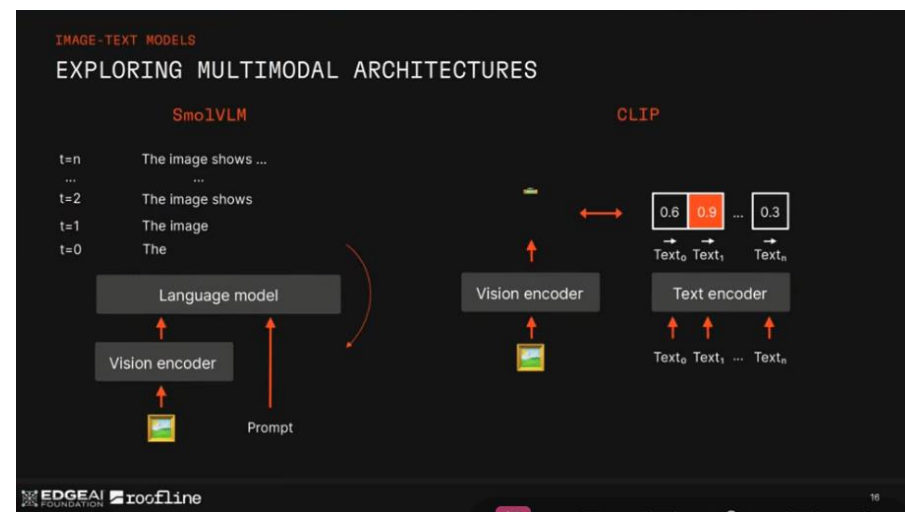
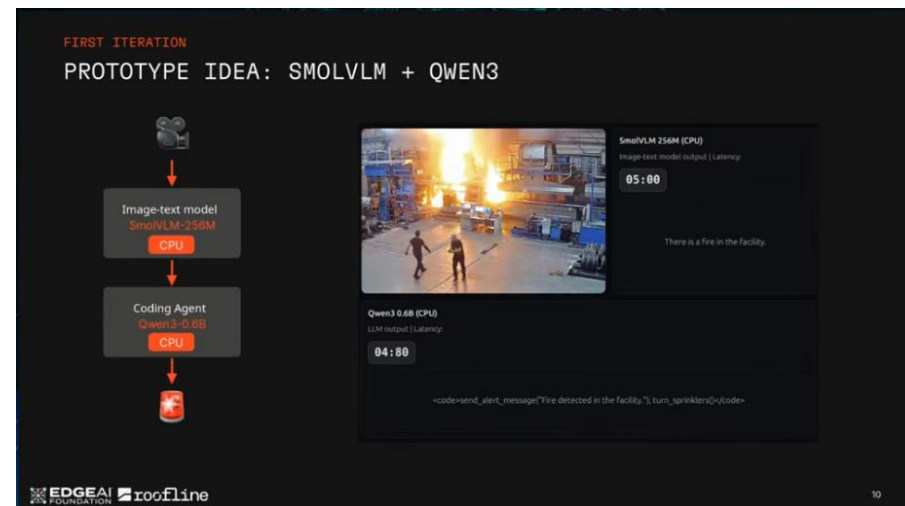
Roofline 提供工具可支援各種框架產生的模型，不同編譯器和不同硬體平台。

Roofline – 加速邊緣代理系統創新速度



案例研究：工廠內火災警報系統，使用 SmoIVLM 分析影像產生警報文字，再由 Qwen3-0.6B 產生程式碼代理，控制系統。

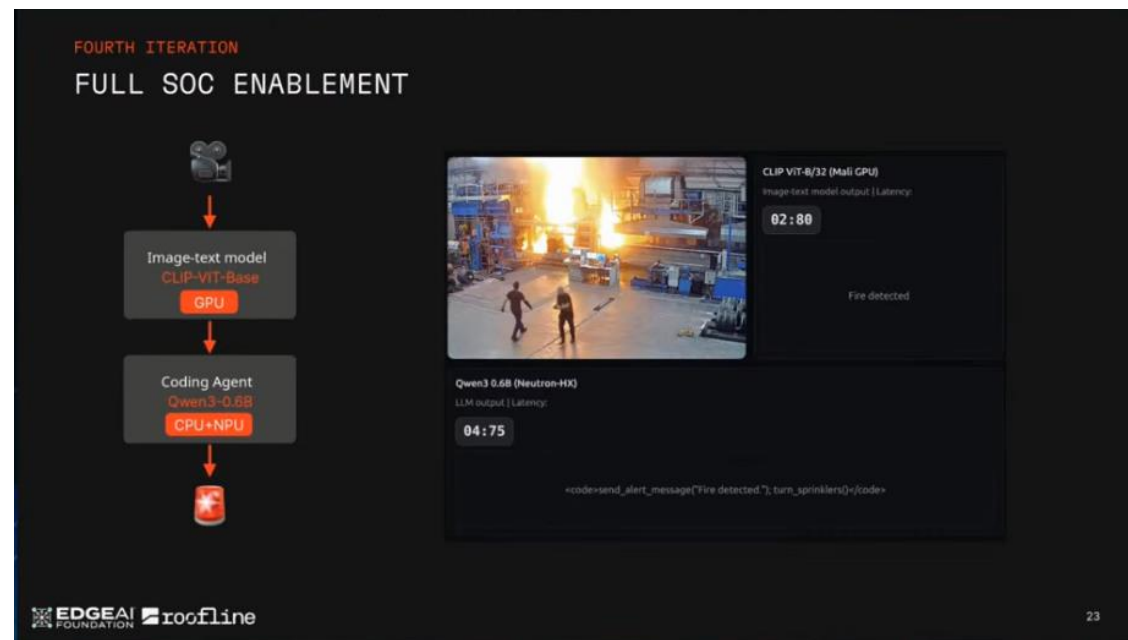
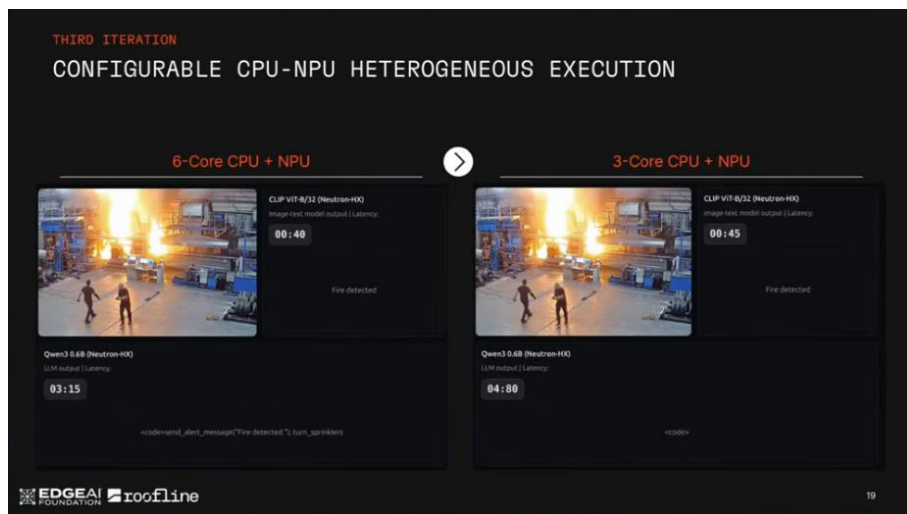
NXP i.MX95 – 6x Arm Cortex-A55, 1x Arm Mali G310 GPU, 1x Arm Cortex-M7, 1x Arm Cortex-M33, NXP eIQ NPU, 1376KB SRAM.



SmoIVLM-256M 切換到 CLIP-ViT-Patch32

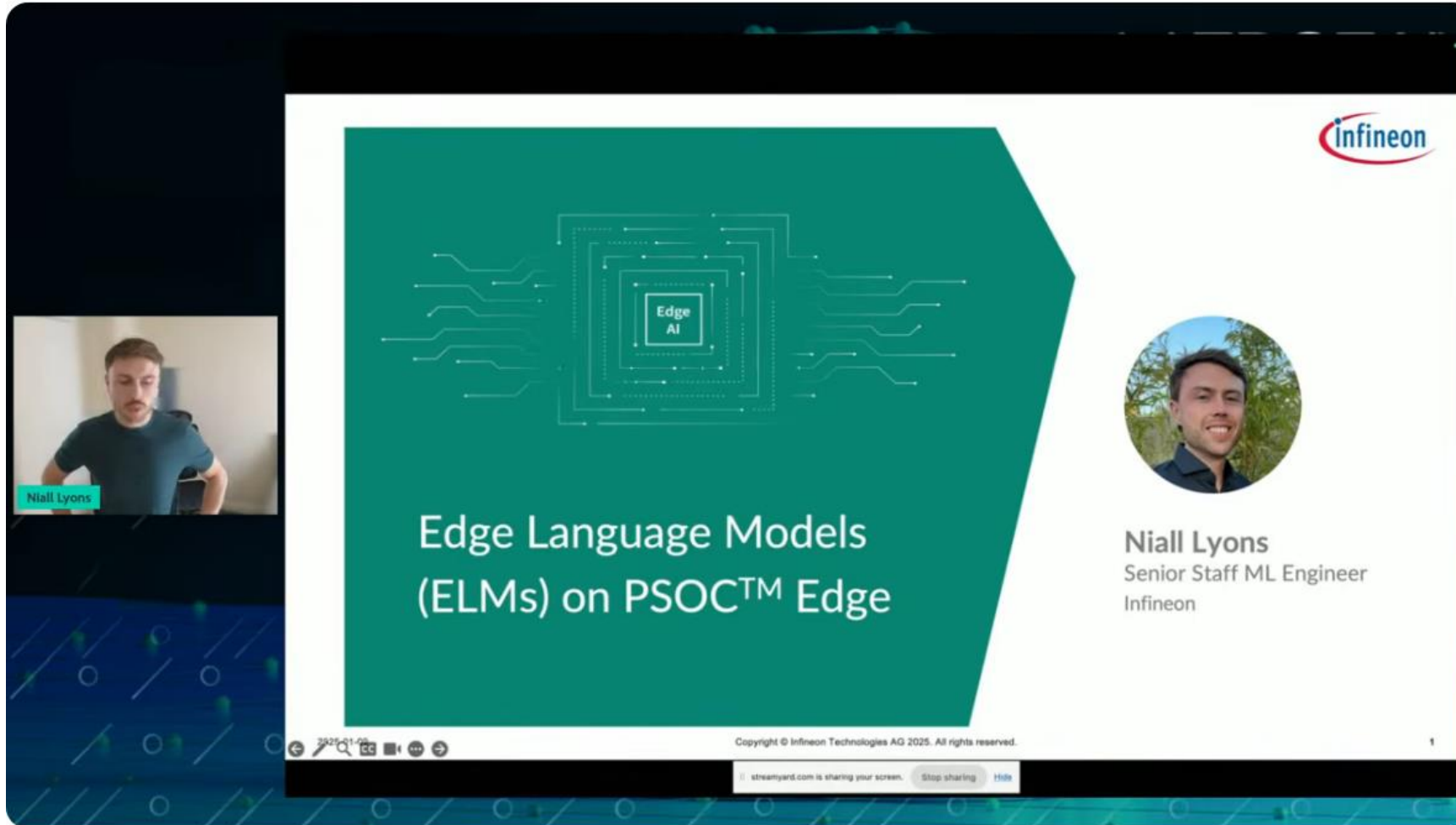
7x Speed-up

Roofline – 加速邊緣代理系統創新速度



推論速度優化：3 Core CPU + NPU, 6 Core CPU 6 + NPU，進展到 CLIP-ViT-Base 使用 GPU, Qwen3-0.6B 使用 NPU。

Infineon – 邊緣語言模型在 PSoC Edge



<https://youtu.be/kPJgEWDROtM?t=8230>



KIT_PSE84_EVAL



KIT_PSE84_AI



PSoC Edge E84

Arm Cortex-M55 @400MHz
NPU NN-Lite +Ethos-U55-128
SRAM 6MB, RRAM 512KB
Arm Cortex-M33 @200MHz

Infineon – 邊緣語言模型在 PSoC Edge



Part I: Redefining Intelligence - Traditional AI vs GenAI

任務特定
系統



Task Specific Systems

Built for singular purposes



狹義智慧



Narrow Intelligence

Expert in one domain, useless in others



漸進式
進展



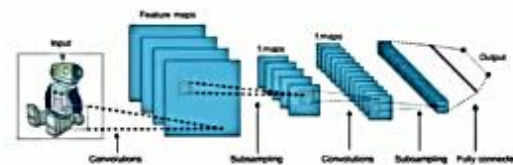
Incremental Progress

Steady but limited improvements

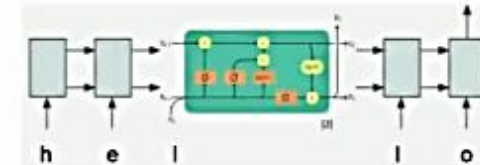


Traditional AI: The Specialist Era (1990-2020)

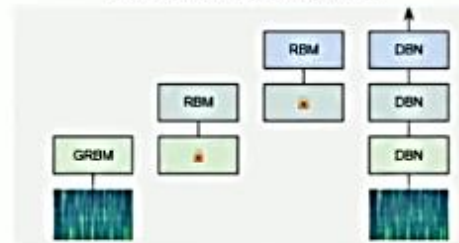
Computer Vision
Convolutional NNs (+ResNets)



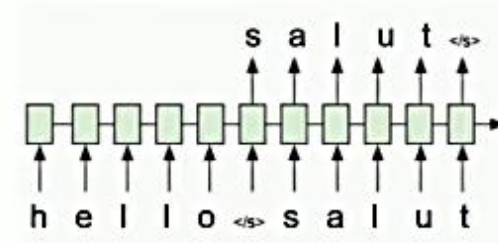
Natural Language Processing
Recurrent NNs (+LSTMs)



Speech
Deep Belief Nets (+non DL)



Translation
Seq2Seq



Infineon – 邊緣語言模型在 PSoC Edge



Part I: Redefining Intelligence - Traditional AI vs GenAI

通用問題
解法



Universal Problem Solvers

One architecture, infinite applications



真智慧



True Intelligence

Human-like adaptability and reasoning



指數級
突破



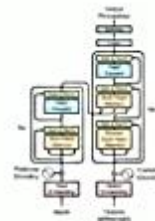
Exponential Breakthroughs

Performance barriers shattered



GenAI: The Generalist Takeover (2020-Present)

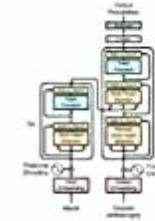
Computer Vision



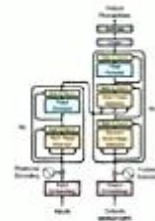
Natural Language Processing



Speech



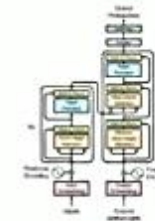
Translation



Graphs/Science



Reinforcement Learning



Infineon – 邊緣語言模型在 PSoC Edge

Edge is Present Where Work Happens

Task Specific Models

- Appliance**
Help and triage on device
- Wearable**
Glanceable guidance
- Industrial**
Offline Workflows
- Healthcare**
Local by default

Why a Phone isn't Enough

 Hands-free Gloves, wet hands, PPE	 Restricted Phones banned (clean rooms, factory floor)	 Reliability Works without coverage, no server outage
 Cost of Ownership No phone ownership, app installs or IT overhead	 Generic Interaction with Device Data is limited or complicated	

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Introducing Edge Language Models: Cloud-Class UX at MCU Power

ELMs are Enough

-  Customized with general intelligence models – interact with relevant data
-  Fully on-device: zero network hops
-  Zero data egress; no per-query fees***
-  Consistent latency without Wi-Fi/4G
-  Device-class power milliwatts, not watts
-  Low cost, low footprint compute, intelligence in everyday devices

 Works with Smart Devices	 Works on machinery	 Reliability – no down time
 Cost effective	 Customized	

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主要用途：智能設備、穿戴式、工業、健康照護
手機不易滿足：釋放雙手、禁用場合、可靠度、擁有成本、通用性

主要特色：通用模型客製化、裝置運算(不用連網)、零數據輸出(隱私)、穩定延遲、低功耗、低成本、低佔用空間計算
邊緣語言模型可滿足：智慧裝置、機器設備、可靠度、成本有效、客製化

Infineon – 邊緣語言模型在 PSoC Edge

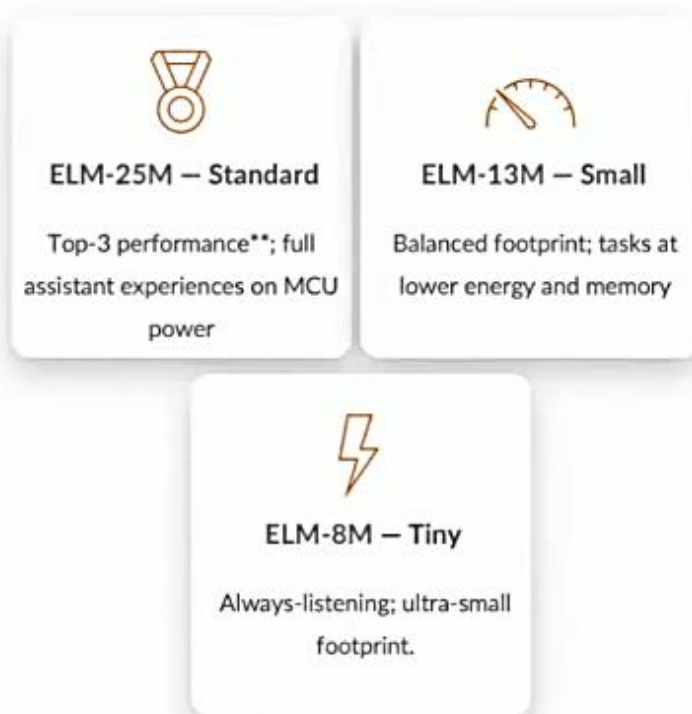
Introducing Infineon's Edge LM family: Straightforward Adoption Path



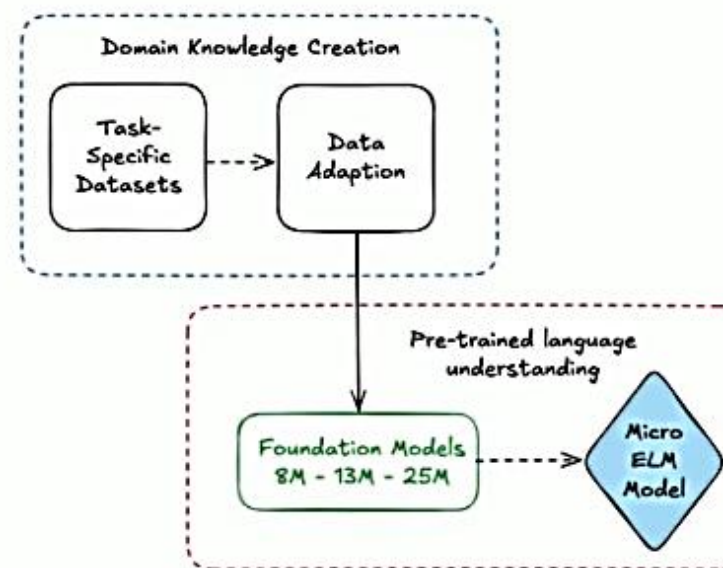
Standard
ELM-25M

Small
ELM-13M

Tiny
ELM-8M



Deployment Flow



White Paper: https://www.infineon.com/gated/edge-language-models-on-psoc-edge_2cc525ee-0e6d-4cde-bb66-bd055f338c5a

Infineon – 邊緣語言模型在 PSoC Edge

Building Product-ready Tiny Intelligence

From Foundation to Product

Fast, Predictable, On-device



Foundation Model Family

8M • 13M • 25M – pick the footprint that fits



Variable knowledge

Short / standard / extended context, all on-device.



Customization toolkit

Distill and adapt to your domain and voice.



Turnkey option

Infineon tunes and validates for you.

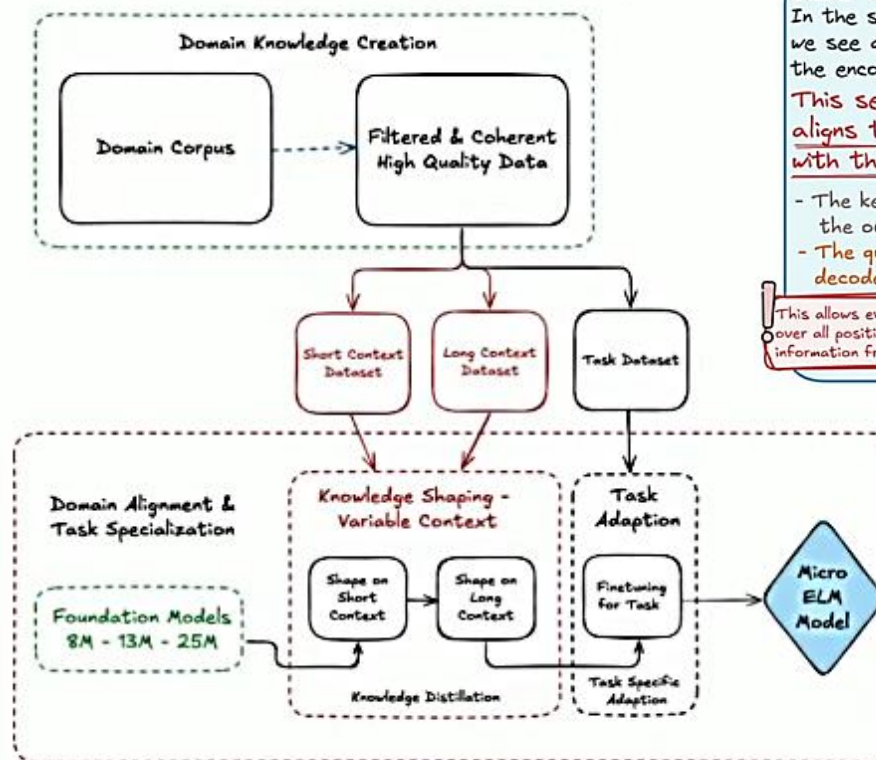


Time-to-market advantage

Ship sooner than building from scratch—lower risk, lower cost.



Edge LM Enablement: Curate • Shape • Tune • Ship



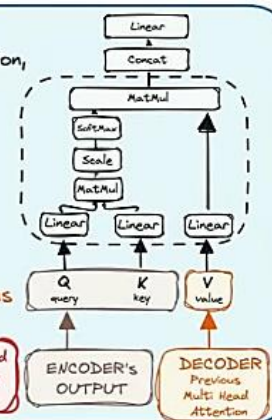
2.2. Cross-Attention

In the second multi-headed attention, we see a cross-attention between the encoder and the decoder.

This setup effectively aligns the encoder's input with the decoder's

- The keys and values come from the output of the encoder.
- The queries come from the previous decoder layer

This allows every position in the decoder to attend over all positions in the input sequence, integrating information from the encoder and the decoder.



Infineon – 邊緣語言模型在 PSoC Edge

PSOC™ Edge Enables LMs: Standing

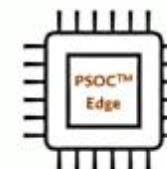


Top 3 Huggingface Industry Metrics 3/447 Edge Device Models (only sub-360M models on the leaderboard)*

Rank	Model	Average	IFEval	BBH	MATH	GPQA	MUSR	MMLU-PRO	CO ₂ Cost
1	ibm-granite/granite-3.1-2b-instruct	21.71%	62.86%	21.82%	15.26%	5.26%	4.87%	20.21%	1.01 kg
2	ibm-granite/granite-3.2-2b-instruct	21.25%	61.52%	21.67%	14.43%	5.37%	4.70%	19.82%	0.51 kg
3	infineon/ifx-elm-25m	21.01%	1.1%	25.8%	0%	33.3%	46.7%	20.1%	0.03 kg
4	nvidia/AceMath-1.5B-Instruct	20.19%	32.12%	16.76%	52.87%	3.24%	4.32%	11.82%	1.66 kg
5	Qwen/Qwen2.5-1.5B-Instruct	18.43%	44.76%	18.15%	22.05%	0.78%	3.19%	19.99%	1.37 kg
18	infineon/ifx-elm-13m	13.0%	0.3%	16.1%	0%	24.1%	28.1%	9.8%	0.03 kg
100	infineon/ifx-elm-8m	3.7%	0.1%	4.0%	0%	8.0%	9.31%	1.12%	0.03 kg



ChatGPT delivers
response in
.3 watt-hour



PSOC™ Edge
delivers response in
.007 watt-hour

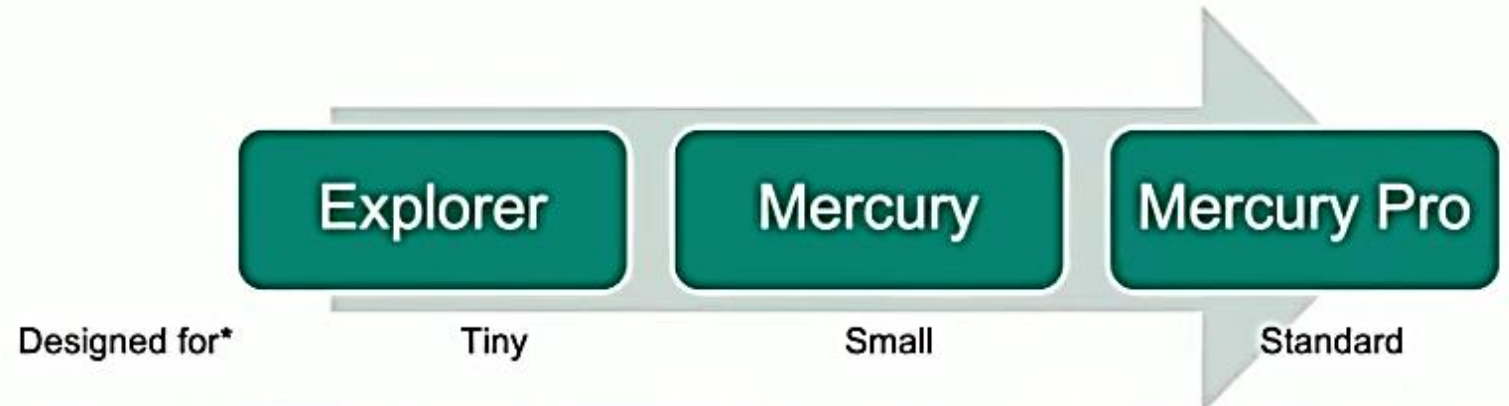


98%
energy savings

Infineon – 邊緣語言模型在 PSoC Edge



ELM-s optimized for PSoC™ Edge Roadmap



Model	ELM-8M (Tiny)	ELM-13M (Small)	ELM-25M (Standard)
Param Count	~8 million	~13 million	~25 million
Device Class	Small appliances, sensors, simple wearables	Advanced wearables, standard appliances, IoT hubs	High-end appliances, vehicle systems, industrial controllers
Typical Tasks	Voice commands, simple Q&A, basic dialog control (one or two turns)	Instruction following, classification, short summaries, multi-turn Q&A (brief)	Conversational assistant, extended responses, complex reasoning (edge scope), long-form Q&A
Context	Short (256–512 tokens)	Standard (512–1024 tokens)	Extended (1024–2048 tokens)
Memory	Minimal RAM (<<10 MB),	Moderate RAM (~10–20 MB)	Higher RAM (~20–40 MB)

7.6 Tokens/sec, TTFT < 4.5sec

2025-01-09

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*All models can run Explorer

18

Wipro Intelligence - 聲音轉譯使用Intel AI PC

EDGE AI FOUNDATION

wipro intelligence™
proof over promise.

Audio Translation on Intel AI PC

Anirban Bhattacharjee – Solution Architect (Embedded /Edge AI)

Acknowledgement

- Wipro team
- Intel team

OpenVINO

Offline video/audio translation

Converts recorded media into another language with high accuracy by processing complete audio/video content and regenerating translated speech.

Real-time audio translation

Enables live, low-latency speech-to-speech communication by continuously translating spoken input into another language during conversation.

On edge deployment on Intel AI PC with no network connectivity

Low latency

Data privacy

➤ 離線式影音轉譯

➤ 即時聲音轉譯

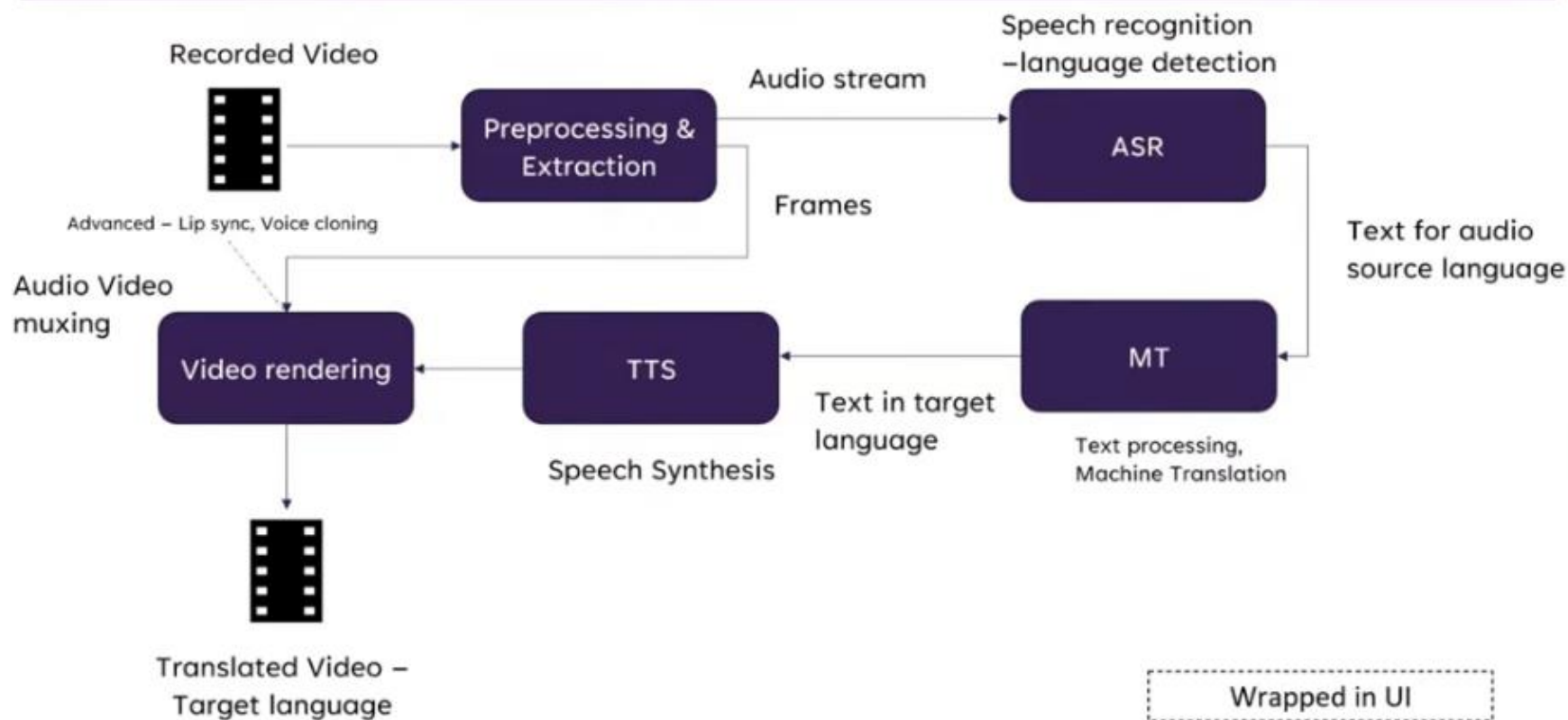
低延遲
高隱私

<https://youtu.be/kPJgEWDROtM?t=2536>

Wipro Intelligence - 聲音轉譯使用Intel AI PC

Processing Pipeline 離線式影音轉譯工作流程

wipro intelligence™
proof over promise.



Internal - General Use

5

未來工作

- 多語系擴展
- NPU支援
- 自動對嘴
- 音色複製
- 分散式處理
- 背影雜音處理

Wipro Intelligence - 聲音轉譯使用Intel AI PC

AI models and compute distribution

wipro intelligence™
proof over promise.

Subsystem	Model	Device	Precision	Disk Size	Memory Usage
ASR	Whisper Turbo	Intel GPU (Open VINO)	INT4	~500 MB	~500–700 MB
Translation EN→HE	MarianMT opus-mt-en-he	Intel GPU (Open VINO)	FP16	~200 MB	~200–300 MB
Translation HE→EN	MarianMT opus-mt-he-en	Intel GPU (Open VINO)	FP16	~200 MB	~200–300 MB
TTS English	VITS mms-tts-eng	CPU (PyTorch)	FP32	~250 MB	~200–300 MB
TTS Hebrew	VITS mms-tts-he	CPU (PyTorch)	FP32	~250 MB	~200–300 MB

“NPU was not used because the pipeline relies on dynamic, autoregressive transformer models and streaming execution, which are not so efficient on current NPUs optimized for static, fixed-shape inference.”

This design leverages open-source models and optimized inference frameworks to enable high-accuracy offline video and audio translation with efficient resource utilization on edge and desktop platforms. : Open VINO, Hugging Face Transformers, Optimum Intel, VITS, Gradio, FFmpeg

- Whisper : transformer-based speech recognition for multilingual transcription and language detection
- Marian / Helsinki-NLP: lightweight sequence-translation
- VITS: flow end-to-end
- Open VINO CPU/GPU performance

Reduced memory footprint

Peak memory , average memory, per stage memory (ASR, MT,TTS)

Balanced compute utilization

Device utilization percentage

性能評估標準

- 降低佔用記憶體
- 翻譯速度量測
- 平衡計算工具
- 速度因素計算

Translation time measurement

Translation time measures the duration required for converting transcribed text into the target language

Speed factor calculation

Speed Factor=
Total Processing Time /
Video Duration

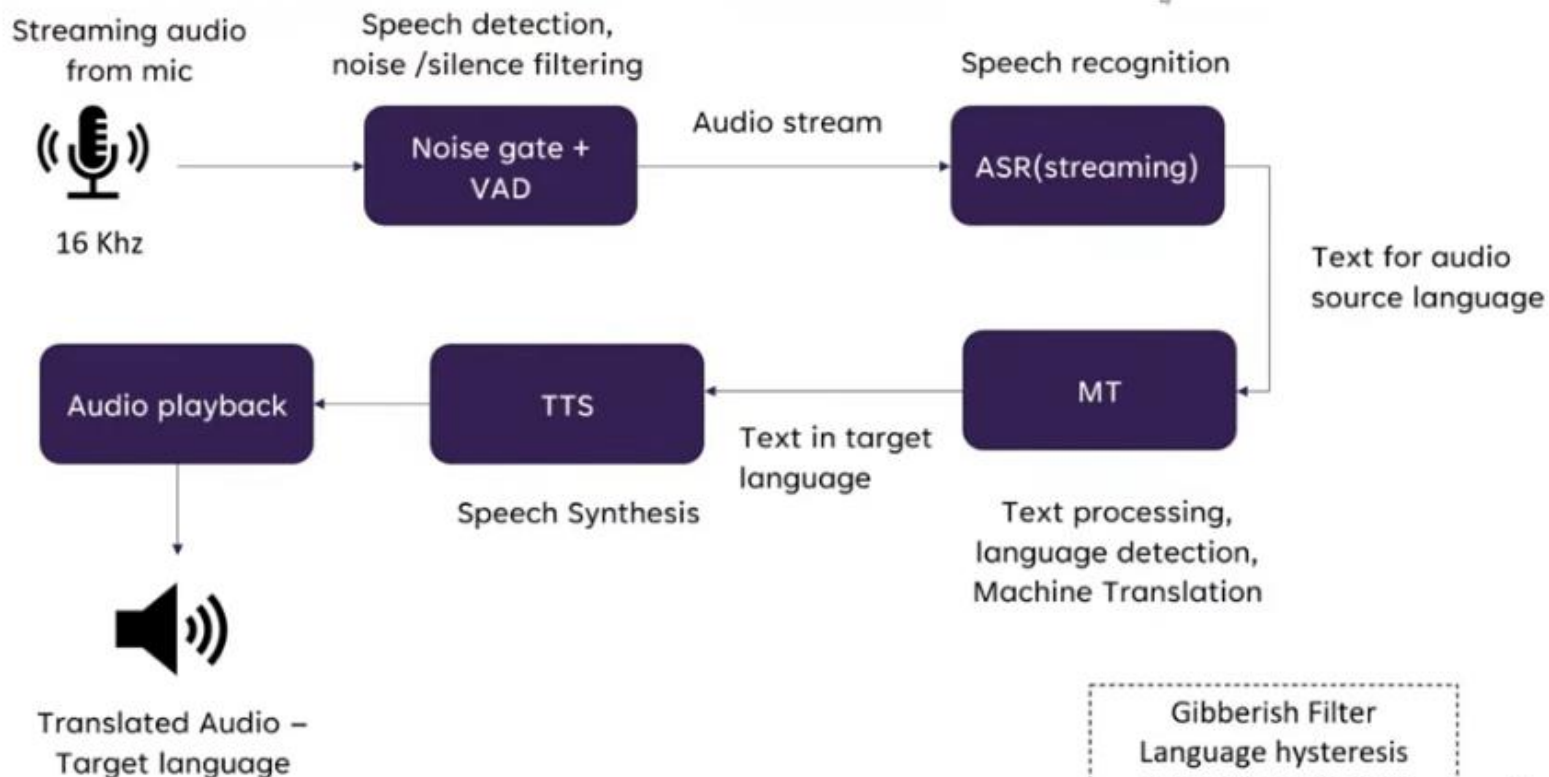
Internal - General Use

Wipro Intelligence - 聲音轉譯使用Intel AI PC

Processing Pipeline

即時聲音轉譯工作流程

wipro intelligence
proof over promise.



Component	Execution Unit
Audio Capture	Mic thread
VAD	Audio processing thread
ASR	GPU worker thread
Translation	GPU worker thread
TTS	CPU worker thread
Playback	Output thread

Pipeline Stage	Typical Latency
VAD + buffering	< 5 ms
ASR — Whisper Turbo INT4	70–120 ms
Translation — MarianMT beam=2	30–60 ms
TTS — VITS on CPU	50–80 ms
Playback scheduling	< 10 ms
End-to-end total	~200–300 ms

Internal - General Use

12

Sreenivasa Reddy – 3.2%幻覺率殺了AI



3.2%
HALLUCINATION RATE

That Almost Killed Our AI
And How I & My team Fixed It

→ **~0.9%**

Sreenivasa Reddy Hulebeedu Reddy
IEEE Senior Member | ACM Member
5th Generative EdgeAI Forum | April 21–22, 2026



WHY EDGE MAKES THIS HARDER



Limited Compute

Can't throw GPU clusters at the problem. Every inference cycle counts.

Latency Sensitivity

Cloud round-trips add 100–300ms. Users abandon after 3 seconds.



No Cloud Fallback

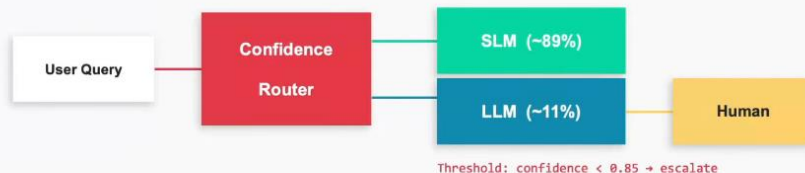
Autonomous agents must decide correctly without phoning home.

<https://youtu.be/kPJgEWDRotM?t=4287>

Sreenivasa Reddy – 3.2%幻覺率殺了AI

01 CONFIDENCE-BASED ROUTING

Route by Confidence, Not Model Size



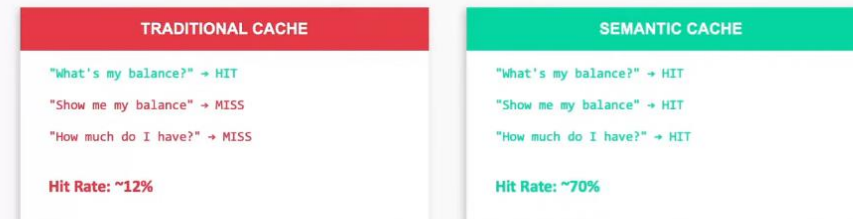
KEY INSIGHT

We stopped chasing bigger models and built intelligent routing instead.

~65–70% cost reduction | Better accuracy
SLMs handle ~89% of remaining queries after cache

02 SEMANTIC CACHING

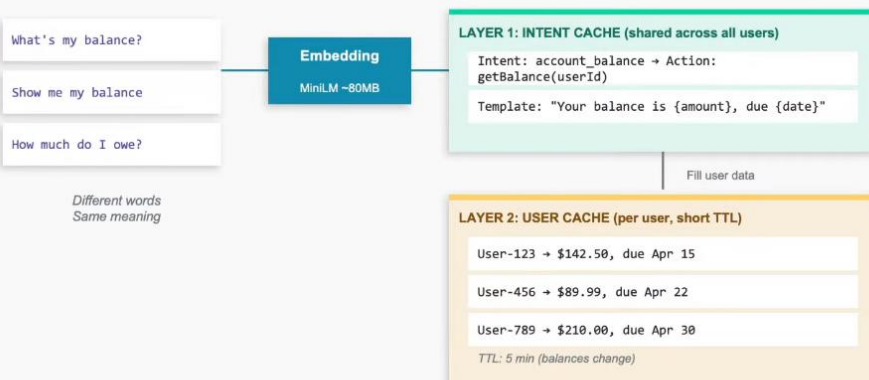
Cache Meaning, Not Strings



WHY THIS MATTERS AT THE EDGE

~70% of requests served with zero model inference.
No GPU needed. No cloud round-trip.
Sub-second response on edge hardware.

HOW THE CACHE HANDLES DIFFERENT USERS



~70% hit rate comes from Layer 1 (shared intent cache) | Generic queries = same answer for everyone

03 INTENT-FIRST ARCHITECTURE

Decompose Intent Before Generation

"I want to cancel"

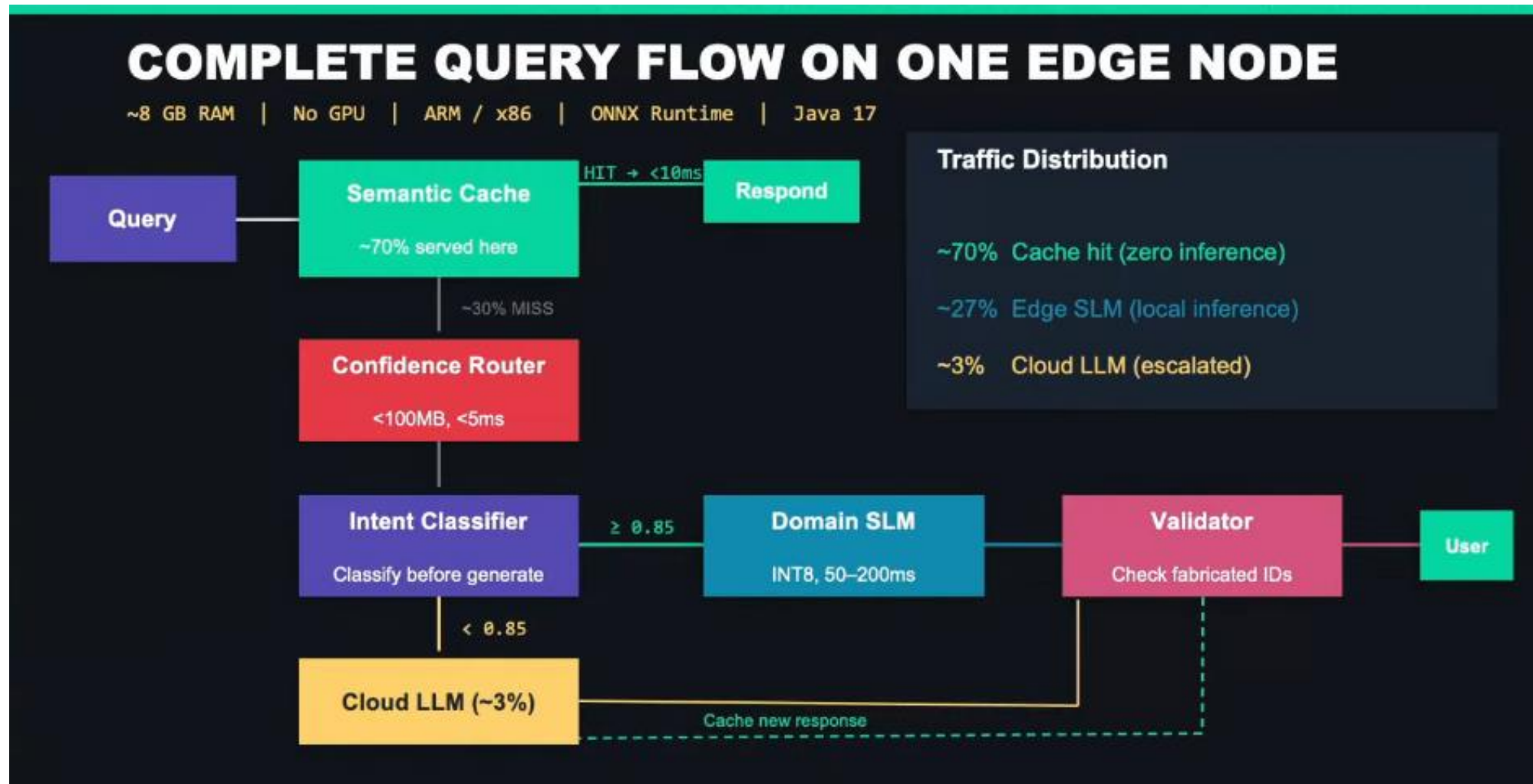
Cancel appointment? Cancel subscription? Cancel last action?



EDGE ADVANTAGE

Smaller specialized models outperform larger generalist ones.
SLMs handle ~89% of remaining queries.
Bounded autonomy — agents know what they know.

Sreenivasa Reddy – 3.2%幻覺率殺了AI



Sreenivasa Reddy – 3.2%幻覺率殺了AI

EDGE HARDWARE REQUIREMENTS		
Confidence Router + Intent Classifier	Semantic Cache Layer	Domain SLMs (per model)
CPU: ARM Cortex-A / x86-64	CPU: Any	CPU: 4~8 cores ARM/x86
RAM: 512 MB	RAM: <2 GB total	RAM: 4 GB per model
Models: <100 MB total	Embedding: ~130 MB	Model: 500 MB-2 GB (INT8)
Latency: <5 ms	Lookup: <10 ms	Inference: 50-200 ms
GPU/NPU: Not required	GPU/NPU: Not required	NPU: Optional (2-3x)
TOTAL EDGE FOOTPRINT: ~8 GB RAM No GPU ARM or x86 BOM < \$200		

DEPLOYMENT TOPOLOGIES COMPARED		
CLOUD-ONLY	EDGE-HYBRID ★	EDGE-NATIVE
Latency: 200-500ms	Latency: <50ms (~97%)	Latency: <50ms (100%)
Offline: No	Offline: Yes (~97%)	Offline: Yes (100%)
Cost/query: ~\$0.003	Cost/query: ~\$0.0004	Cost/query: ~\$0.0001
Edge traffic: 0%	Edge traffic: ~97%	Edge traffic: 100%
Full capability, full dependency	RECOMMENDED — Best of both	Max autonomy, bounded capability

硬體要求：三個步驟可分別採用不同硬體來滿足推論所需算力、延遲、精度、耗能。如上表所示可搭配 Arm Cortex-A CPU, 4~8 核 Arm / x86 CPU 或 GPU。

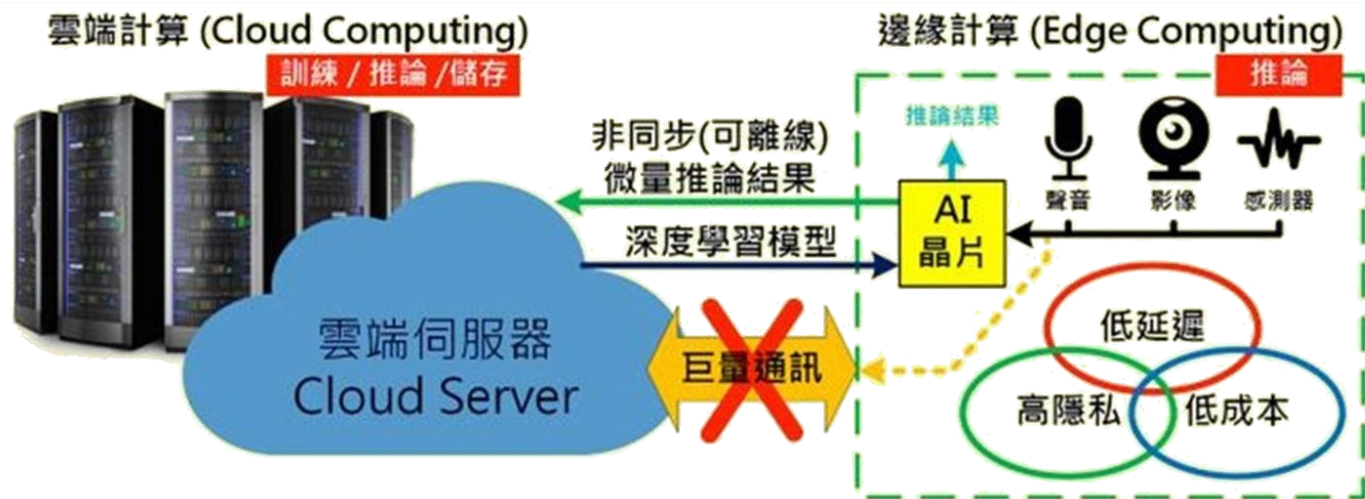
部署拓樸比較：可以得到較大模型服務有更好的精度，但延遲較長、價格較高，全部地端（邊緣）視硬體等級會得到較快的反應及較低推論成本，但受限硬體亦會造成精度不佳。可採混合式作法來平衡需求。

Sreenivasa Reddy – 3.2%幻覺率殺了AI

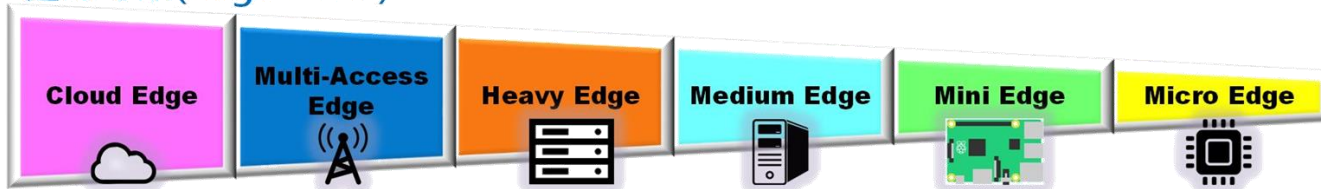
PRODUCTION RESULTS			
METRIC	BEFORE	AFTER	IMPROVEMENT
Hallucination Rate	~3%	~0.9%	~72% ↓
Inference Cost	High	~65-70% lower	~65-70% ↓
Cache Hit Rate	~12%	~70%	~6x ↑
Customer Escalations	~12%	~7%	~42% ↓
System Availability	99.1%	99.95%	↑
Quality Issue MTTR	4+ hrs	~8 min	~30x ↓



Edge AI

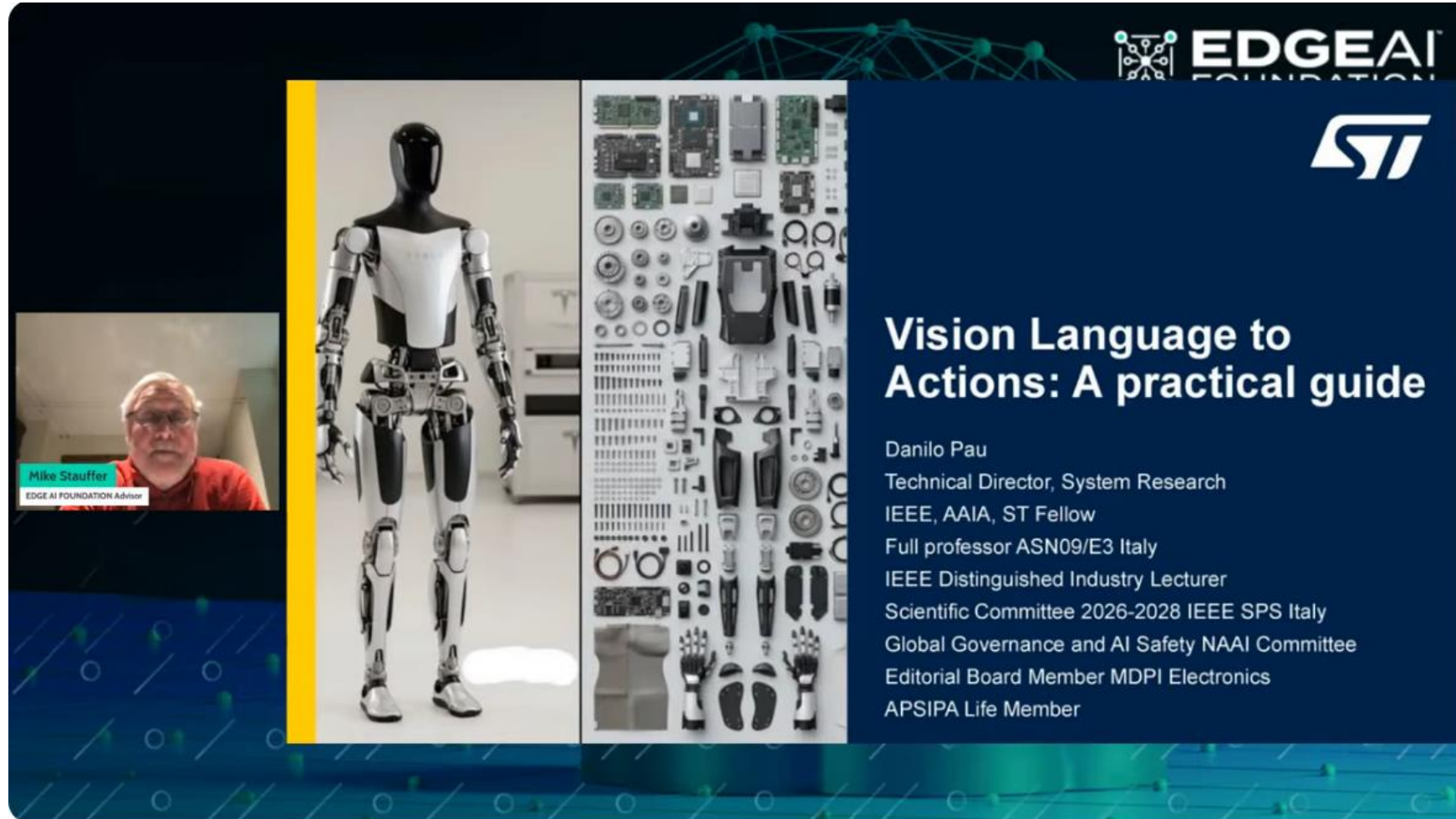


邊緣等級(Edge Level)



12.3. 代理式人工智慧與分散式系統

STM – 視覺語言到行動：實務指南



<https://youtu.be/kPJgEWDROtM?t=6210>

STM – 視覺語言到行動：實務指南

The teleoperation paradigm



- The approach
 - Bypass the “**autonomy challenge**” by keeping the human in the loop
 - The robot is **the physical avatar** deployed for safe, precise interaction in complex, unstructured environments (medical, emergency etc)
- **Master Maneuvering System (MMS)**
 - **Immersive seat:** equipped with wearable controls, data gloves, and HMD
 - **Master alarms:** grants full 1-to-1 range of motion over the robot axes and fingers
 - **Master foot:** allows operator to walk in place to drive lateral and forward mobility
 - **Self-interference prevention:** automatically ensures robot and user do not disrupt each other's movements

STM – 視覺語言到行動：實務指南

全自動
VS.
遠端遙控



	Figure 03	Toyota T-HR3
AI approach	VLA end-to-end neural net	Human in the loop
Primary training approach	Nvidia Isaac Lab	Human guidance
Key hardware differentiator for human-like delicacy	• 3 grams tactile, • 7th Gen Hands 16 dof, • mmWave low latency data comms	• Cr-N Alloy strain gauges • Torque Servo Modules • Force feedbacks
Target use	True general purpose (Home & Commercial)	Safe assistant (Medical, emergency, home)

感知操作環境

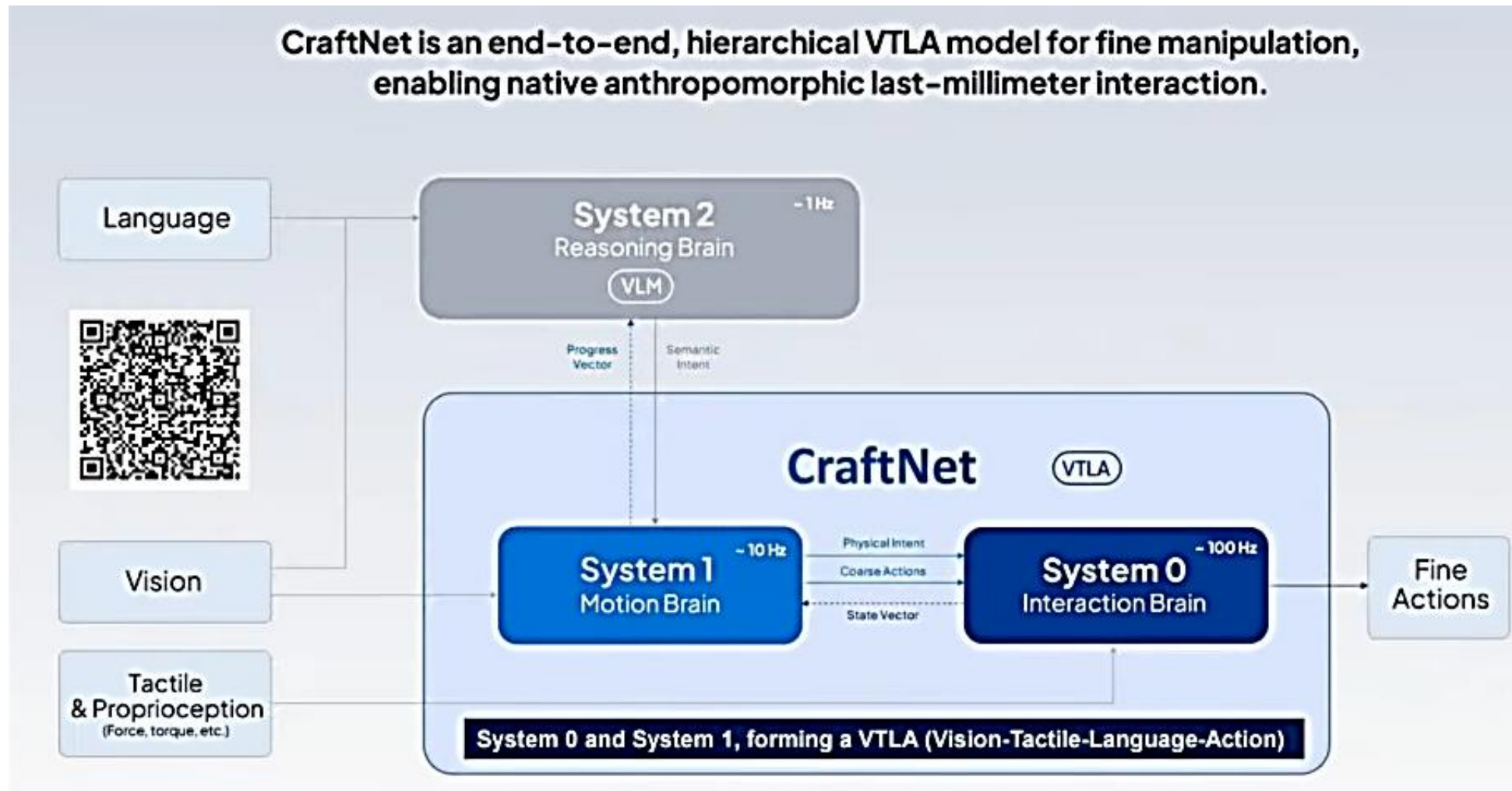
具備自我覺察並理解作業環境



透過執行決策來採取行動

基於所獲得的理解進行推理並做出決策

STM – 視覺語言到行動：實務指南



VTLA模型

視覺
(Vision)

觸覺
(Tactile)

語言
(Language)

動作
(Action)

STM – 視覺語言到行動：實務指南

S2:
感知理解和語言
7B VLM, 10Hz

S2: Scene Understanding and Language

- The semantic reasoning layer: processing scenes, understanding language, and producing latent goals for S1. Example: "Go back to the top rack and pick up the cups"
- Architecture: **7B VLM** working at **10Hz**

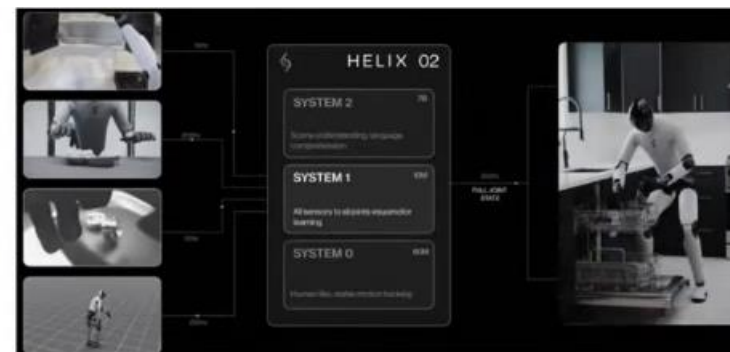


Figure.A1
Helix-2

S1:
來自感測器的機
器人狀態分析
80M , 200Hz

S1: Robot state analysis from Sensors

- On the ground of S2 inputs on main goal, it analyses the inputs from sensors for instance head camera, hand palm camera/TOF, hand tactile/pressure sensors, to provide inputs for S0 to drive complex manipulation.
- Architecture: **80M Transformer** working at **200 Hz**.



S0:
類人，全身控制
10M NN, 1KHz

S0: Human-like, whole-body Control

- Drives human body control on the ground of S1 inputs.
- Uses a 10M-parameter neural network mapping full-body joint state and base motion to joint-level actuator commands at 1 kHz.
- Learns loco-manipulation by human motion imitation from 1,000+ hours of retargeted joint-level data.



Introducing Helix 02: Full-Body Autonomy

STM – 視覺語言到行動：實務指南

The nVIDIA summary



高階推理
(1 ~ 5 Hz)

感知和計畫
(30 Hz)

即時控制框架
(100 ~ 1K Hz)

硬體抽象層



Presented @ 2026 IEEE ISSCC

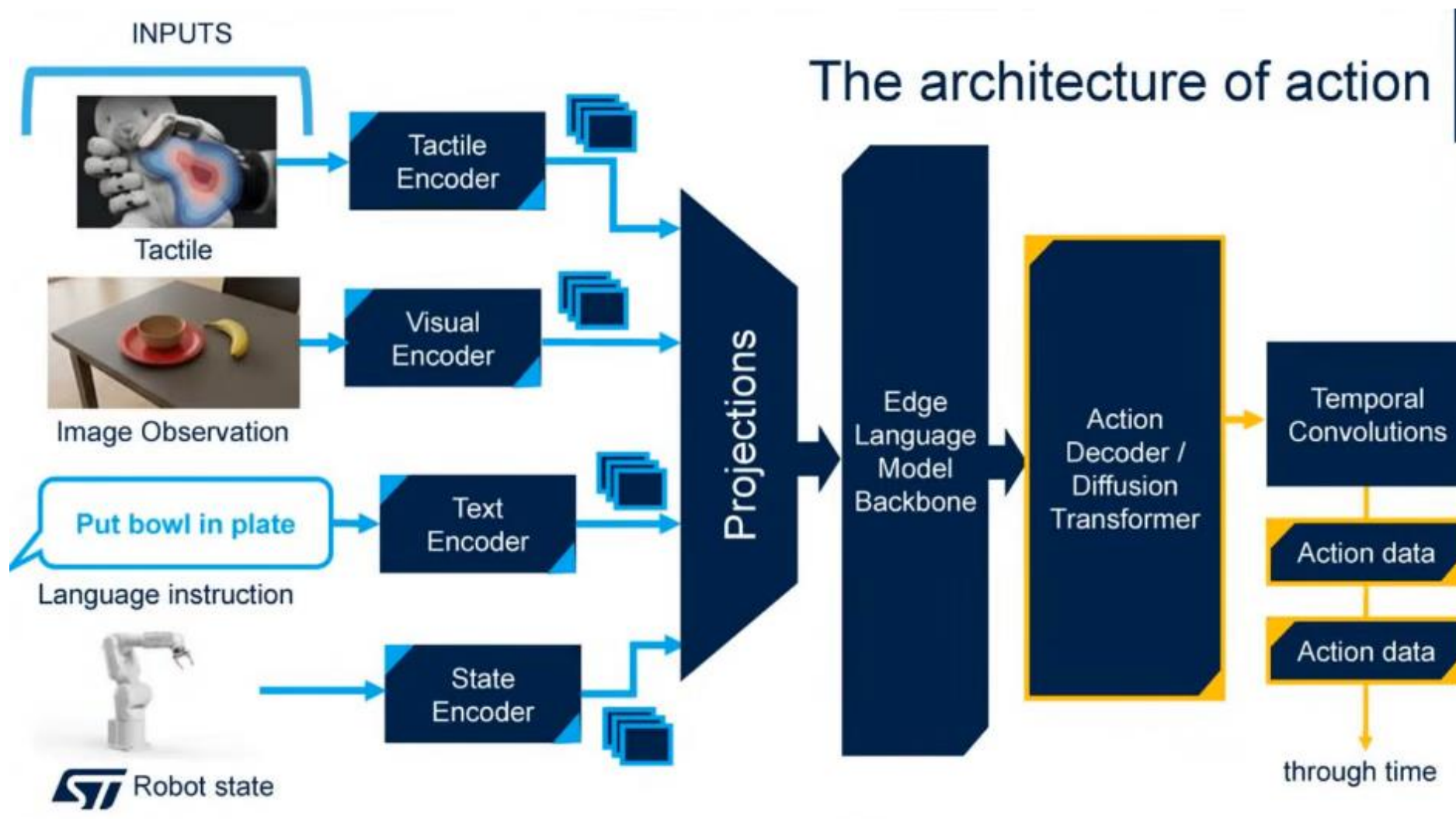
STM – 視覺語言到行動：實務指南

Requirements summary

	System 0	System 1	System 2
HELIX		200 Hz 80 M Transformer	9 Hz 7b VLM
HELIX 2	1000 Hz 10 M	200 Hz 80 M Transformer	9 Hz 7b VLM
CraftNet	100Hz	10Hz	1Hz VLM
nVidia	100-1000 Hz	30 Hz	5 Hx



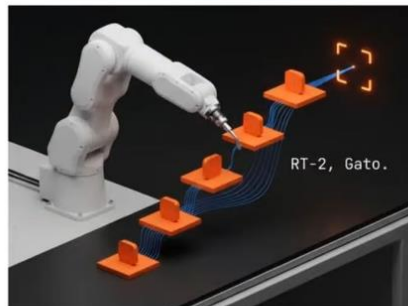
STM – 視覺語言到行動：實務指南



STM – 視覺語言到行動：實務指南

Architectural Divergence: Tokens vs. Diffusion

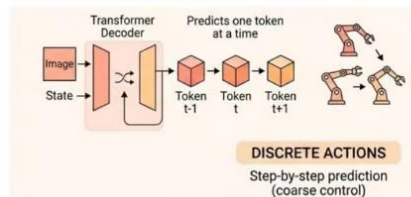
Autoregressive (Discrete Tokens)



RT-2, Gato.

Decoder-Based Autoregressive Generation

- Treats actions as sequences of discrete tokens and predicts them autoregressively, one token at a time
- Learn and generate P_{data}
- step-by-step actions too coarse for fast, precise, dexterous control
- Models : RT-2, OpenVLA



Architectural Paradigm Shift in VLA Models

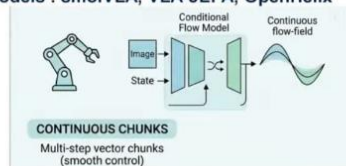
Diffusion (Continuous Denoising)



Octo, OpenVLA.

Conditional Flow Matching based action chunk Generation

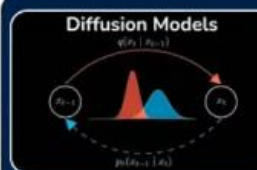
- Treats actions as continuous chunks (multi-step control outputs) and generates these continuous action vectors directly.
- Gradually convert one distribution to another being compute-efficient
- produce smooth, high-frequency actions enabling precise, dexterous control
- Models : smolVLA, VLA-JEPA, OpenHelix



Integration of Multimodal Sensing (Tactile & Force)

- **Beyond Vision:** New architectures are incorporating **tactile and force sensing** to handle "contact-rich" tasks where vision alone fails (e.g., detecting slips or managing insertion force).
- **Tactile-VLA Models:** Models like **Rho-Alpha** and **VLA-Touch** are demonstrating significant success in high-precision tasks such as charger insertion and bimanual manipulation.

Diffusion Models



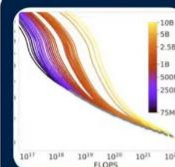
Shift from Autoregressive to Diffusion & Hybrid Models

- **Efficiency & Control:** Researchers are moving away from purely autoregressive policies toward **diffusion-based and hybrid models**. These approaches generate action sequences more efficiently and provide better alignment between abstract reasoning and low-level motor control.
- **Discrete Tokenization:** Some systems, like **Cortex VLA**, are tokenizing both actions and perception into a language-like vocabulary, allowing them to leverage the reasoning power of large language models for robot control.)



Long-Horizon Reasoning and World Models

- **Goal-Oriented Missions:** Robots are evolving from executing single instructions to fulfilling broad missions (e.g., "inspect this facility") by inferring context and planning multi-step routes.
- **Visual Foresight:** Emerging models "foresee" environmental changes before acting. A major research goal for 2026 is achieving **1-hour coherent world model predictions** to allow robots to simulate "what-if" scenarios internally.



Challenging the Scaling Laws: moving to "Small-Scale"

While the "scaling laws" for VLA models are being tested, there is a strong trend toward **lightweight efficiency**. Models like **Evo-1** (0.77B parameters) are achieving state-of-the-art results (94.8% on LIBERO) with significantly fewer parameters than larger foundations.



Generalization Across Robot Morphologies

- **"One Brain, Any Robot":** The industry is pushing toward generalist policies that can be deployed across different embodiments (e.g., humanoids, mobile manipulators, and robotic arms) with minimal fine-tuning.
- **Commercial Rollouts:** Companies like **XPENG** are beginning to integrate VLA 2.0 into humanoid robots and autonomous vehicles for mass production scheduled for late 2026.

Key Trend in 2026

Intel – 全棧機器人VLA (OpenVINO)



<https://youtu.be/kPJgEWDRotM?t=11497>

Intel – 全棧機器人VLA (OpenVINO)

[Learn more](#)


Physical AI Studio

Summary

An open-source application designed to simplify Physical AI model development on Intel hardware. It streamlines the VLA model lifecycle—from robot calibration and data collection using imitation learning, through model training, to inference on Intel platforms.

Key Features	Supported Policies	Supported Robots
- Simple and modular API and CLI for training, inference, and benchmarking Vision-Language-Action (VLA) policies. - Export models to OpenVINO, ONNX, or Torch formats for accelerated inference. - Benchmark policies on standardized environments like LIBERO and PushT. - Unified inference API across all export backends.	- ACT - SmoIVLA - Pi0 - Pi0.5 - GR00T N1	- SO-ARM101 - AlohaMini - LeKiwi - WidowX AI

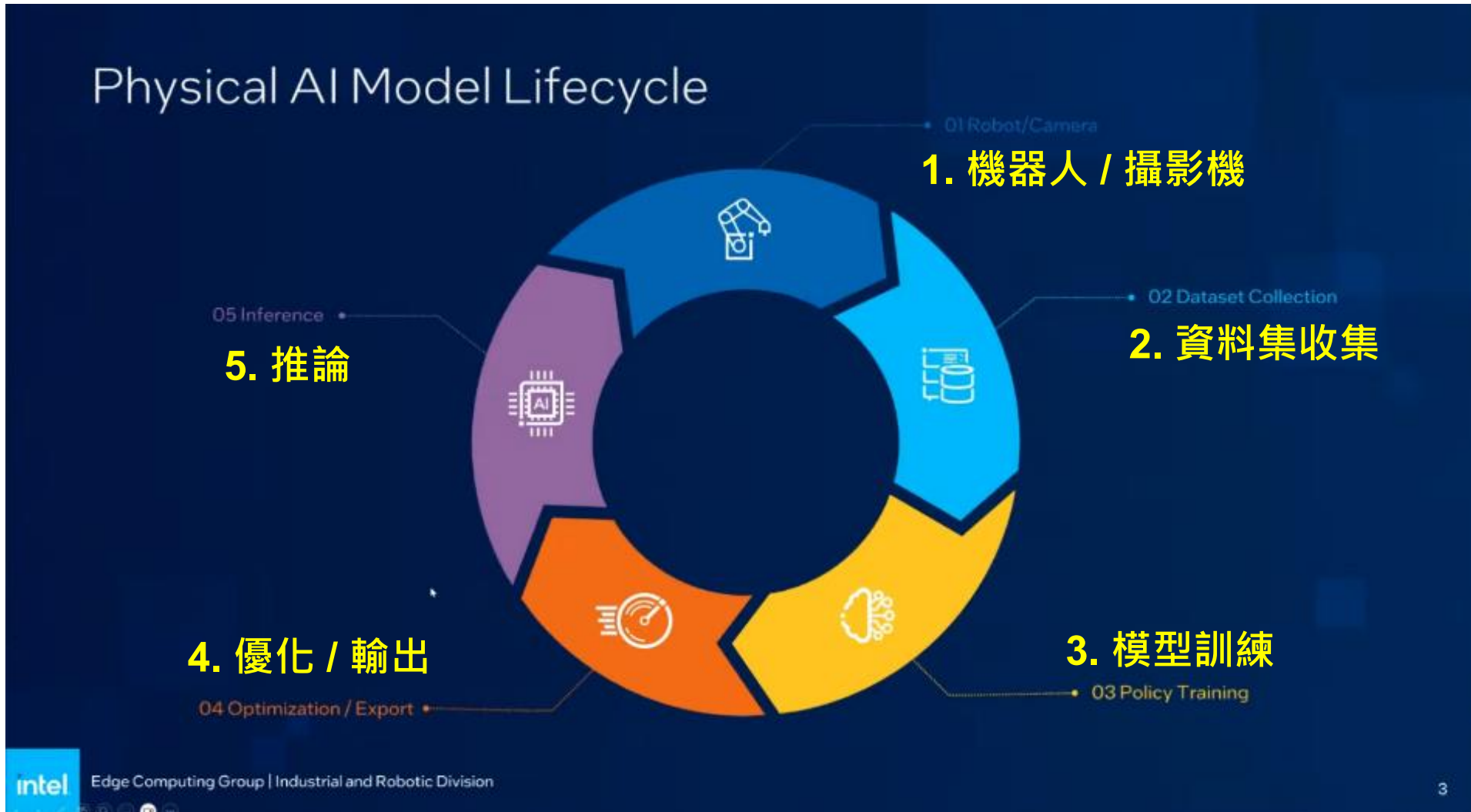




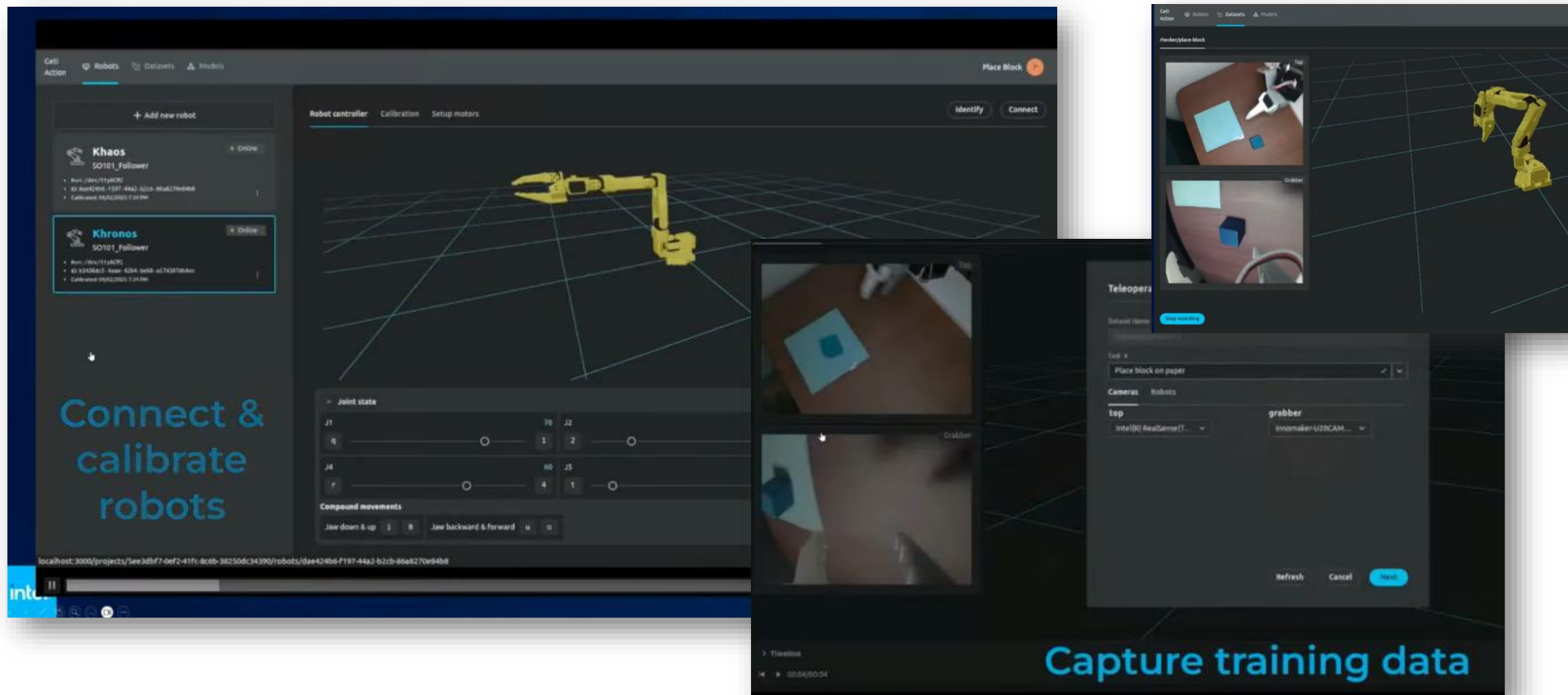


Edge Computing Group | Industrial and Robotic Division

Intel – 全棧機器人VLA (OpenVINO)



Intel – 全棧機器人VLA (OpenVINO)



Intel – 全棧機器人VLA (OpenVINO)

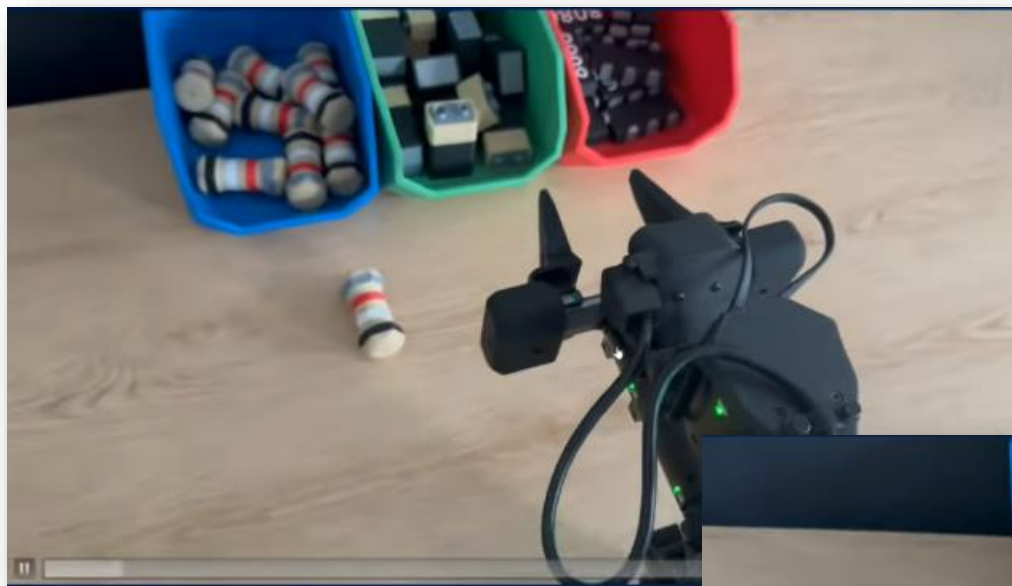
The screenshot displays the Intel OpenVINO VLA interface, which is used for training and inference of Vision-Language-Action (VLA) models. The interface is divided into several sections:

- Top Bar:** Contains navigation tabs for 'Get Action', 'Robots', 'Datasets', and 'Models'. A 'Place Block' button is visible on the right.
- Left Panel:**
 - Checker/place-block:** A video feed showing a robotic arm interacting with a block on a table.
 - Timeline:** A graph showing various joint angles (Shoulder Flex, Shoulder Ext, Elbow Flex, Wrist Flex, Wrist Roll) over time.
- Center Area:**
 - View training data:** A large area for visualizing training data, featuring a 3D model of a robotic arm.
 - Models:** A table listing trained models. The table has columns for 'MODEL NAME', 'TRAINED', and 'ARCHITECTURE'.

MODEL NAME	TRAINED	ARCHITECTURE
Model #1	11/3/2025, 12:57:03 PM	act
- Right Panel:**
 - Model Run Model #1:** A video feed showing the model's output, with a 3D model of the robotic arm and the word 'Infer' overlaid.

Fine-tune VLA model

Intel – 全棧機器人VLA (OpenVINO)



找尋電阻

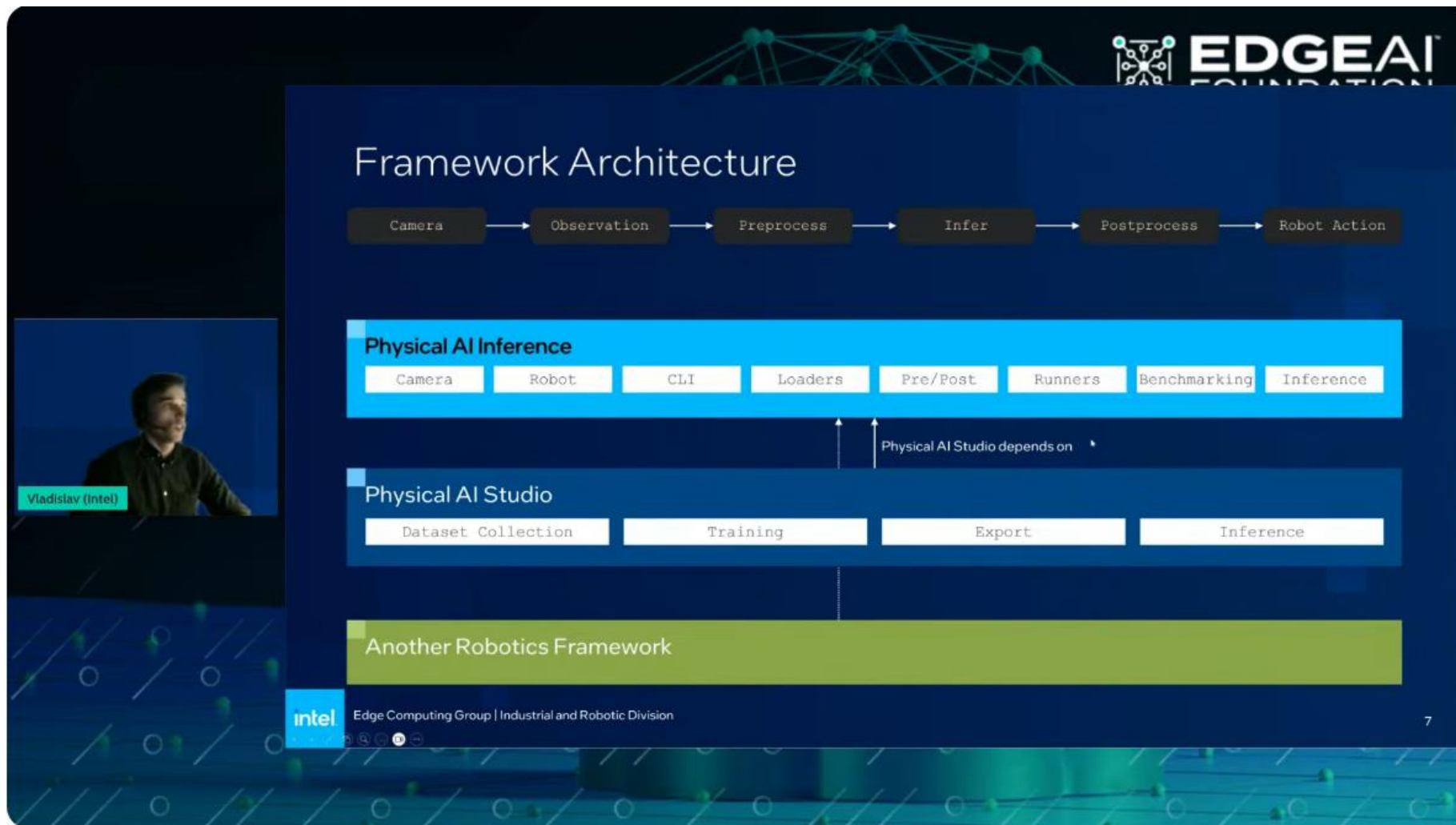


置放電阻

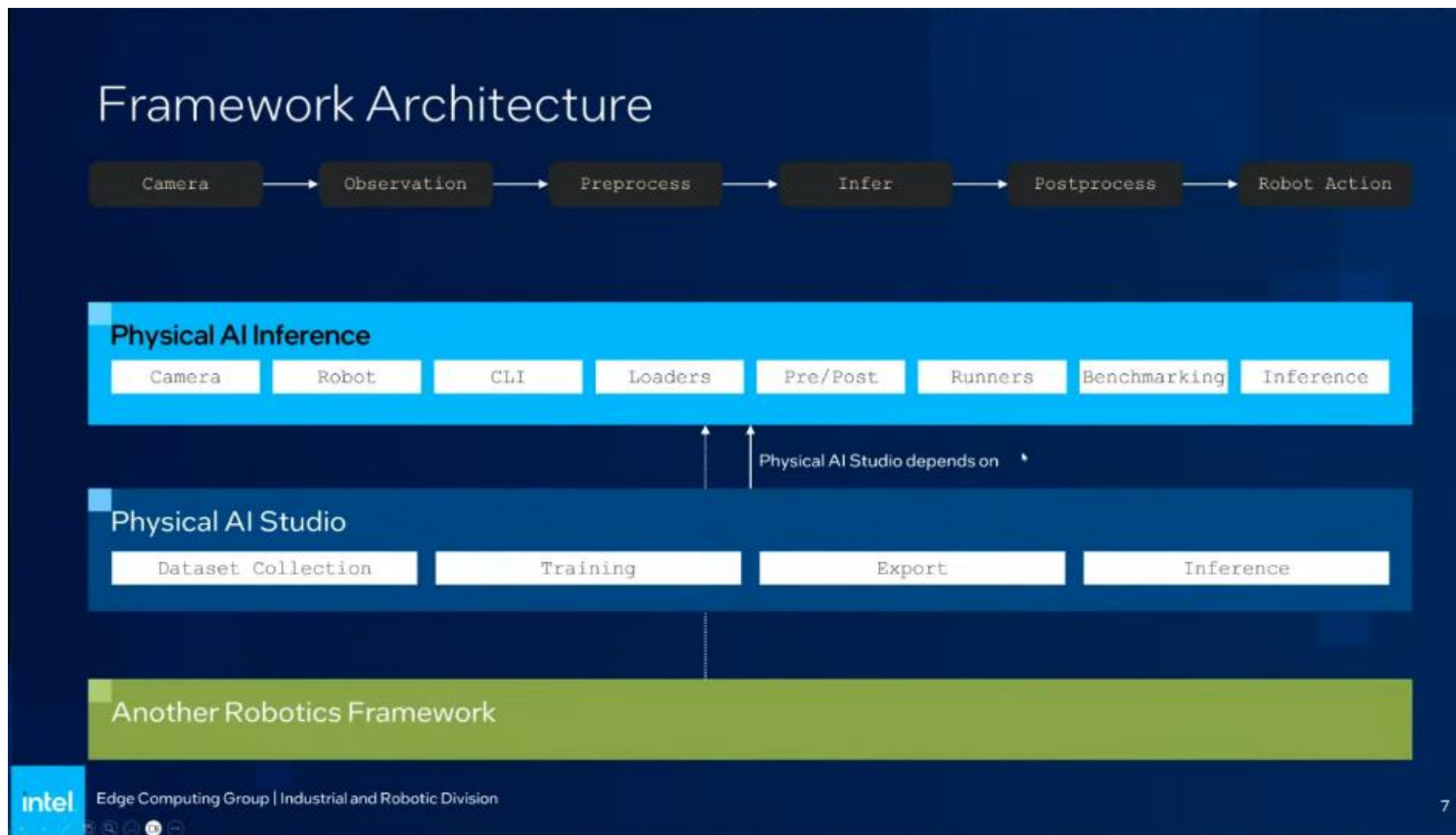


抓取電阻

Intel – 全棧機器人VLA (OpenVINO)



Intel – 全棧機器人VLA (OpenVINO)



Intel – 全棧機器人VLA (OpenVINO)

Physical AI Inference on Intel Hardware

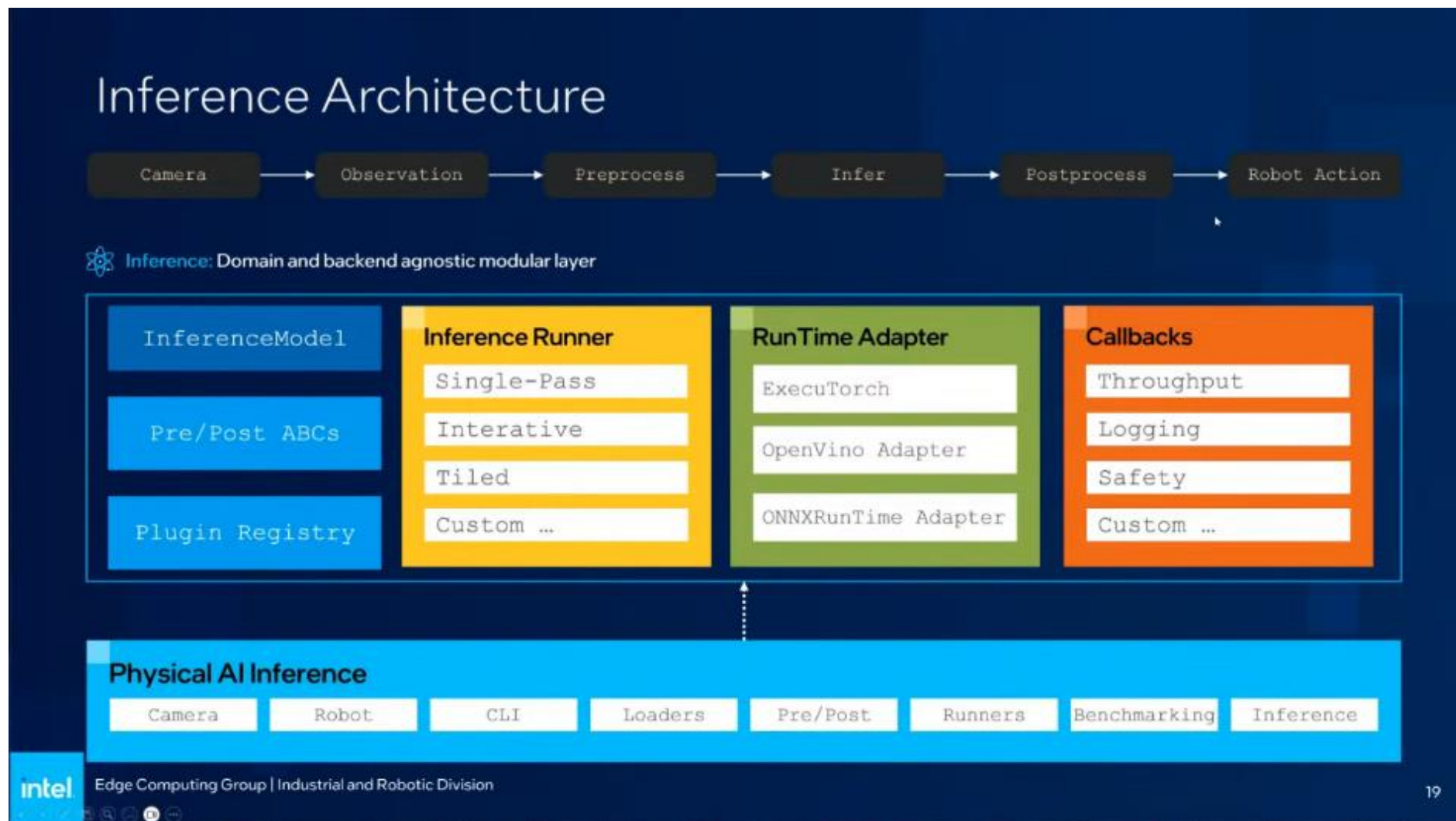
The Challenge

- Training to Inference a variety of hardware is DIFFICULT!
- Every customer/user re-implements similar export/deployment plumbing
 - Camera capture & Observation construction
 - Safety checks & action clamping
 - Episode Loop & Reset Logic
 - Device lifecycle & cleanup
 - Runtime Backend selection & model loading
 - Model Inference

The Proposal (Physical AI Framework)

- Standardize inference and deployment pipelines.
 - Inference Core (Domain agnostic)
 - Built-in format loaders (manifest.json)
 - Built-in runners (SinglePass, Iterative, ActionChunking)
 - Observation pipeline, safety runtime, episodes
 - Device management, CLI, validation
- Most models need zero external code
- One API. Same behavior
- Run on Intel hardware

Intel – 全棧機器人VLA (OpenVINO)



Intel – 全棧機器人VLA (OpenVINO)

Proposed: Shared Framework, Built-in Runners

 Proposed: Shared inference framework, built-in runners:



o Framework provides

- Observation pipeline (camera -> obs dict)
- Safety runtime (clamp, velocity, e-stop)
- Episode orchestration (run N, reset, log)
- Device management (lifecycle, cleanup)
- Built-in format loaders (metadata.yaml, manifest.json)
- Built-in runners (SinglePass, Iterative, ActionChunking)
- CLI / API

o Users still own (only if exotic):

- Custom Runner (exotic forward pass)
- Custom Preprocessor (exotic input shaping)
- Custom Postprocessor (exotic output shaping)
- SDK driver implementations

Most models: zero custom code needed.

Intel – 全棧機器人VLA (OpenVINO)

Backend Support

Hardware	Backend	Adapter
Intel CPU/GPU	OpenVINO	OpenVINOAdapter
Cross Platform	ONNX	ONNXAdapter
Torch Native	ExecuTorch	ExecuTorchAdapter
NVIDIA	TensorRT	TensorRTAdapter

```

Inference via CLI
> physai run --model ./act_policy --backend openvino --device cpu
    
```

Intel – 全棧機器人VLA (OpenVINO)



1

Training to Inference on a variety of hardware is difficult

2

Physical AI to standardize inference and deployment pipelines.

3

One API, same behaviour on frameworks, backends

4

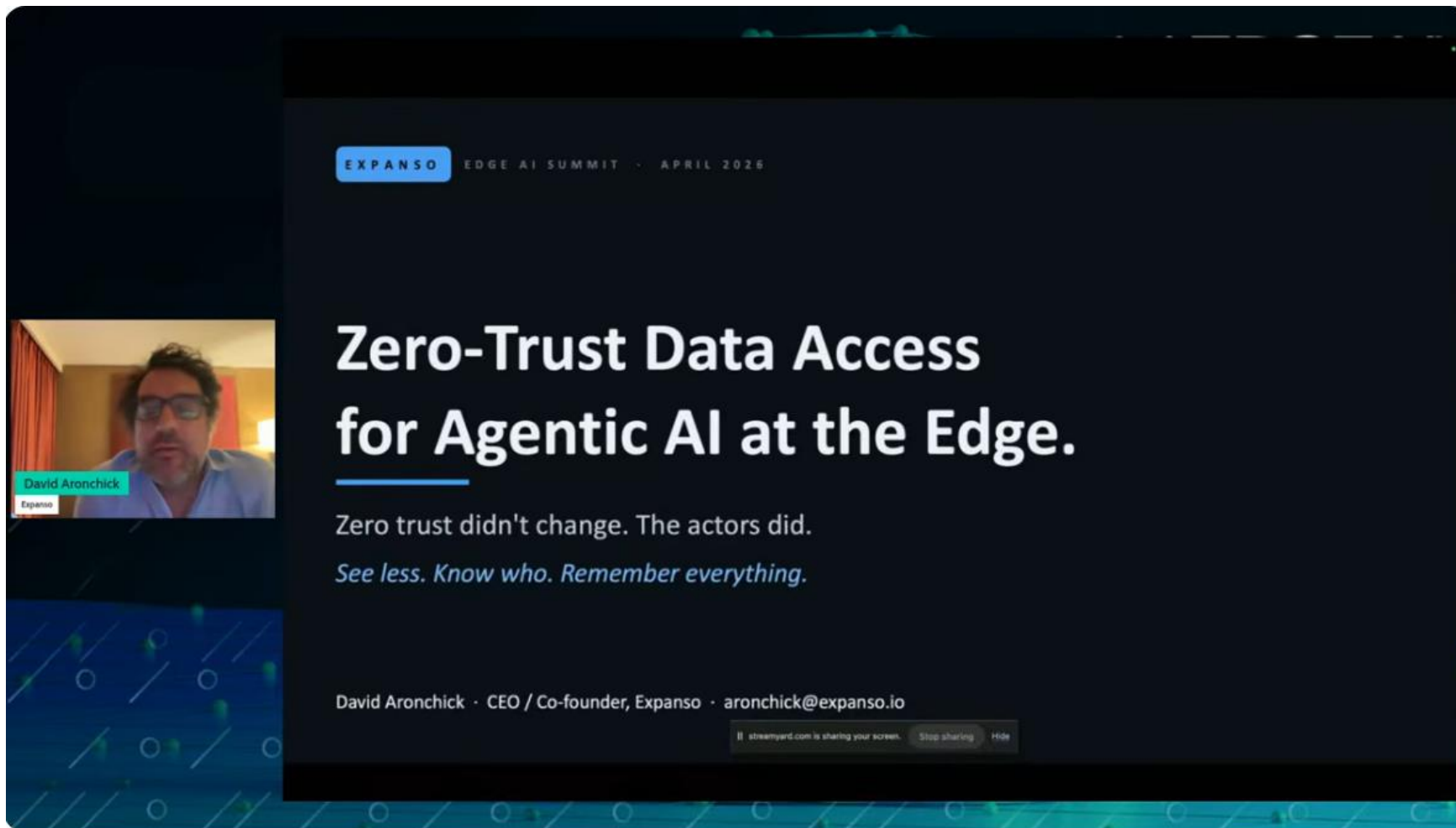
Runs on Intel hardware

intel

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29

Expanso - 邊緣代理人工智慧的零信任資料存取



零信任 (Zero Trust)
是一種網路資安策略，
核心精神為「永不信任，始終驗證」。

三大核心支柱

- 身份鑑別
- 設備鑑別
- 信任推斷

<https://youtu.be/kPJgEWDRotM?t=9858>

Expanso - 邊緣代理人工智慧的零信任資料存取

The screenshot displays the Expanso interface with a dark background. It shows a Q&A session where a user asks to run a QC on a morning batch, the AI responds with success and metrics, and an agent uploads raw feed data. Below this, a summary of data access is shown with three boxes: 'Where did the data go?' (UNKNOWN), 'What identity signed the upload?' (UNKNOWN), and 'Can we revoke it?' (NO). The interface also includes a status bar at the bottom indicating 'EXPANSO - ZERO-TRUST DATA ACCESS FOR AGENTIC AI' and a screen sharing notification from streamyard.com.

Q: Run QC on the morning batch.

AI: Batch passes. 847 items inspected. • CONFIDENCE: 0.93 • LATENCY: 142ms

A: Agent uploaded 12 GB of raw feed to a public summarizer.

And when the incident review arrives...

Question	Answer
Where did the data go?	UNKNOWN
What identity signed the upload?	UNKNOWN
Can we revoke it?	NO

EXPANSO - ZERO-TRUST DATA ACCESS FOR AGENTIC AI

streamyard.com is sharing your screen. Stop sharing Hide

執行工作

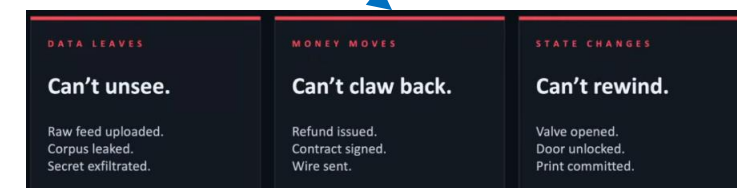
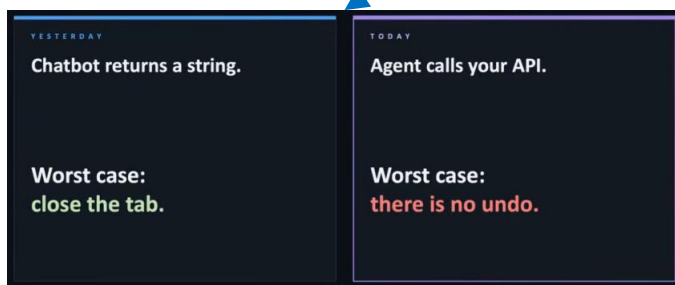
AI 代理

代理上傳

資料去哪？
上傳身份？
能撤銷否？

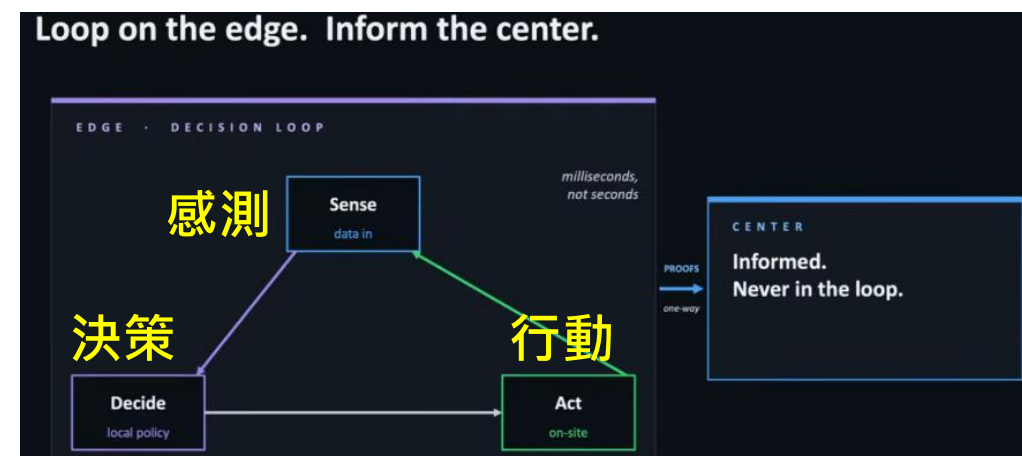
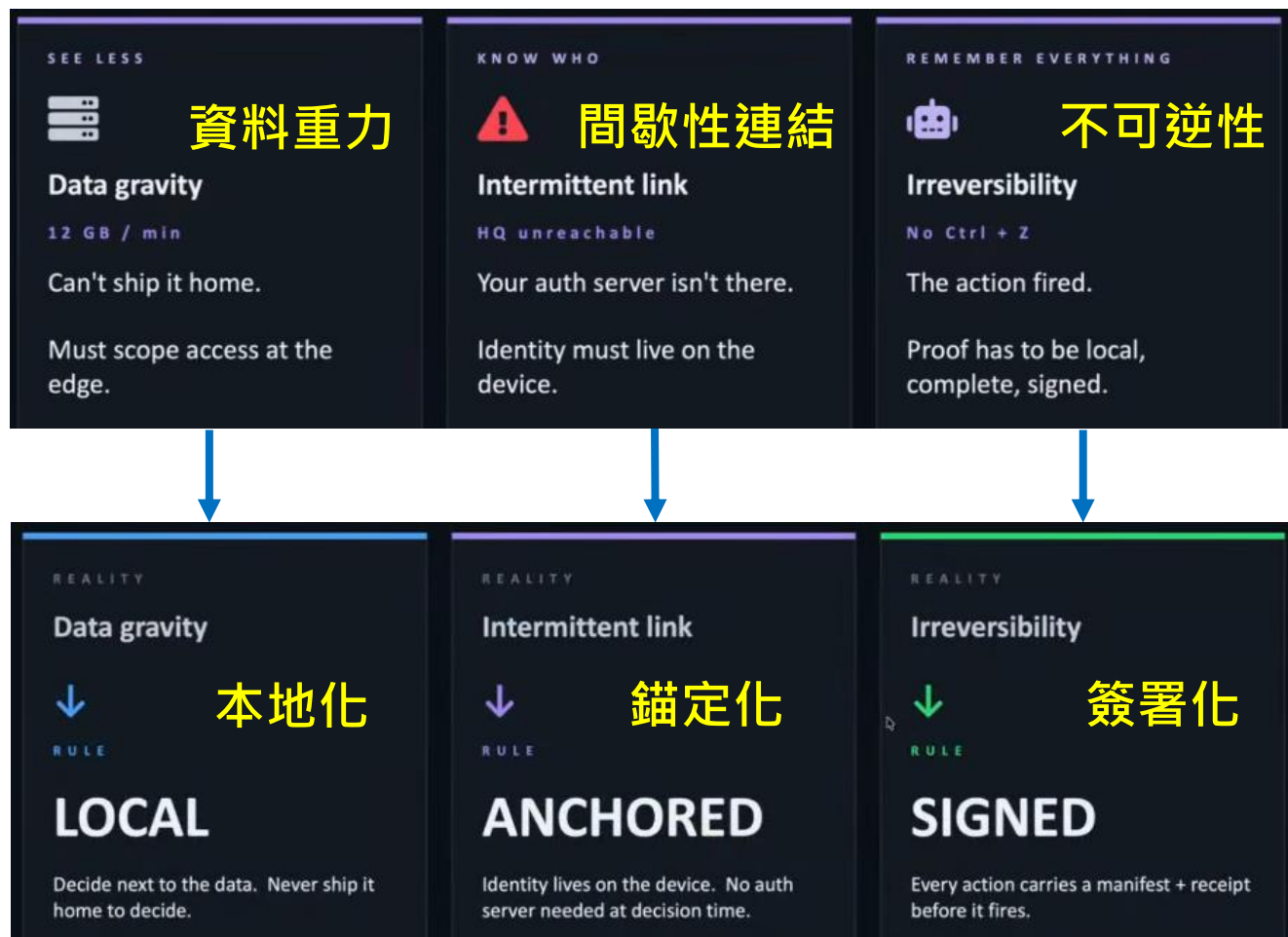
Expanso - 邊緣代理人工智慧的零信任資料存取

三件代理人能作到而聊天機器人作不到的事



Expanso - 邊緣代理人工智慧的零信任資料存取

邊緣強制三個規則本地化



Expanso - 邊緣代理人工智慧的零信任資料存取

Zero trust didn't change. The actors did.

 SEE LESS What data? Only what policy allows.	 KNOW WHO Which actor? The binary, on this device.	 REMEMBER EVERYTHING What happened? Signed. Replayable.
---	--	--

Three rules. Three primitives.

- SEE LESS
01 Gate the Data
 The agent never holds raw data.
- KNOW WHO
02 Bind the Identity
 Hardware, not a .env.
- REMEMBER EVERYTHING
03 Prove the Action
 Signed before. Signed after.

BEFORE <pre>SELECT * FROM prod.readings</pre> Agent holds everything. <i>Firehose. No scope. No receipt.</i>	AFTER <pre>SELECT qc FROM scoped_view</pre> Agent holds what policy allows. <i>Scoped. Redacted. Auditable.</i>
---	--

BEFORE <pre>API_KEY= sk-abc123...</pre> A secret in a file. <i>Copy the .env, become the agent.</i>	AFTER SVID bound to device TPM A key bound to the silicon. (or at least the instance) <i>Attested at boot. Not portable.</i>
--	---

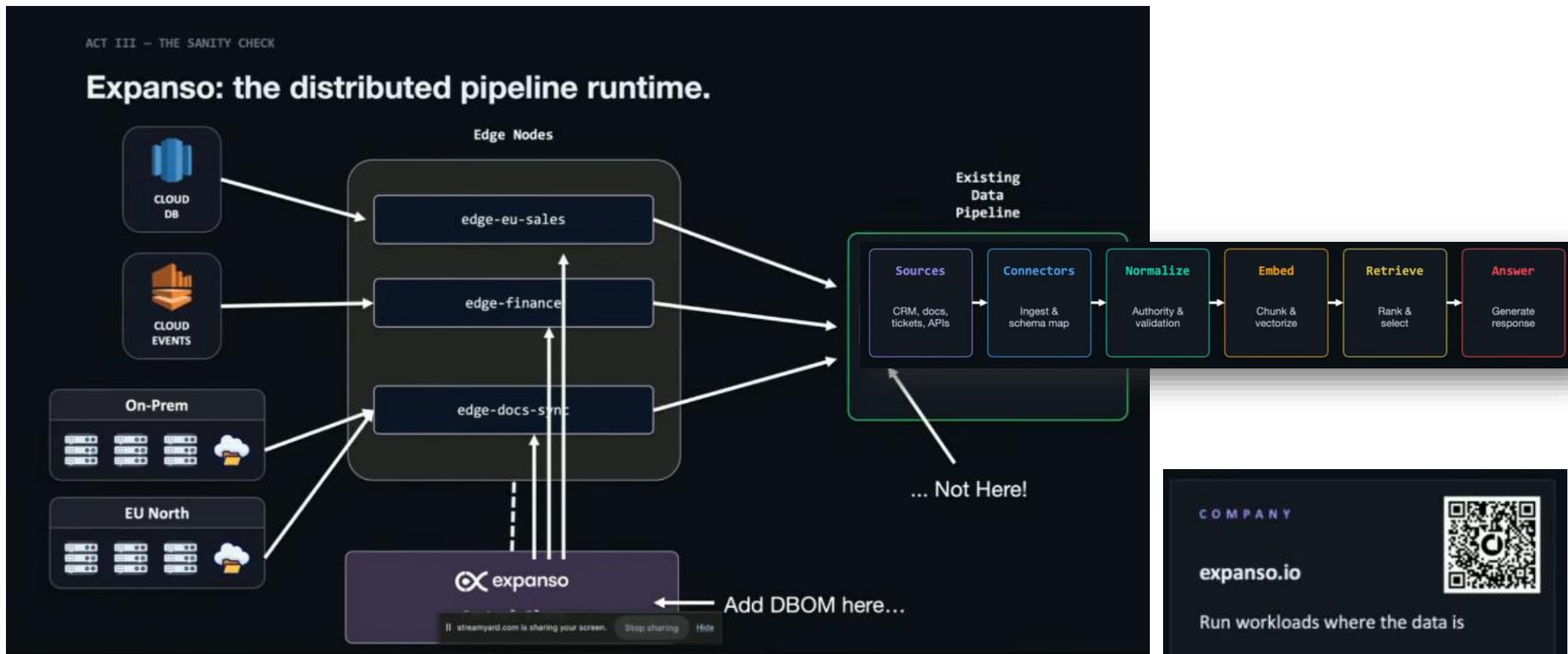
<pre>access-receipt.json - Makoto-style - signed, local, tamper-evident { "agent": "line7-qc-v1a@factory-eu-03", "device": "TPM:0x91af...", "read": "camera.station-3", "read-date": "2026-04-18T14:12:07Z", "policy": "qc.read.images@v3", "justification": { "prompt_hash": "sha256:a8f1...", "tools_used": ["vision.detect", "rules.qc.v2"] }, "sig": "ed25519:device+fleet, 0x1b2c..." }</pre>	What this proves Agent identity bound to device. Data boundary station-3 only. Policy version v3 in force. Decision trace prompt + tools.
--	--

取得資料

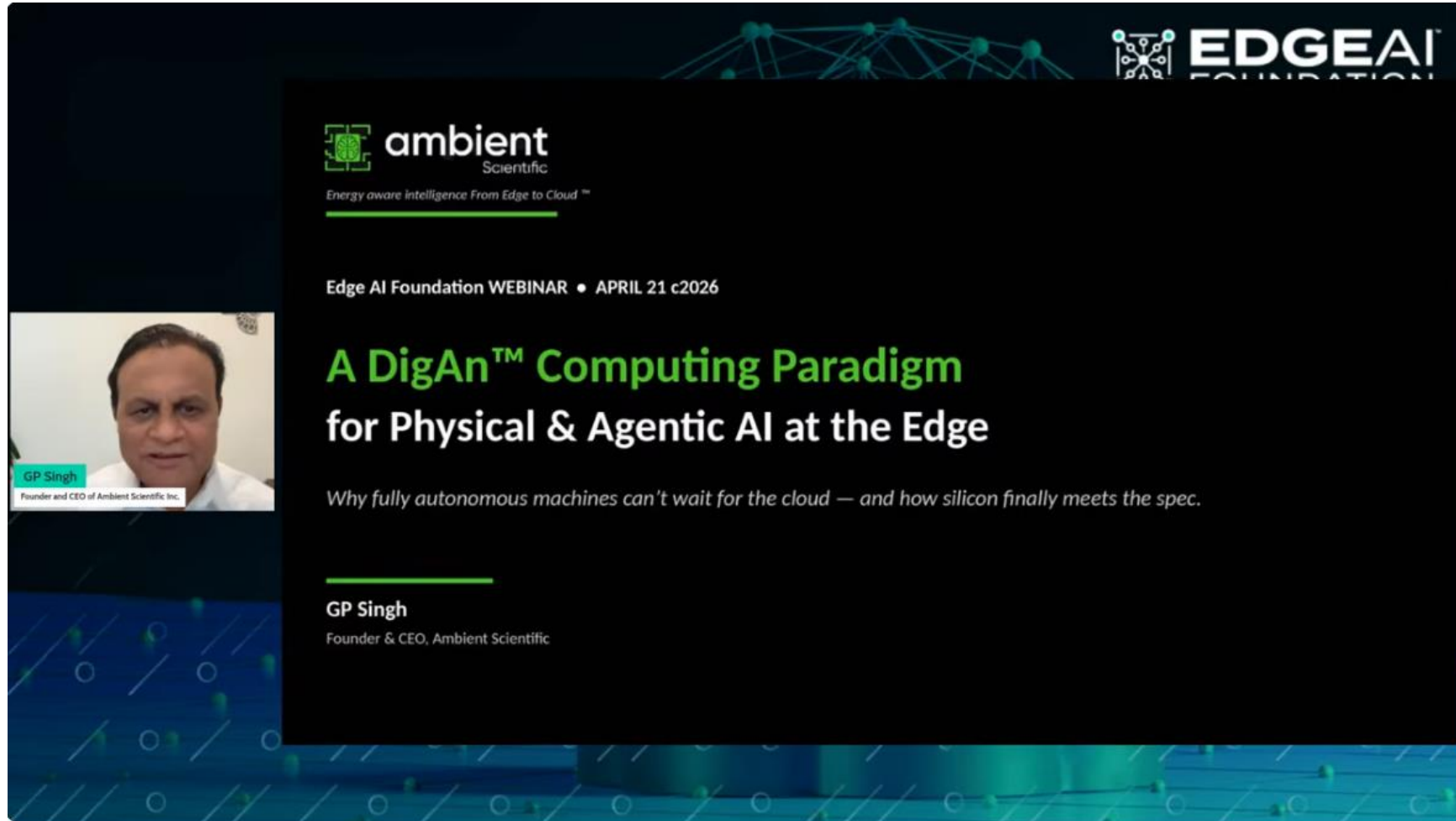
綁定身份

證明動作

Expanso - 邊緣代理人工智慧的零信任資料存取



Ambient – DigAn 混合信號計算範式



<https://youtu.be/kPJgEWDRotM?t=13382>

Ambient – DigAn 混合信號計算範式

Three shapes of modern AI

Before we talk silicon, let's agree on what we're actually trying to run.

Physical AI

Senses & acts in the world.

Vision, audio, motion, tactile intelligence. The model turns sensor streams into decisions in milliseconds.

- Hearables understanding your environment
- Robotic arms grasping unseen objects
- Drones avoiding obstacles in flight

Agentic AI

Plans, reasons, and takes actions.

An LLM-class model that holds context, calls tools, and orchestrates multi-step work on a user's behalf.

- On-device assistant drafting replies
- Scheduling, researching, summarizing
- Private data stays under user control

Autonomous AI

Closes the loop without a human.

Physical + Agentic fused: perceives, decides, and acts continuously, safely, and alone. The fully autonomous machine.

- Humanoid robots doing real work
- Self-driving vehicles & delivery bots
- Autonomous drones & industrial systems

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Different jobs. Different compute bills. One shared constraint.

	Physical AI	Agentic AI	Fully Autonomous
Primary job	Perceive → respond	Understand → plan → act	Perceive + plan + act, alone
Time horizon	Milliseconds	Seconds to minutes	Continuous, safety-critical
Typical model	CNNs, small transformers	LLMs, tool-use agents	All of the above, concurrently
Bottleneck	Throughput at mW power	Memory & context, local	Everything — at once
What ruins it	Jitter, frame drops	Latency, cost per query	Any single-point failure

The shared constraint: all three want to run close to the sensor, on a device with a finite power budget.

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實體人工智慧：感知世界並採取行動
代理人工智慧：計畫、推理並展開行動
自主人工智慧：無需人工干預完成閉迴路控制

分別對**實體人工智慧**、**代理人工智慧**、**自主人工智慧**之主要工作、時間跨度、典型模型、瓶頸、崩潰因素進行比較。

Ambient – DigAn 混合信號計算範式



Tomorrow's physical AI needs 10–100× more compute.

Existing GPU / CPU silicon is already at its thermal limit. Scale it 10–100× and it cannot fit inside a watch, a drone, or a humanoid — too hot to hold, too heavy to carry.

10× → 100×

more on-device performance demanded by next-gen vision, language, and autonomy workloads.

Conventional GPU / CPU path

30 W Jetson today → 300–3,000 W tomorrow

- Thermal envelope blown — too hot to hold.
- Form factor broken — won't fit in a watch, drone, or humanoid.
- Battery life collapses — minutes, not days.
- No node-shrink path left. Dennard scaling ended a decade ago.

DigAn™ ASI path

50× PPA + real-time DVFS → 10–100× in the same envelope

- Scales performance through architecture — not more watts.
- Stays inside mW–W budgets of wearables, drones, robots.
- Always-on, battery-friendly; no active cooling required.
- Only DigAn-based ASI chips can deliver 10–100× on-device.

Only DigAn-based ASI silicon can deliver 10–100× more compute inside a small physical-AI device.

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Ambient – DigAn 混合信號計算範式



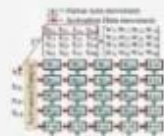
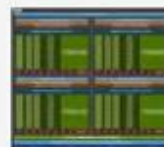
Logic wins the silicon. Memory gets pushed off-die. Power pays the bill.

GPUs and digital AI chips spend almost all their area on logic — so operands have to come from DRAM, and every fetch burns power.

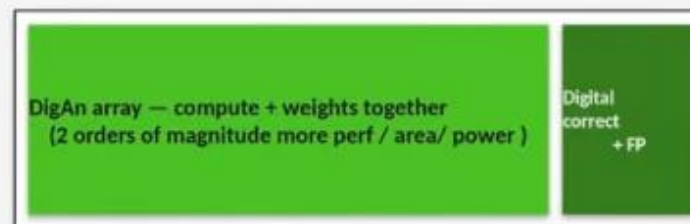
Today's GPU / digital AI chip



- Nearly all silicon is digital logic.
- Weights don't fit on-die — they live in DRAM.
- Every operand means an off-chip hop.
- Operand fetch dominates the power bill.



DigAn™ architecture



- 2 orders of magnitude more performance per mm².
- Weights fit on-die — no external DRAM required.
- Operand fetch power collapses.
- Supports floating-point — drops into existing PyTorch / TensorFlow toolchains and training flows.

Ambient – DigAn 混合信號計算範式



Introducing DigAn™ — the hybrid that ends the trade-off.

Analog efficiency where it's safe. Digital precision where it counts. One architecture, accurate and efficient at the same time.



Analog where it's efficient

Dense MAC executed as currents and charges, inside the array that already holds the weights.
No operand movement, no DRAM hop — orders of magnitude less energy per op.



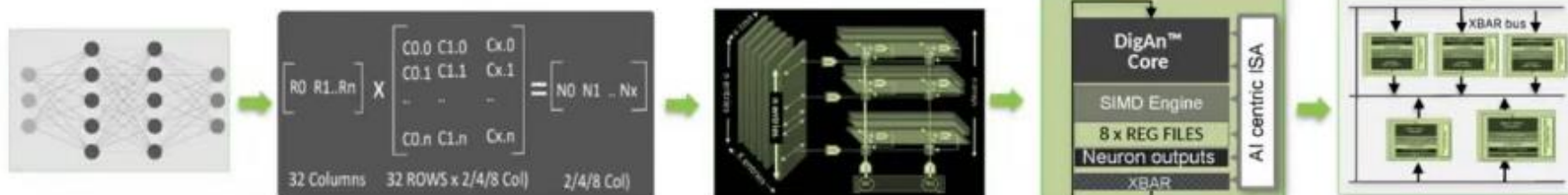
Digital where it's accurate

The moments analog gets unreliable — high-precision accumulation, non-linearities, calibration, control flow — run on digital logic on the same die. Nothing is left to drift.



DigAn™ = both, at once

A purpose-built hybrid architecture: analog-class efficiency, digital-class accuracy. Not approximate computing — exact results, at milliwatt-class power.



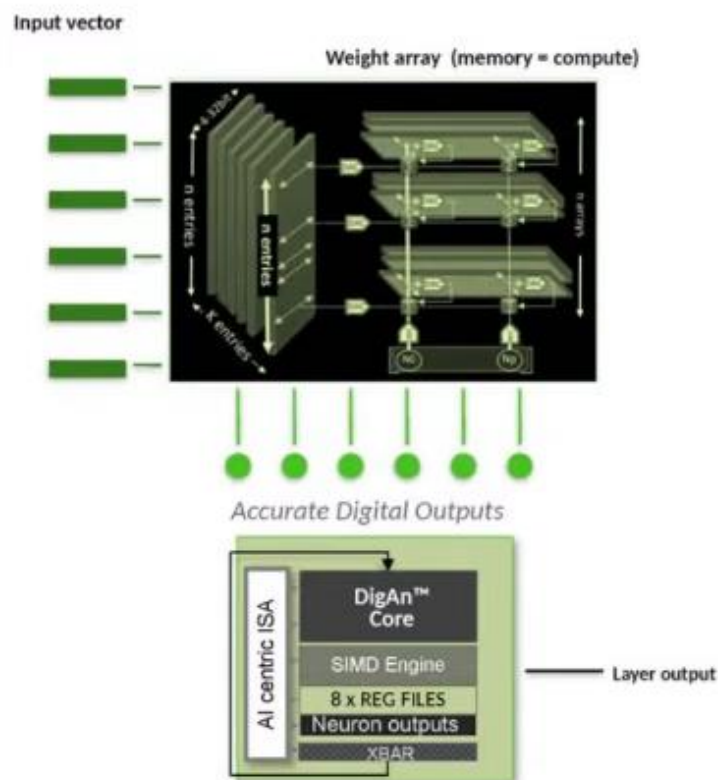
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10 / 17

Ambient – DigAn 混合信號計算範式



Inside DigAn™: compute in memory, accuracy by design.



Three physical steps

- 1 **Encode inputs as analog pulses**
Still use digital inputs. Prepare to compute
- 2 **Multiply & sum by physics.**
Each cell's conductance holds a weight. Ohm's law multiplies. Kirchhoff's law sums.
- 3 **Digital takes it home.**
Accumulate, correct, and apply non-linearities digitally — bit-exact results, layer after layer.

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Ambient – DigAn 混合信號計算範式



The intelligence finally fits inside the machine.

DigAn™ architecture plus patented real-time DVFS — digital-class accuracy, analog-class efficiency, power that tracks the workload instant-by-instant.

DigAn™ — analog efficiency + digital accuracy, on one die

Real-time DVFS (patented) — power & performance track the workload

	Today's GPU stack	Ambient DigAn™ + DVFS	Shift
Size	Rack unit / discrete GPU module	Single die that drops onto a sensor board	100×+ smaller
Power	10s–100s of watts, constant under load	Scales in real time to the instantaneous workload	Orders of magnitude lower
Accuracy	Digital floating point	DigAn™: digital-class, bit-accurate results	Equal, not approximate
Thermals	Active cooling, heatsinks, airflow	Passive cooling, inside sealed enclosures	No fans, no vents
Where it runs	Rack, car trunk, tethered base station	On the drone, in the earbud, inside the robot	Where the work is

50× PPA

Performance per unit power and per unit area — the density that makes physical & agentic AI workable, miniaturized, affordable, and cool to the touch.

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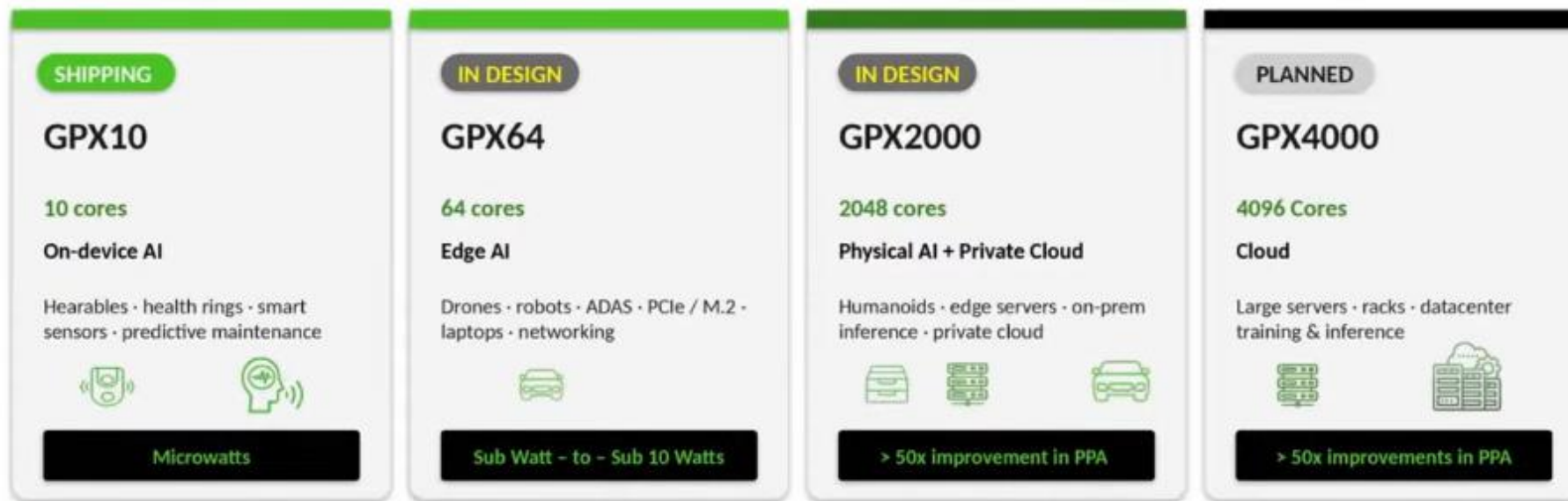
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Ambient – DigAn 混合信號計算範式



One scalable core. A full family — sensor to cloud.

Step-and-repeat DigAn™ cores tile into four product tiers. One compiler, one toolchain (PyTorch, TensorFlow) across every tier.



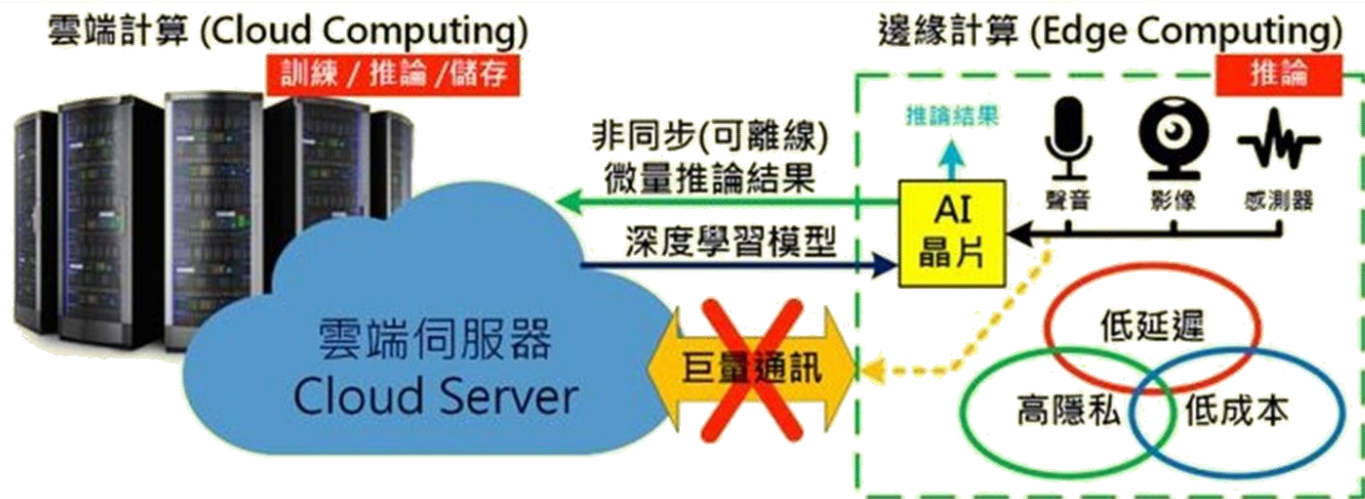
Unified software — PyTorch · TensorFlow · Ambient compiler & runtime. One toolchain from intelligent sensor to cloud

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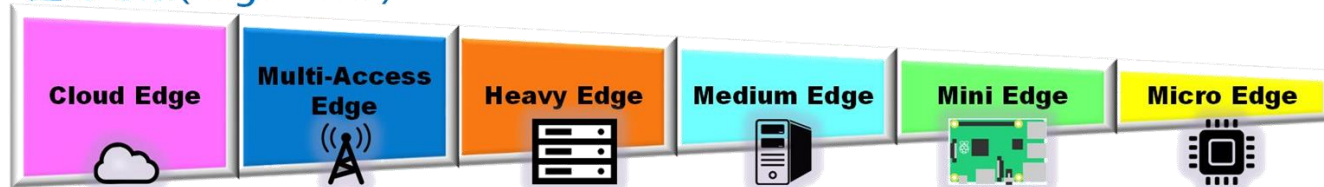
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Edge AI

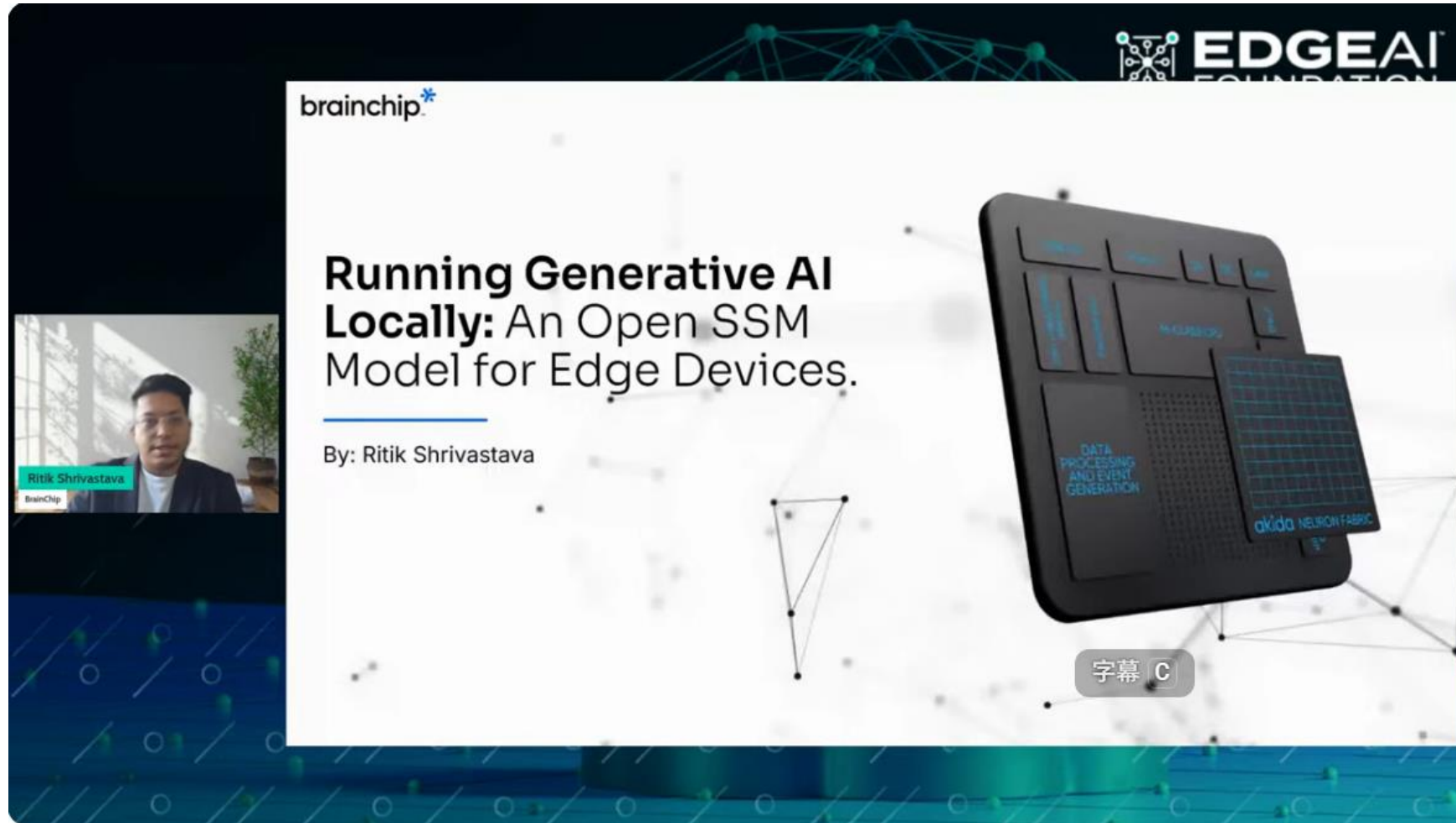


邊緣等級(Edge Level)



12.4.硬體優化與平台資源

Brainchip - 本地運行生成式 AI (SSM)



https://youtu.be/_5lvJjrWCv4?t=307

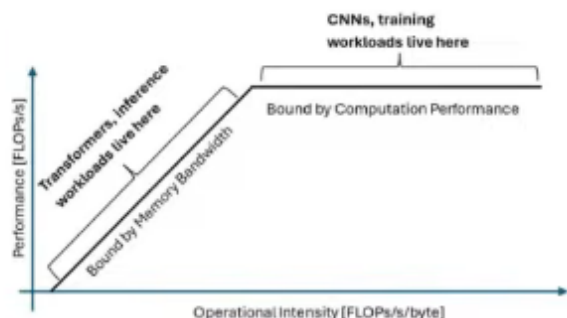
Brainchip - 本地運行生成式 AI (SSM)

brainchip*

Why is Memory Bandwidth Key?

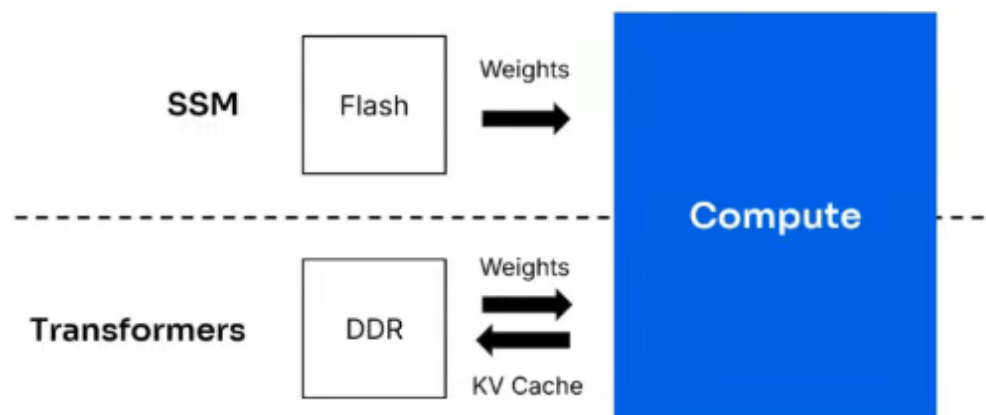
Changes in Model Paradigms

* AI Compute has Changed



* Memory Efficiency

- If 1 MAC uses 1 unit of energy
- **On-chip SRAM** Fetch = 2 units
- **Off chip Flash** fetch = 20 units
- **Off chip DRAM** fetch = 100 units



Brainchip - 本地運行生成式 AI (SSM)

brainchip*
Worldwide Leader in Neuromorphic AI

Solution Targets	   
Akida Tools	  
Offerings	 Silicon  License IP akida Model Zoo
Gen AI	 Keyword Spotting/ Denoising  Anomaly Detection  Voice Assistant (ASR → LLM RAG → TTS)  LLM



Research Leadership in SSM
 Pioneering SSM and dynamical-systems AI since 2020, turning theory into edge reality.



Edge-Optimized SSM Models
 Unlike GPU GenAI, BrainChip builds neuromorphic AI for real-time on-device intelligence.



Scalable Platform
 From IP to solutions powering Gen AI efficiently at the edge.

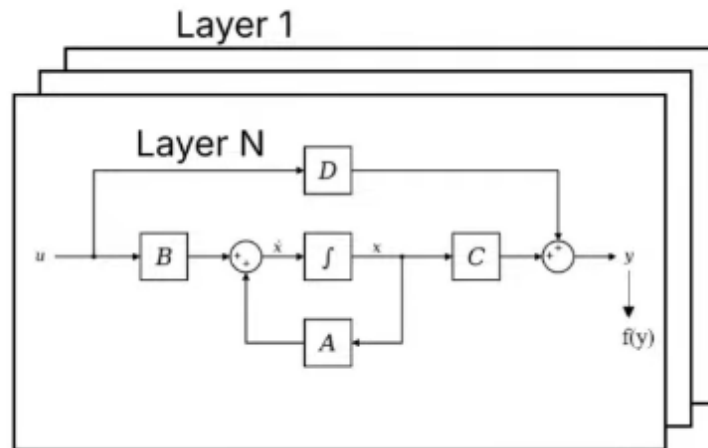
11/18/2025
Confidential
3

Brainchip - 本地運行生成式 AI (SSM)

brainchip*

State Space Models

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du \\ f(y)\end{aligned}$$



Surprise: Inspiration is from sophisticated signal processing but works with sequential tasks!

* Principles:

1. **Recurrence:** $x(t+1)$ is a function of $x(t)$.
2. **Duality:** The recurrent equations can be written as a convolution.
3. **Stable training:** training can occur in the convolutional domain to avoid vanishing gradient in RNNs and to efficiently parallelize computation.
4. **Well-suited for inference:** run in recurrent domain to reduce latency and foot-print.
5. **Insight:** TENNs and SSMs are a stack of generalized Fourier filters running in recurrent mode with non-linearities between layers.

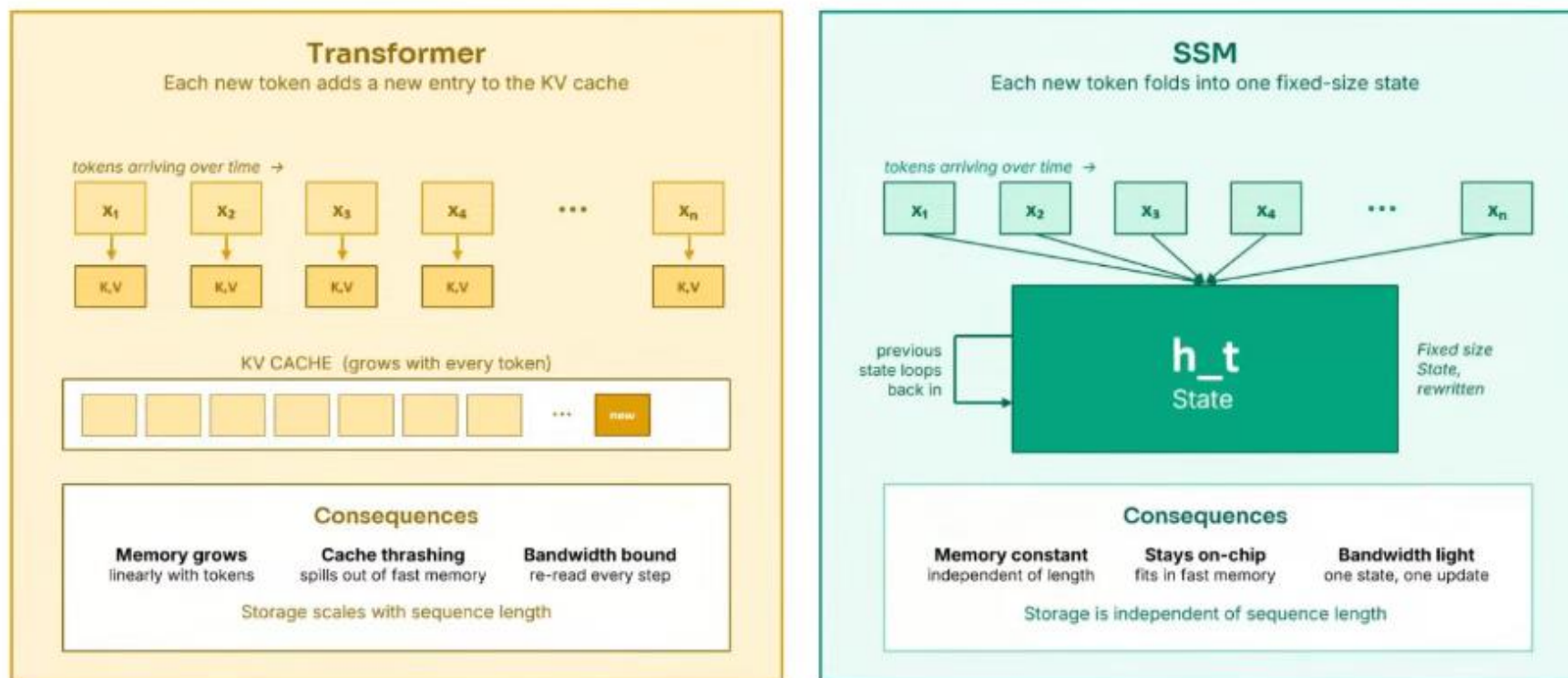
Brainchip - 本地運行生成式 AI (SSM)

brainchip*

SSM vs Transformers (Memory Movement)

Why SSM State is the Key to Hardware Efficiency

Watch what each architecture does as new tokens arrive



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PreCog (Patent Pending)

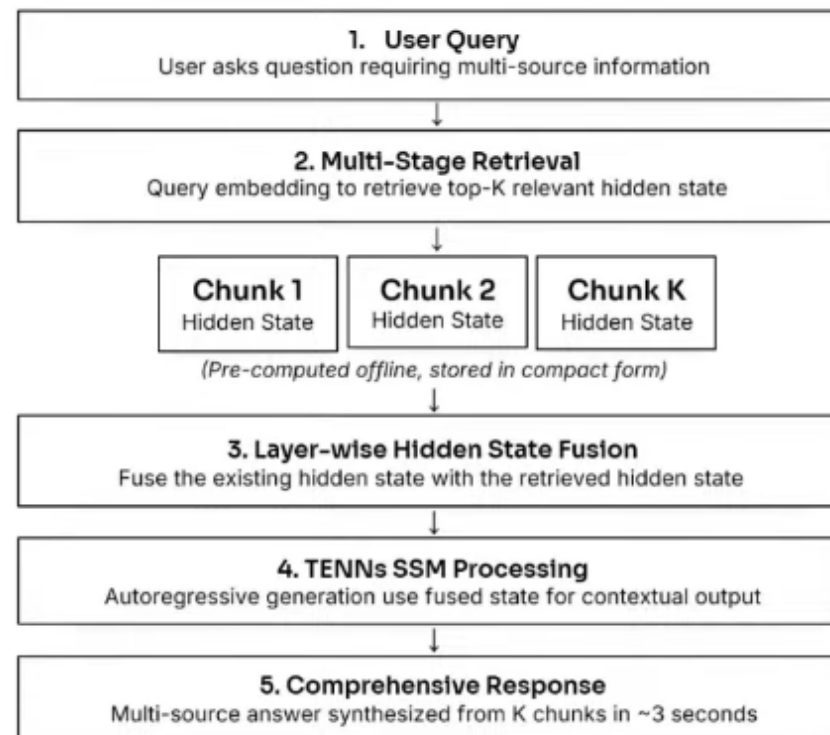
Minimize Preloading Times in RAG

PRECOC pre-computes and stores hidden states from information sources offline, enabling real-time query processing without reloading billions of parameters.

Key Technical Features

- Multi-stage retrieval with semantic + keyword scoring
- Layer-wise hidden state fusion across model layers

Performance Advantages



Brainchip - 本地運行生成式 AI (SSM)

brainchip*
Why SSM at the Edge

SSM's Constant Memory Footprint Makes Real-time Edge LLM Inference Practical Without DDR

Here's why:

Small Transformers
Llama-1B, Phi-3-mini, Qwen-1.5B

KV Cache
Grows with context length

Memory
Spikes unpredictably at inference

Latency
Generally attention is $O(n^2)$ per token

Power
High memory movement from KV cache access

VS

Tenns / SSM
BrainChip 1.2B State Space Model

No KV Cache
Fixed-size hidden state, constant memory

Memory
Predictable and flat at any context length

Latency
Recurrent inference - $O(1)$ per token

Power
Constant-size state reduces memory movement

Same parameter count. Fundamentally different behavior on constrained hardware.

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Brainchip - 本地運行生成式 AI (SSM)

brainchip*

End-to-End Developer Workflow



Goal: Clone the repo, run the evals, fine-tune on your data, deploy on your hardware, all in one open release.

Brainchip - 本地運行生成式 AI (SSM)

brainchip*

From Architecture to Application

TENNs 1.2B LLM being released can run real-time on-device, even on a low-resource target.

Here's where that matters:



**Defense
Forward Operating
Base**

No Cloud Access

Classified doc Q&A, fully airgapped



**Healthcare
Remote Field
Clinic**

Dead Zone

Drug protocols by voice, offline



**Industrial
Factory
Floor**

Behind an Airgap

Equipment manuals, no data leaves

Clone it. Fine-tune for your domain. **Deploy on any standard device.**

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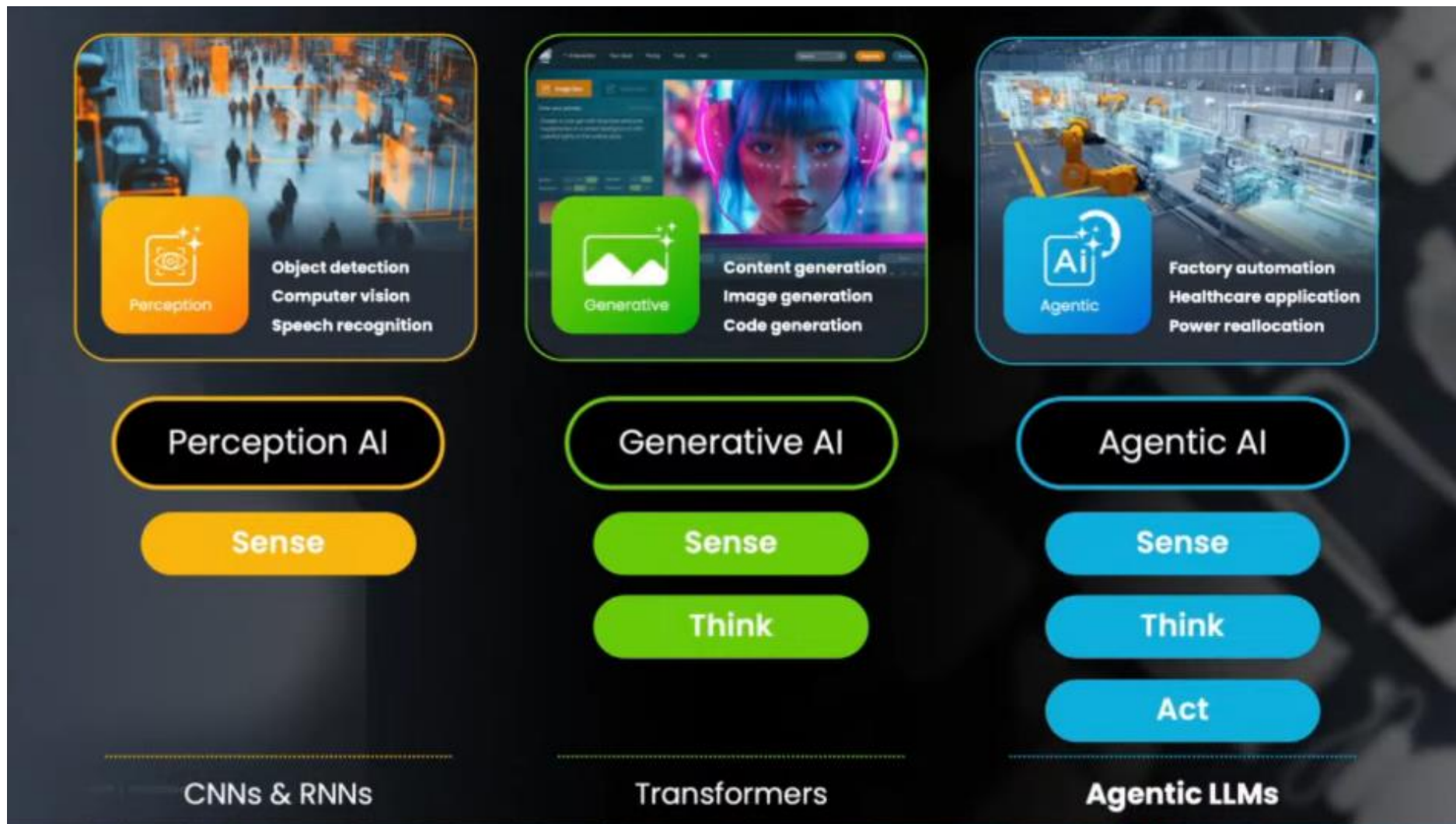
NXP – Ara Discreate NPU



NXP 2025/2 收購
DNPU 供應商
Kinara, 旗下 **Ara-1**,
Ara240 亦 成 為
NXP 產品。

<https://youtu.be/5lvJjrWCv4?t=1538>

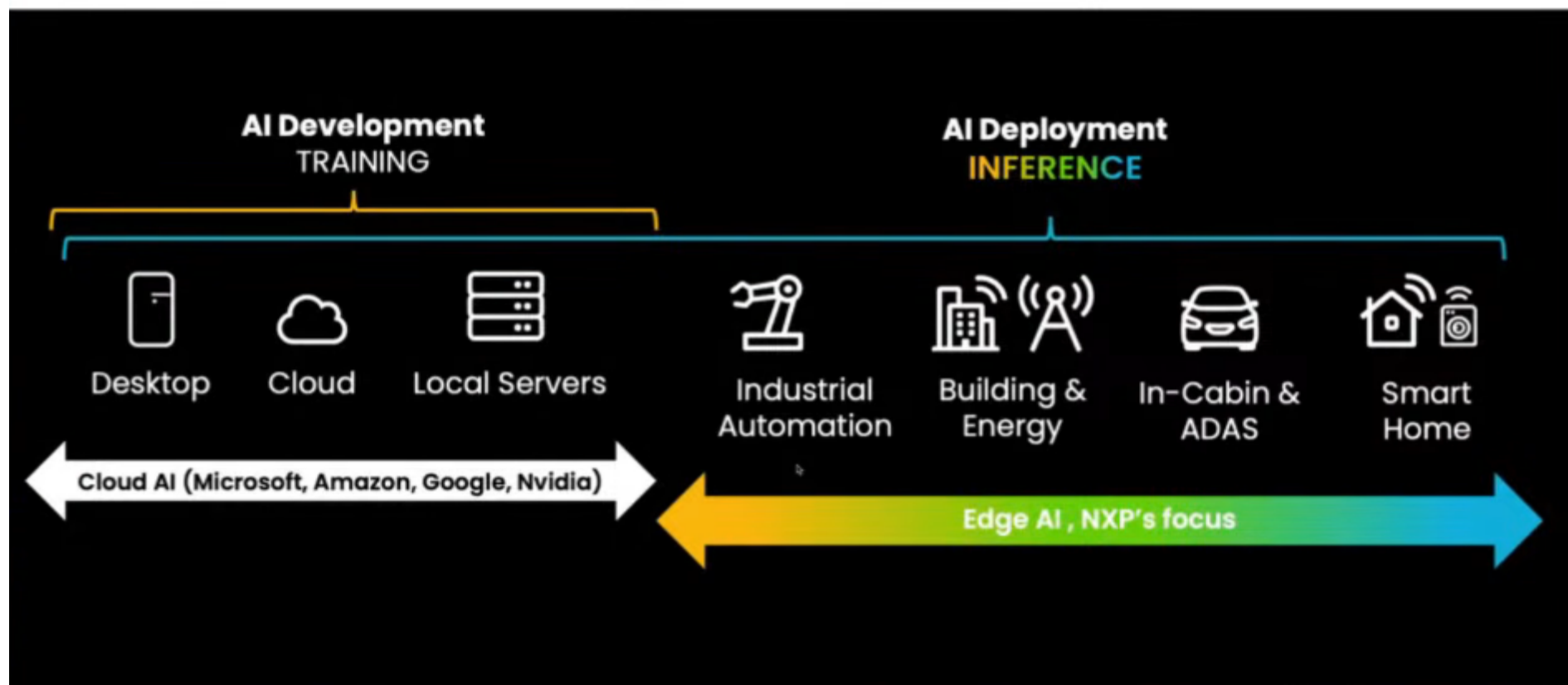
NXP – Ara Discreate NPU



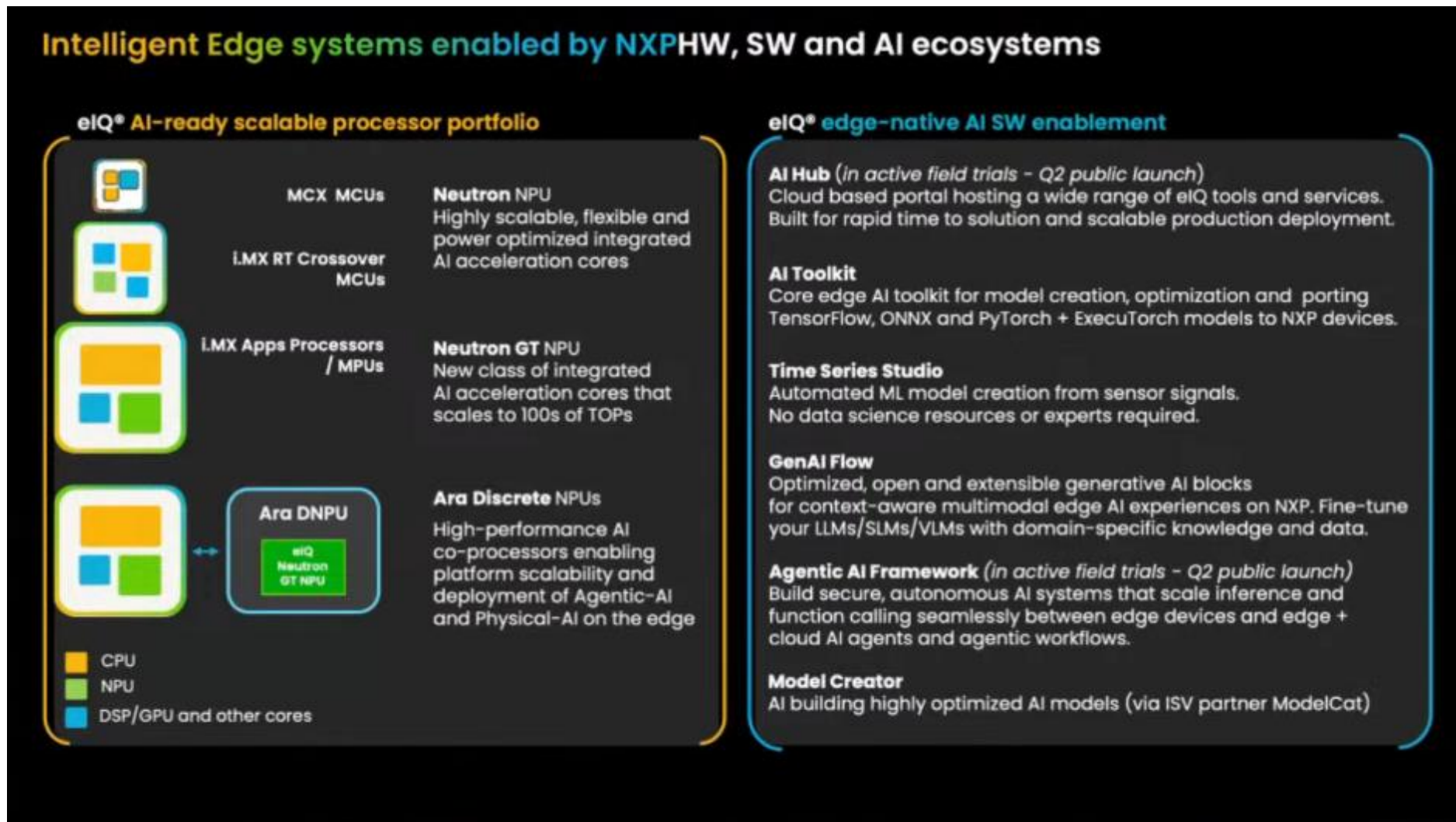
NXP – Ara Discreate NPUs

AI demand is moving from training to **inference at the edge**

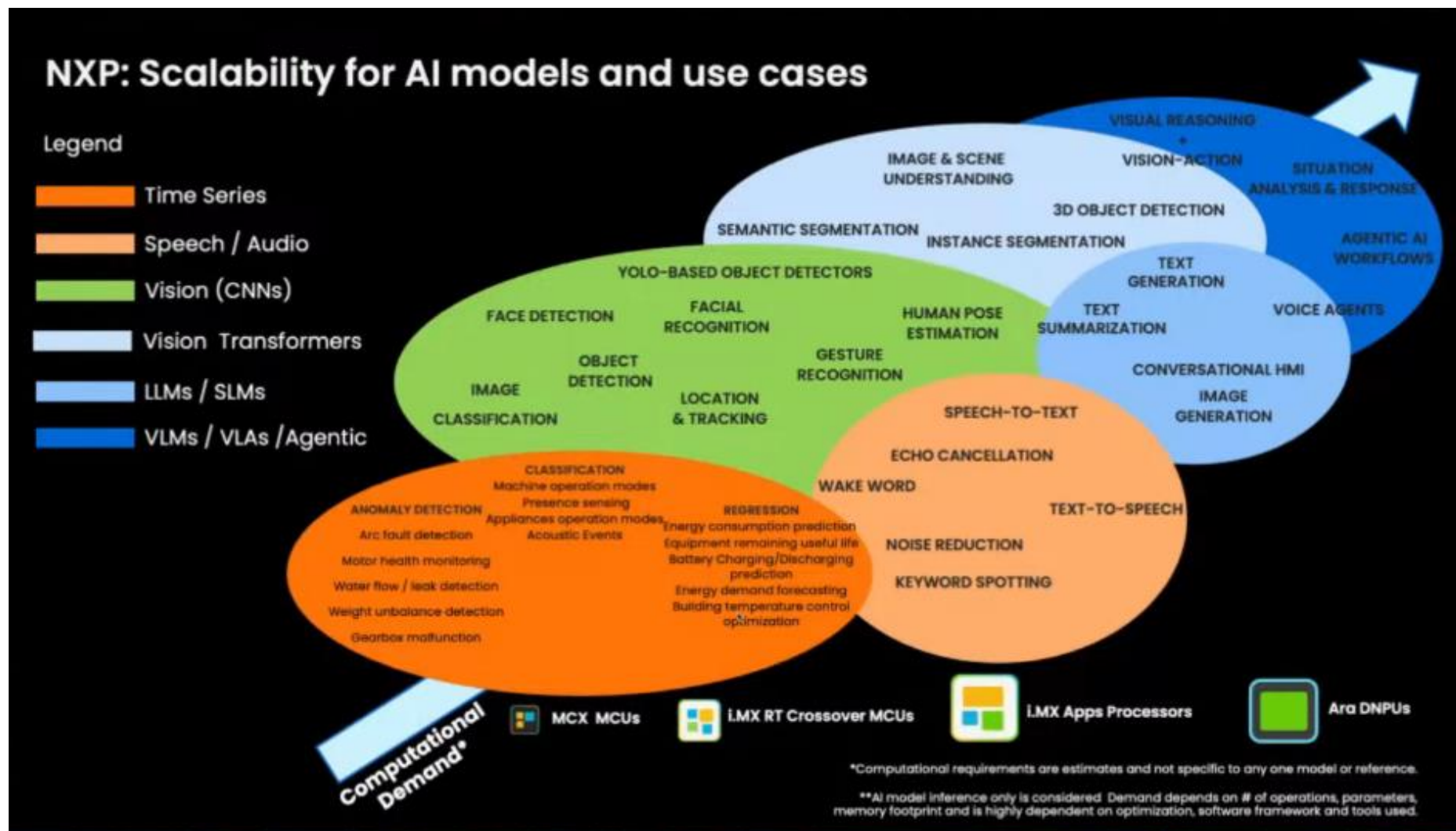
NXP captures the edge AI inference opportunity with secure, scalable systems-first solutions.



NXP – Ara Discreate NPUs



NXP – Ara Discreate NPUs



NXP – Ara Discreate NPUs

Edge AI Success Story : PyTorch/ ExecuTorch + eIQ Neutron NPU

Building trust and confidence with the latest AI ecosystems through “one click” deployments to NXP products.

Executorch: <https://github.com/pytorch/executorch/tree/main/backends/nxp>



Enabled Devices

The i.MX RT700 equipped with the integrated eIQ Neutron NPU supports PyTorch natively, making NXP is 1 of 6 silicon vendors to support ExecuTorch since GA in 2025.

i.MX 93 for Ethos U65 NPU is also available
i.MX95/943/952 NPU support in Q4 2026.



In-house Tools & Open-source

ExecuTorch (ET) support for on-device inference with Neutron NPUs is made possible via the upstreamed eIQ® Neutron software stack and relevant MCUXpresso SDKs/BSPs.

Meta RL is using the Neutron NPU ET backend for faster development cycles with perception edge AI models



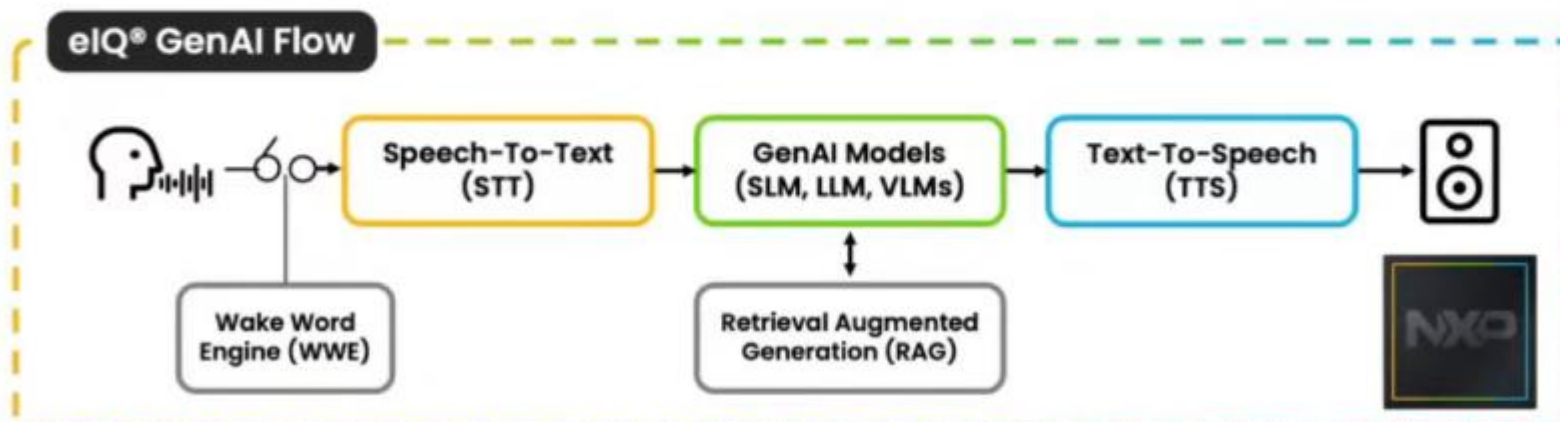
Global Cooperation

Teams cooperating— NXP owning the project & development and Meta with the PyTorch Community.

NXP – Ara Discreate NPUs

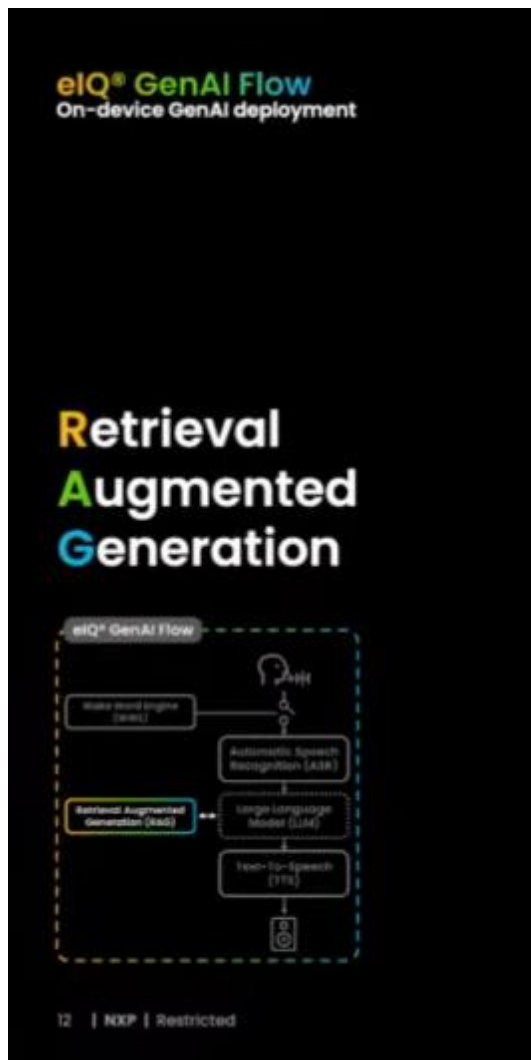
eIQ® GenAI Flow creates AI-powered HMI experiences for the Intelligent Edge
Engage with conversational AI today on i.MX microprocessors!

- **Wake Word Engine (WWE):** A VIT (Voice Intelligent Technology) Wake-Word triggers the ASR.
- **Speech-to-Text (STT):** Automatic Speech Recognition converts spoken input into text.
- **Retrieval-Augmented Generation (RAG):** Enhances the LLM/VLM* with relevant knowledge.
- **Text Generation (LLM + VLM*):** Generates a response based on the retrieved context.
- **Text-to-Speech (TTS):** Converts the response into speech output.



*VLM with GenAI Flow v3.0 only
Automotive demo [here](#)
Smart Home demo [here](#)

NXP – Ara Discreate NPUs



Retrieval Augmented Generation (RAG)

Enhance the knowledge base of LLMs

Model: all-MiniLM-L6-v2^[1]

Parameters: 23 million

Customize and control the LLM:

- **Build** a knowledge database
- Enrich queries with relevant context
- Generate context-aware answers
- Keep your data private and secure

NXP's Innovation 🚀
Human-inspired RAG for better chunking

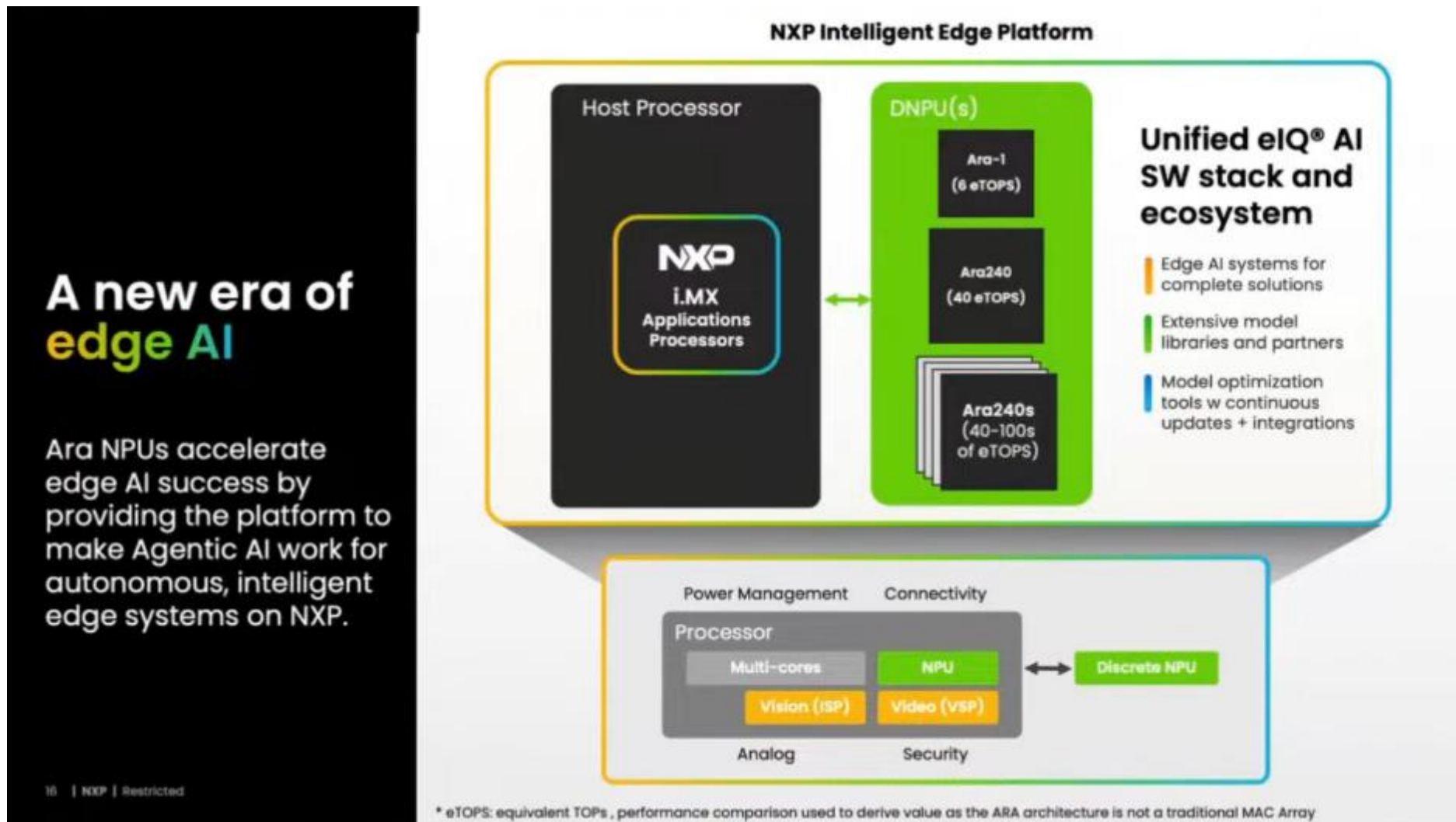
Database generator:

- Advanced PDF document parsing leveraging **Docling**.
- Solutions to break down large textual data using chunking algorithms.
- Embedding model adapted for NXP platforms to generate RAG databases.

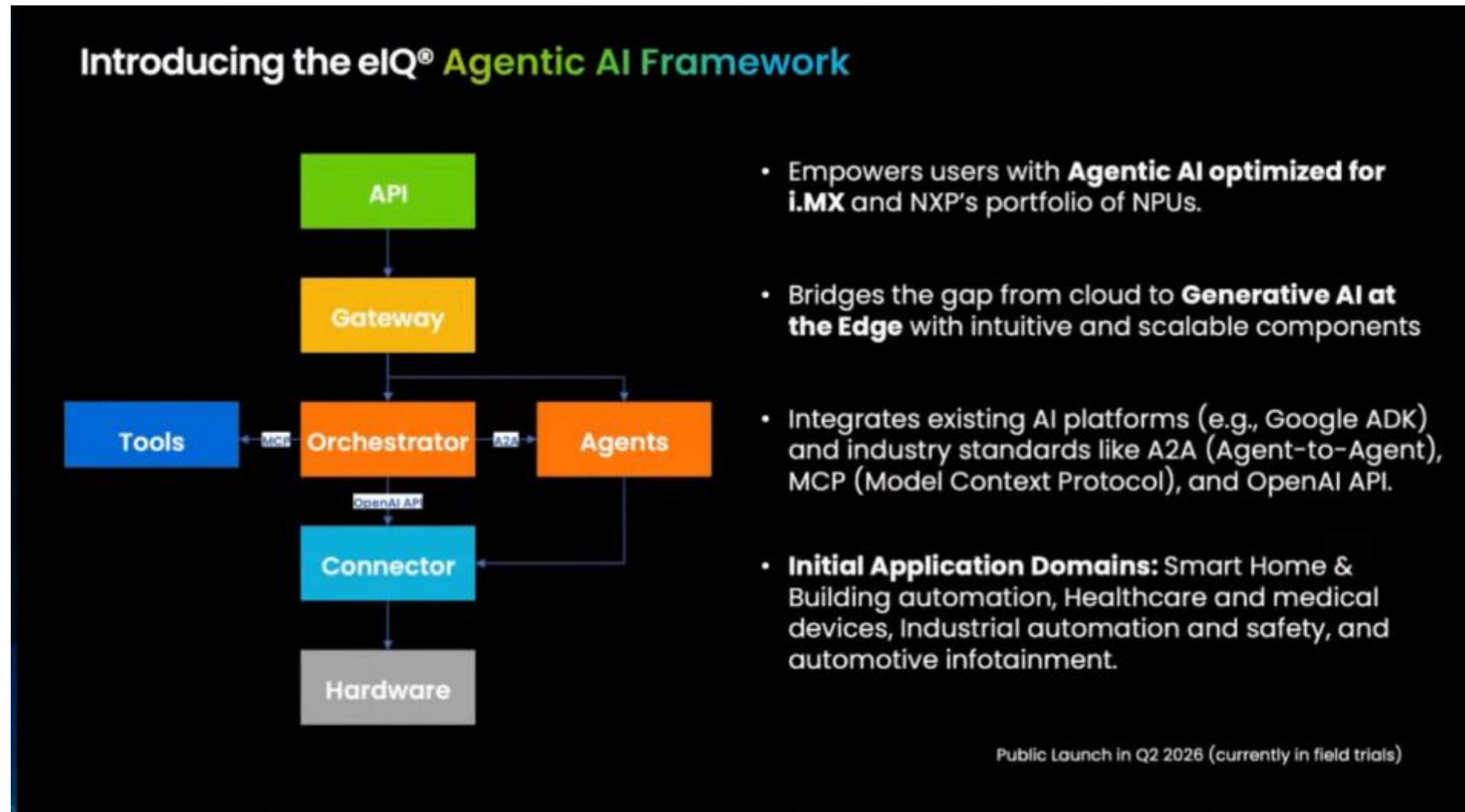


[1] Hugging Face, "sentence-transformers/all-MiniLM-L6-v2 - Hugging Face," huggingface.co/sentence-transformers/all-MiniLM-L6-v2

NXP – Ara Discreate NPUs



NXP – Ara Discreate NPUs





Godel Space – 從軌道到行動災害偵測



https://youtu.be/_5lvJjrWCv4?t=3080

Godel Space – 從軌道到行動災害偵測



Godel Space – 從軌道到行動災害偵測

The Stack: NASA Prithvi + NVIDIA Jetson



Prithvi-EO-2.0-300M

300M parameter Vision Transformer,
NASA + IBM, Open-weight. Pre-trained on
Harmonized Landsat Sentinel-2. 4 time
steps + 6-band imagery.

Fine-Tuned Tasks

Burn scar detection · Flood mapping ·
Change detection

Edge Hardware

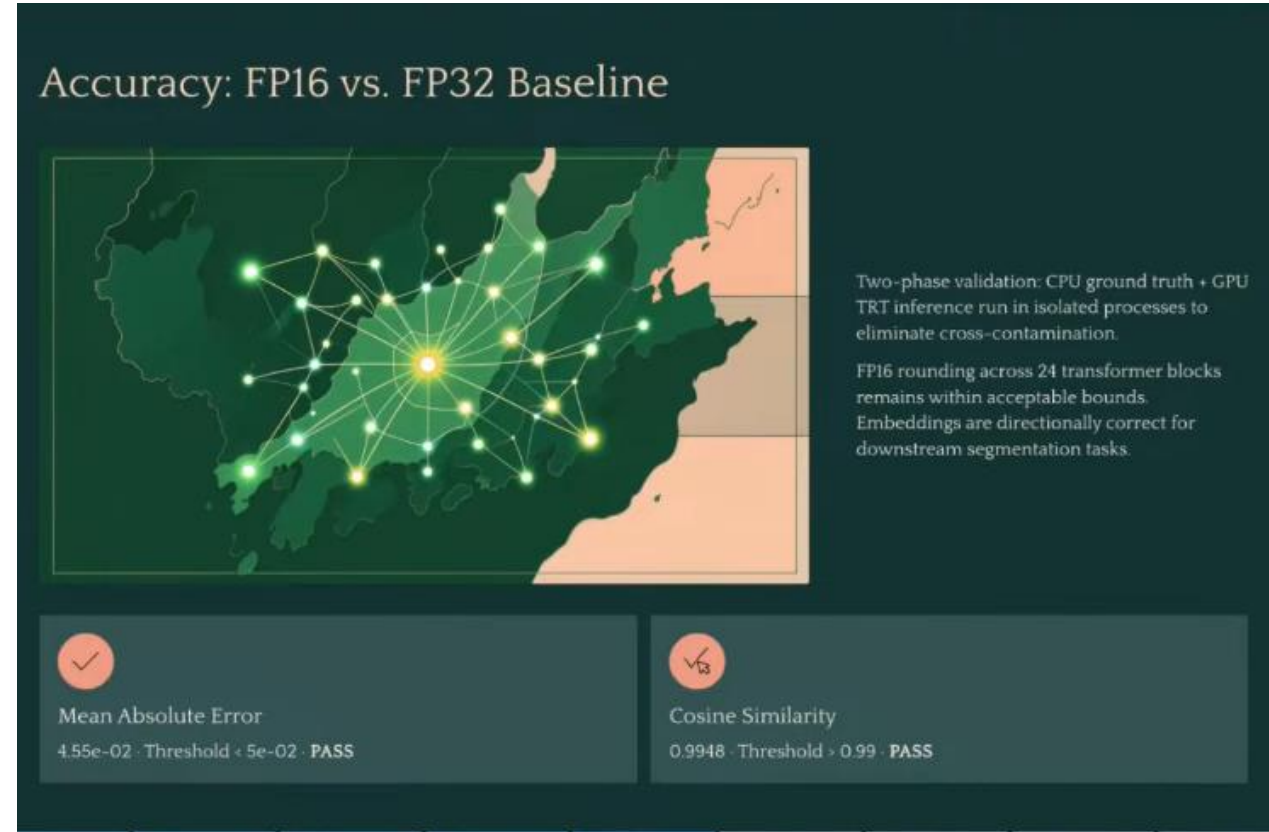
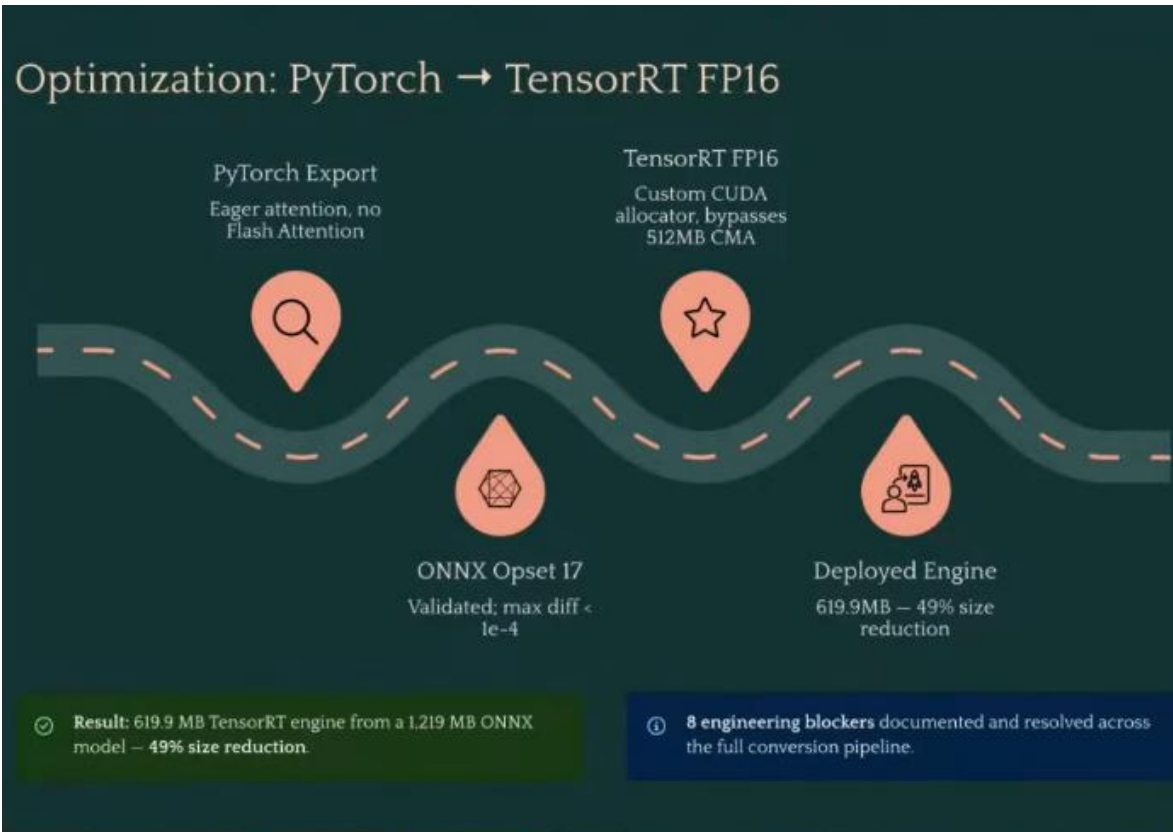
NVIDIA Jetson Orin Nano Super · 8GB
unified memory · 67 TOPS INT8 · JetPack
6.2 · TensorRT 10.3 · CUDA 12.6

Flight Path

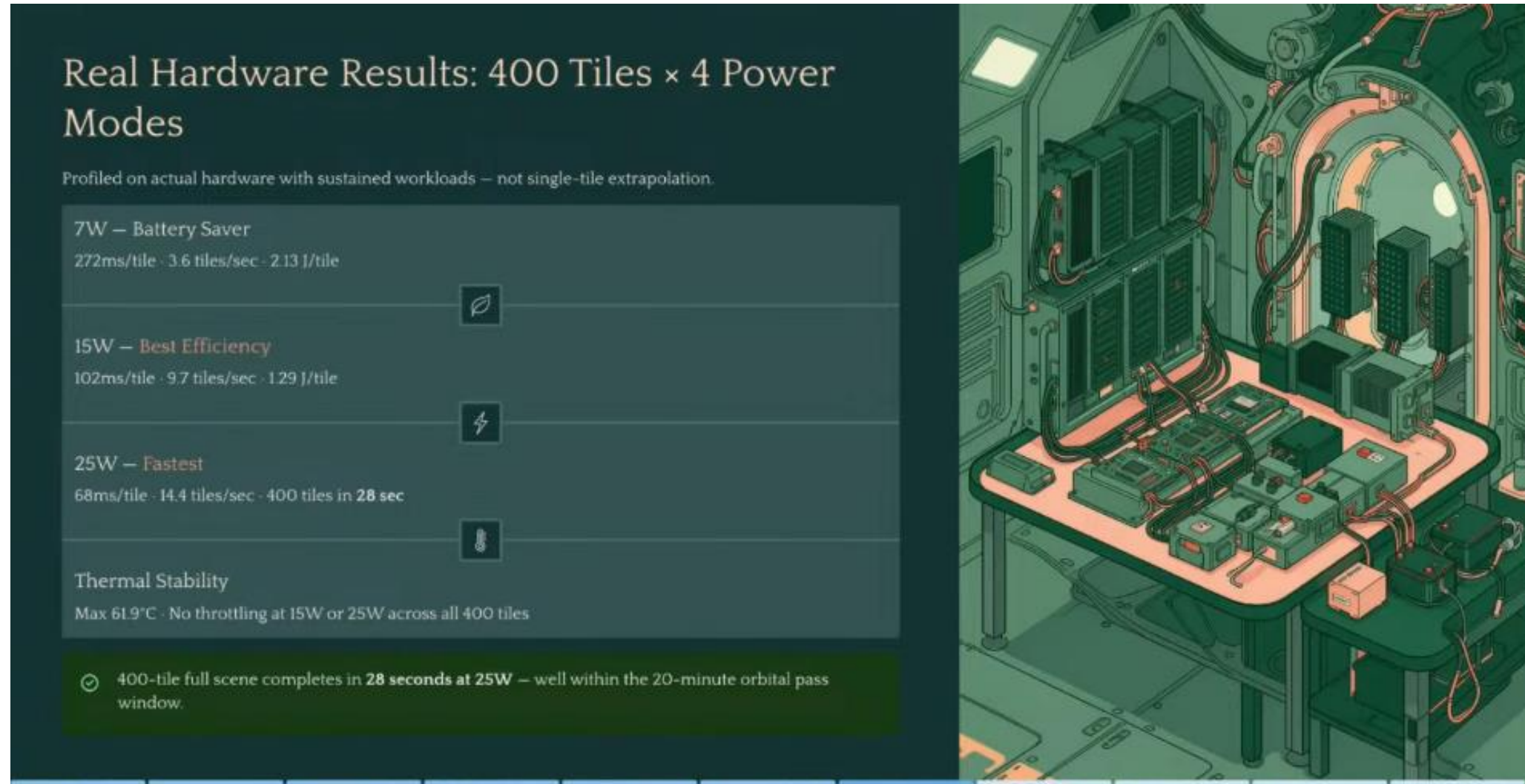
Jetson AGX Orin 64GB – 275 TOPS,
radiation-tolerant variants for orbital
deployment



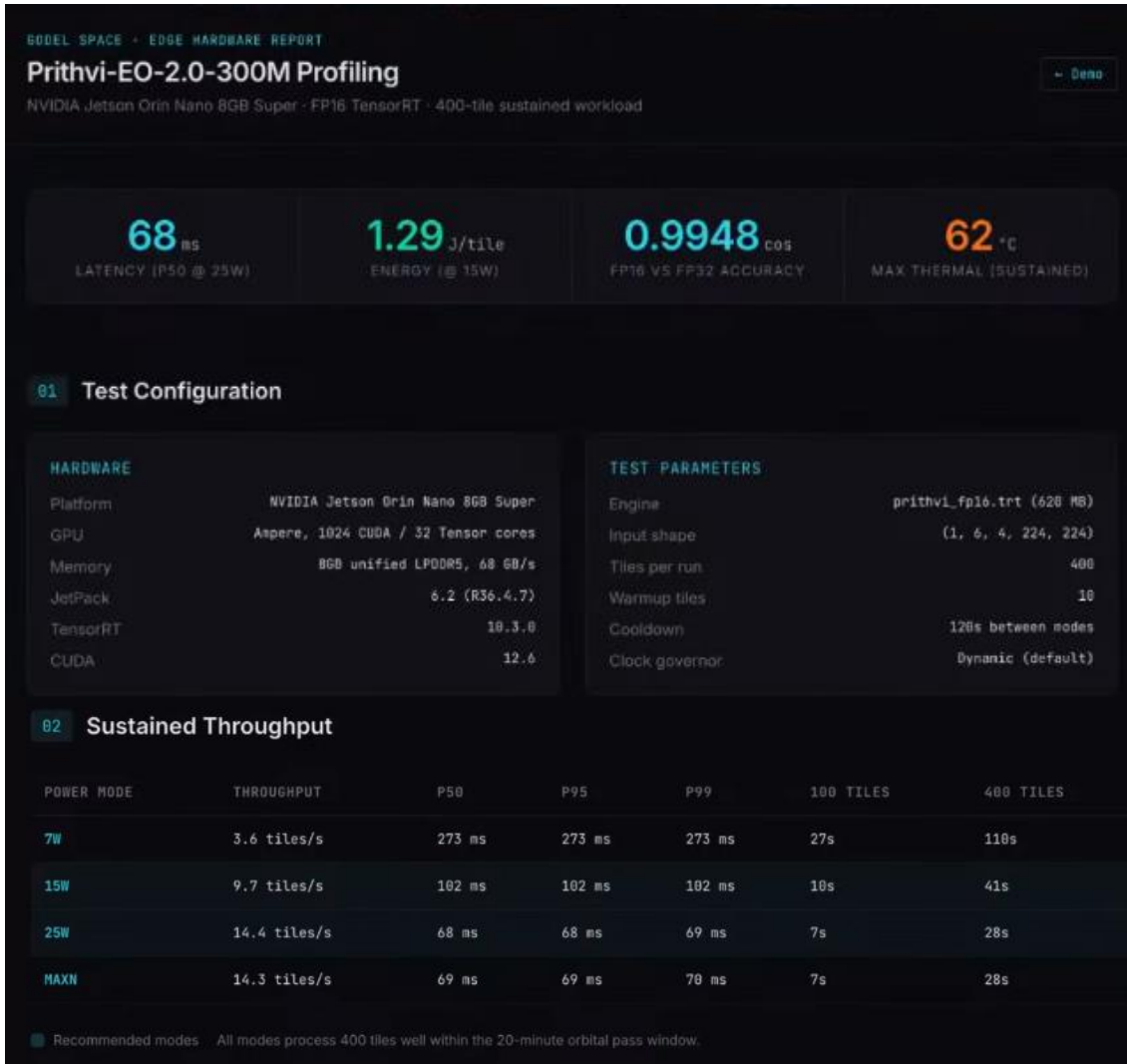
Godel Space – 從軌道到行動災害偵測



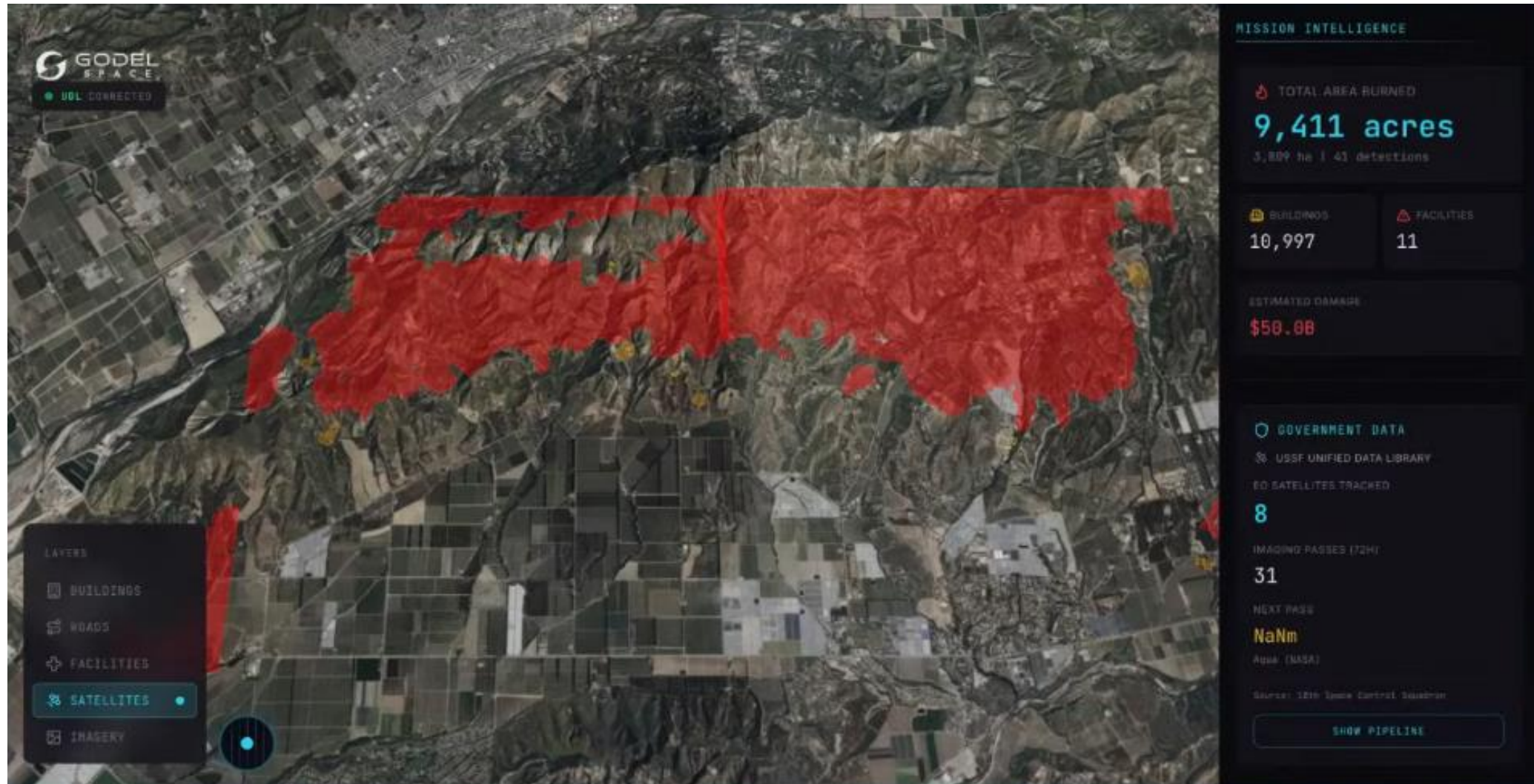
Godel Space – 從軌道到行動災害偵測



Godel Space – 從軌道到行動災害偵測



Godel Space – 從軌道到行動災害偵測



Edge AI Innovation - 邊緣全部招回

EDGE AI GENAI FORUM - 5TH EDITION - DAY 2

Total Recall at the Edge

A Deterministic Universal Memory Layer for Agentic AI

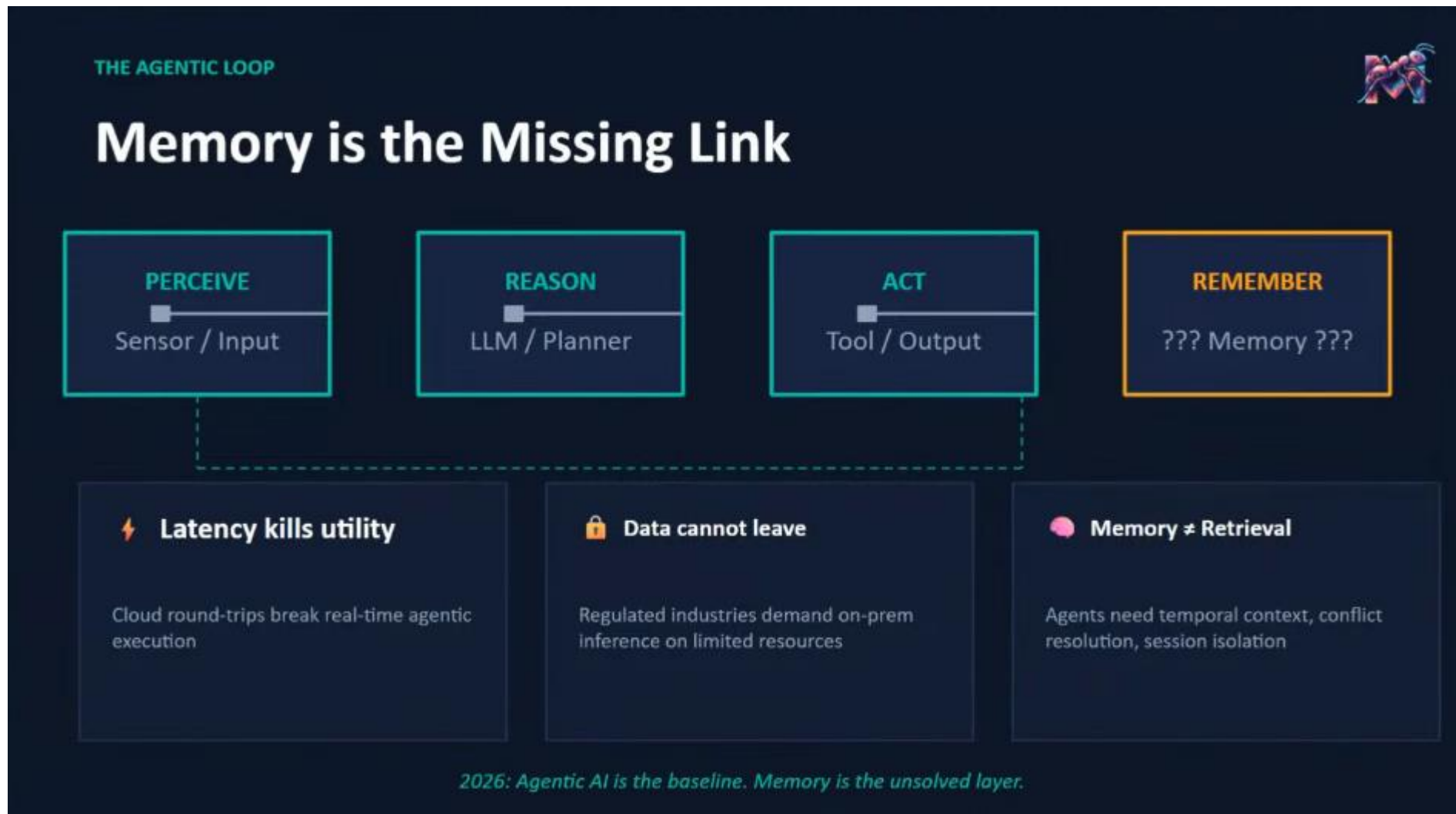
Dr. Majid Fekri | CTO & Co-Founder, EdgeAI Innovations®
Tara Khani | CEO & Co-Founder, EdgeAI Innovations®
moorcheh.ai · memanto.ai

EDGE AI INNOVATIONS INC.

EDGE AI FOUNDATION

https://youtu.be/_5lvJjrWCv4?t=4731

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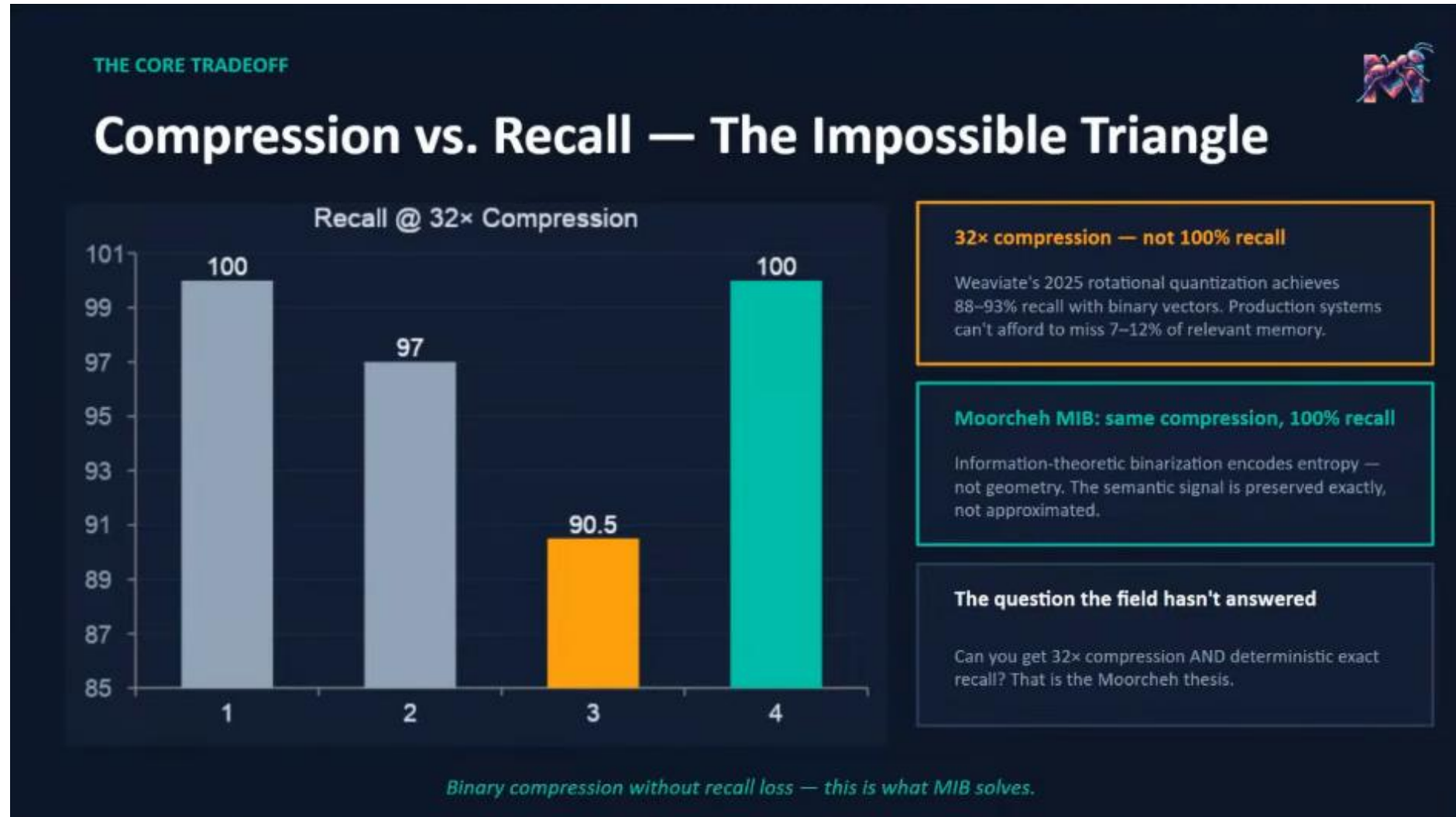
STATE OF THE ART

Retrieval at the Edge — What Exists Today

Qdrant Edge <i>Private Beta · Jul 2025</i> Still HNSW-based. GPU-optional. Same memory constraints, smaller footprint.	Binary Quant. <i>Weaviate 2025</i> Rotational quantization → 32× compression but 88–93% recall. Not deterministic.	d-HNSW / RDMA <i>ACM HotStorage 2025</i> Disaggregated memory via RDMA. Complex infra. Not edge-native.
GraphRAG <i>Microsoft / Community</i> Knowledge graphs for reasoning. Requires LLM calls per retrieval. Latency compounds.	On-Device RAG <i>Various · 2025</i> Bandwidth-independent. Accuracy suffers with small local models.	Hybrid CPU+SQ <i>Ruvector / SochDB</i> Multi-tier: binary filter → scalar HNSW → full precision rerank. Complex pipeline.

GAP: No solution achieves deterministic exact recall + CPU-native + zero idle cost — simultaneously.

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THE ARCHITECTURE · EDM + ITS

Why Every Modern CPU is Enough

CPU ARCHITECTURE

- L1 Cache (32KB)**
Active registers
- L2 Cache (256KB)**
Hot data
- L3 Cache (8–32MB)**
← MIB vectors live here
- DRAM (GB)**
HNSW graph lives here

*MIB vectors (96 bytes each) fit in L3 cache.
No DRAM fetches. No GPU needed.*

EDM — Efficient Distance Metric

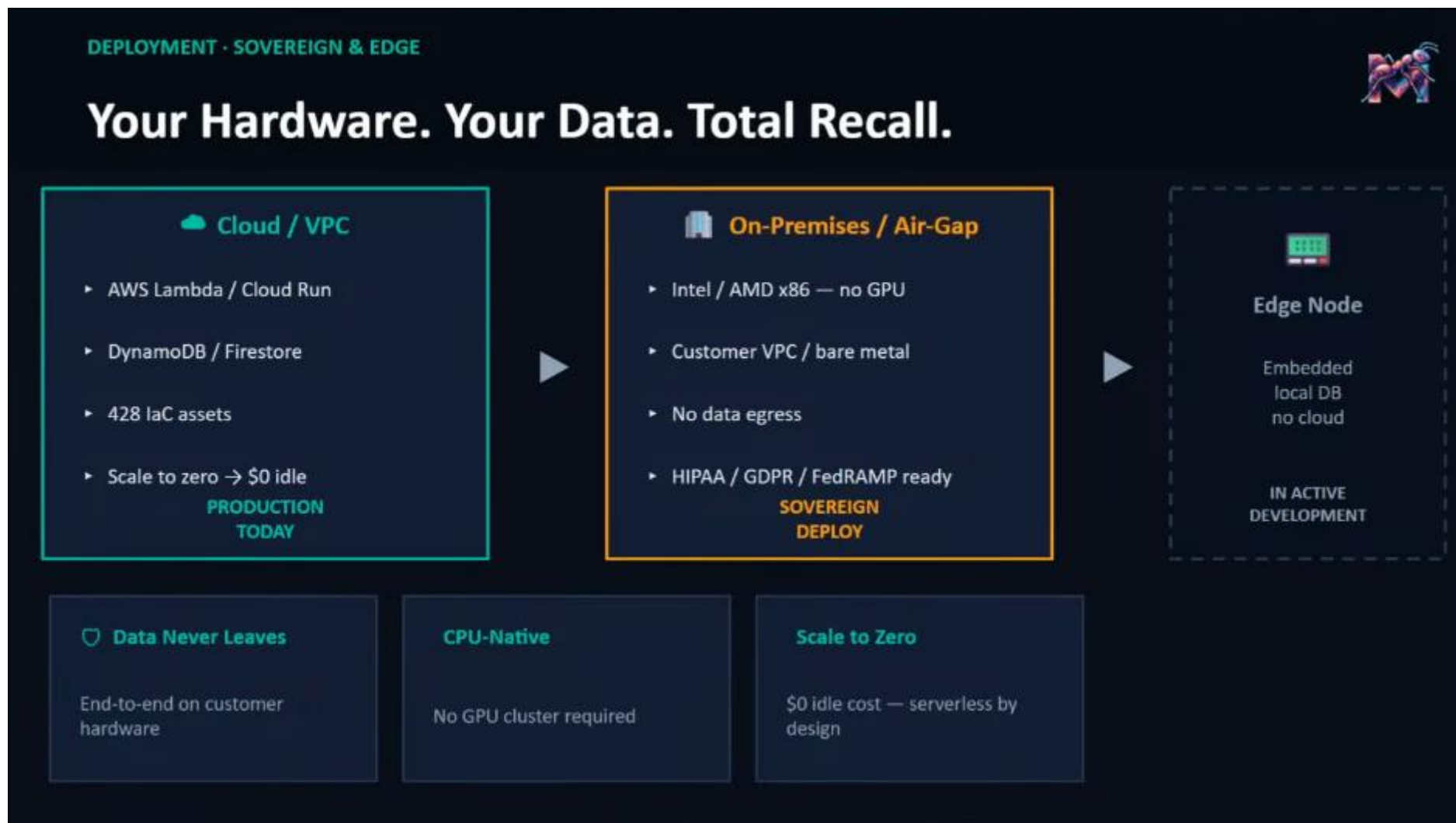
- Replaces float distance with bitwise POPCOUNT
- Native CPU instruction since 2008 (SSE4.2)
- Zero floating-point operations
- 1000× faster than cosine similarity

ITS — Information-Theoretic Score

- Normalized 0–1 relevance score
- Deterministic: same query → same result
- No probability threshold tuning needed
- Mathematically grounded, not heuristic

Result: 2,000+ QPS · CPU-only · No GPU cluster required

Edge AI Innovation - 邊緣全部招回



Edge AI Innovation - 邊緣全部招回

THE PRODUCT

一般代理記憶層

Memanto — Universal Agentic Memory Layer





Remember()

Store episodic events, decisions, preferences, commitments.
Zero LLM calls at ingestion.
Sub-millisecond write latency.



Recall()

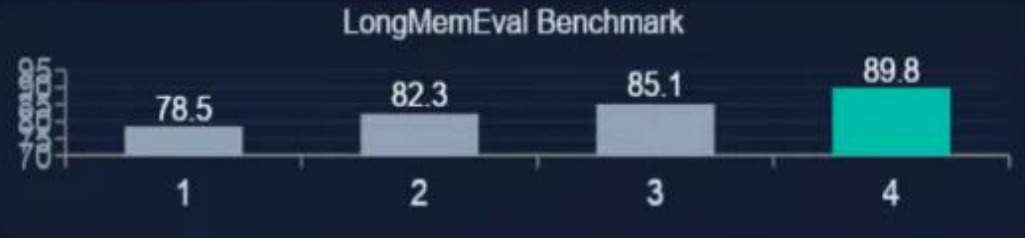
ITS-powered deterministic retrieval.
Single query, no multi-hop.
<90ms end-to-end.



Answer()

Memory-augmented generation.
Conflict resolution built-in.
Temporal versioning automatic.

LongMemEval Benchmark



Step	Score
1	78.5
2	82.3
3	85.1
4	89.8

89.8%

LongMemEval

13

Memory Types

87.1%

LoCoMo

0

LLM calls at ingestion

🚀 Official Launch: April 30, 2026 · San Francisco

Edge AI Innovation - 邊緣全部招回

Remember.
Recall.
Answer.
On Your Hardware.

MEMANTO
REMEMBER · RECALL · ANSWER

- moorcheh.ai**
Search engine & Sovereign deployment
- memanto.ai**
Agentic memory SDK · Launching Apr 30 · SF
- arXiv white papers**
Benchmark methodology & results

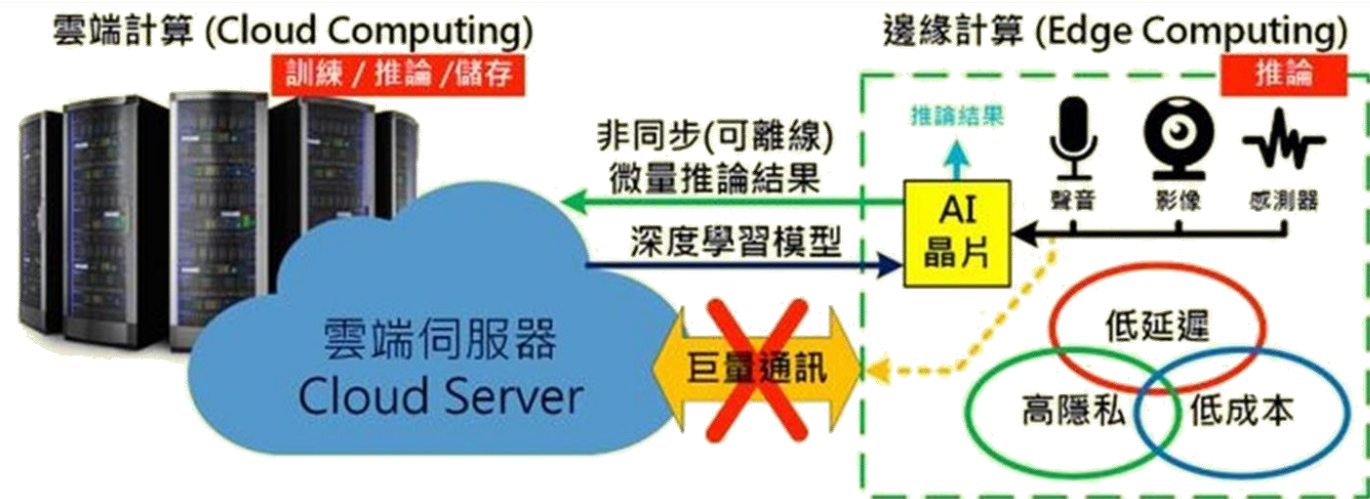
14,000+
SDK Downloads

Patent Pending
MOOR-001NP (USPTO)

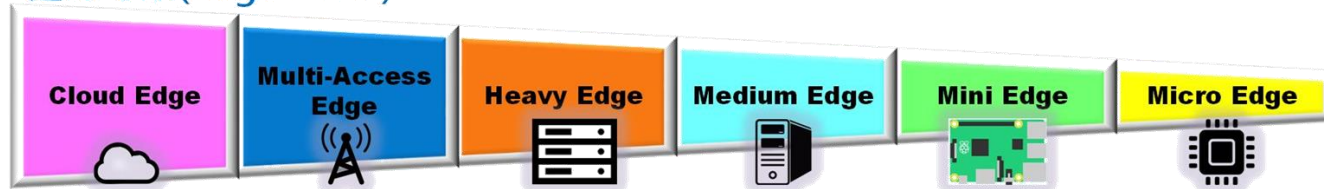
Dr. Majid Fekri · CTO & Co-Founder, EdgeAI Innovations · moorcheh.ai



Edge AI

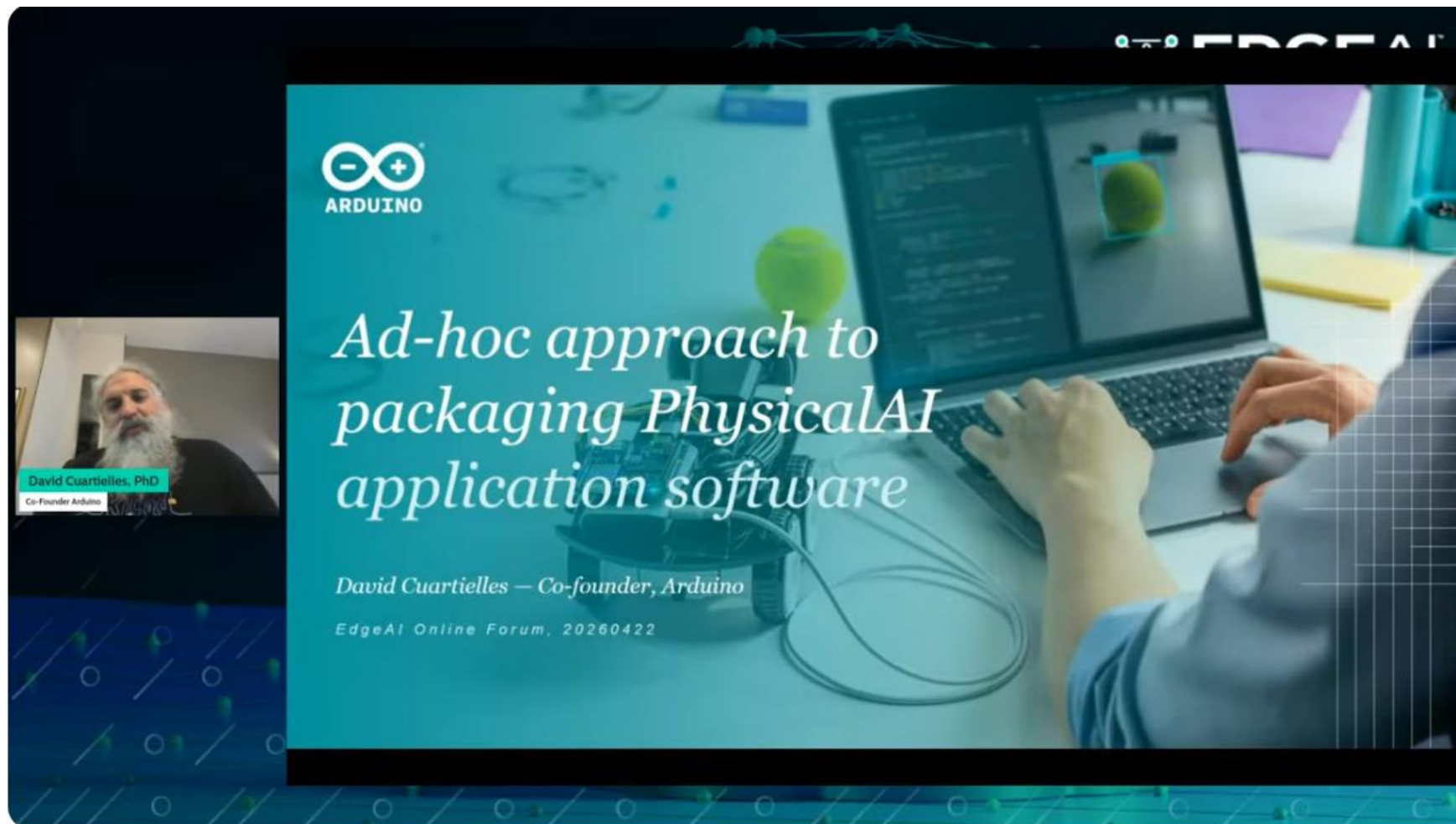


邊緣等級(Edge Level)



12.5. 學術與研究基礎

Arduino - 實體AI應用軟體打包



https://youtu.be/_5lvJjrWCv4?t=6323

Arduino - 實體AI應用軟體打包

The shift

The edge used to run one tiny model.

Now it runs several — and it must move things.

01

Classical embedded AI

One tiny model, one MCU, one task.

02

Generative AI at the edge

Local LLMs and foundation models on capable SoCs.

03

PhysicalAI

Many models, orchestrated, acting on the physical world in real time.



Arduino - 實體AI應用軟體打包

The problem

Today's pipelines package one thing at a time

Containers

assume a Linux userland.

*Great for the SoC.
Useless on the MCU.*

Firmware flashing

assumes one binary.

*Useless when Python drives a model
that drives a motor.*

Model zoos

assume convert + flash.

*They ignore the glue layer that makes it
run.*

We need a format that bundles containers + sketches + models + glue — as one artifact.



Arduino - 實體AI應用軟體打包

Our bet — Arduino × Qualcomm

Qualcomm

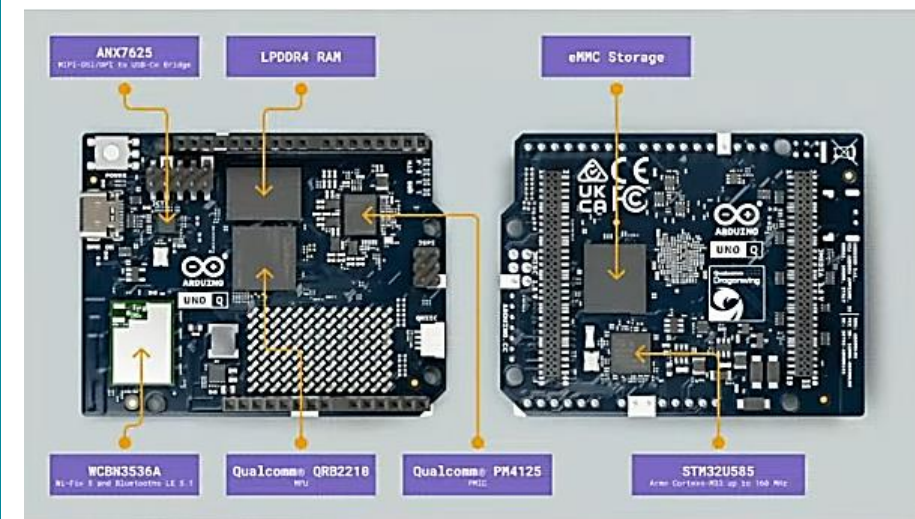
A dual-brain board: Arduino UNO Q

Announced October 2025, with Qualcomm.

“Dual-brain platform
that combines
high-performance
computing with
deterministic real-time
control.”

SoC gets the models.

MCU owns the microseconds.



Qualcomm Dragonwing QRB2210 (MPU) · STM32U585 (MCU) · LPDDR4 RAM · eMMC · Wi-Fi + BT

Arduino - 實體AI應用軟體打包

Scaling up — Arduino × Qualcomm

Qualcomm

Arduino VENTUNO Q

Where AI takes action.



ARDUINO®
VENTUNO™ Q

40 TOPS NPU
Qualcomm Dragonwing IQ8

16 GB RAM
concurrent multi-model

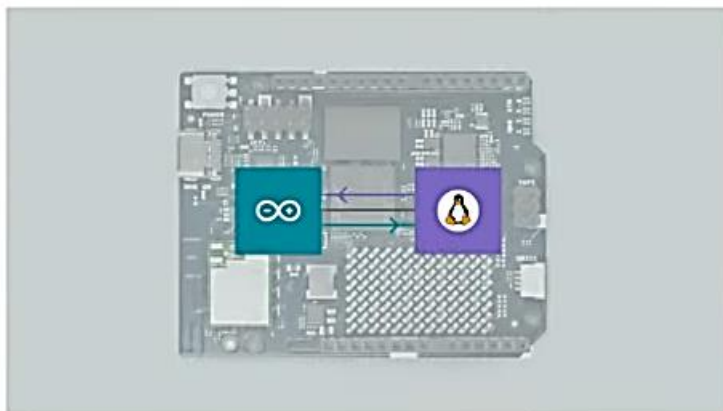
<1 ms
STM32H5 real-time motion

Wi-Fi 6 · BT 5.3 · CAN-FD
2.5 Gb Eth · 3× MIPI · ROS 2

Arduino - 實體AI應用軟體打包

The hidden protagonist

The Bridge — RPC between Linux and the MCU



Python on Linux calls a function that runs on the MCU. And back.



```
app.py — on the SoC

from arduino.bridge import board

# read a sensor living on the MCU
temp = board.sensor.read_temp()

# act on the physical world
if temp > 30:
    board.servo(1).sweep(0, 180)
```

A hybrid PhysicalAI app, in four layers

SoC / Linux

Containerized services

Models, web UI, message broker, databases.

SoC / Linux

Python / JS orchestration

The conductor — glue, policy, state, and timing.

Seam

Bridge RPC

The contract between the two brains.

MCU / RTOS

Sketch + TinyML

Actuation, sensors, hard real-time loops.

This stack is the unit we want to ship — as one thing.

Arduino - 實體AI應用軟體打包

The toolchain — Arduino × Edge Impulse

Arduino App Lab



Browser-based
no installers, works locally

Open source
GPL-3.0 — break it, fork it, ship it

Offline-capable
trust the device, not the cloud

Edge Impulse native
train a model, deploy into App Lab

01

PLAY

Run a Brick.

Zero code.

Curated examples out of the box.

FOR

first-time users · demos

02

TINKER

Edit the code.

Python + sketches.

Open a Brick, modify Python on Linux or the sketch.

FOR

educators · makers

03

TRAIN

Bring your model.

Edge Impulse · Qualcomm AI Hub.

Fine-tune, or drop your own LLM into a Brick.

FOR

ML engineers · product teams

04

MASTER

Escape hatch.

Your own toolchain.

Deploy with your own frameworks, outside App Lab.

FOR

integrators · advanced teams

Generative AI

Local LLMs
Vision-language models
On-device reasoning

Audio Intelligence

Keyword spotting
Speech-to-text · text-to-speech
Acoustic event recognition

Vision AI

Detection · classification
Segmentation · tracking
Pose · depth

Robotics

Navigation · SLAM
Sensor fusion
Action-triggering policies

POWERED BY

end-to-end ML Ops



EDGE IMPULSE

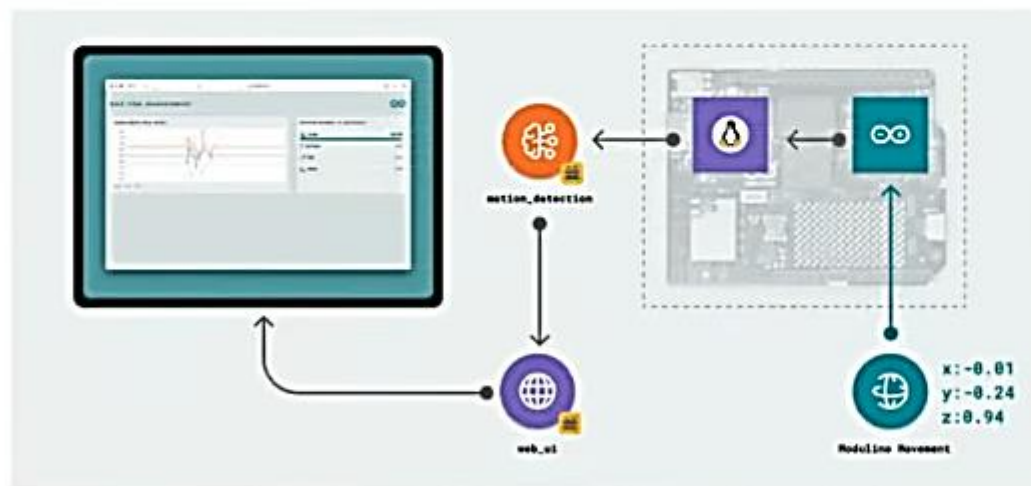
Qualcomm AI Hub optimized model catalog

Arduino - 實體AI應用軟體打包

The composable unit

Bricks — stack them, don't glue them

“Compose your project by stacking together ready-to-use Arduino Bricks.”



example: a motion_detection Brick — sensor → logic → web UI, on one board



Under the hood, each Brick is:

- **Container**
a self-contained Linux service
- **Or sketch**
a self-contained MCU firmware
- **Plus an interface**
the RPC contract it exposes
- **Plus metadata**
what board, what permissions

This is the ad-hoc standard — in practice.

Arduino - 實體AI應用軟體打包

A rideable go-kart.



Robotic arm.



Smart mirror.

ALIF/Grovety – 微型文字轉語音應用



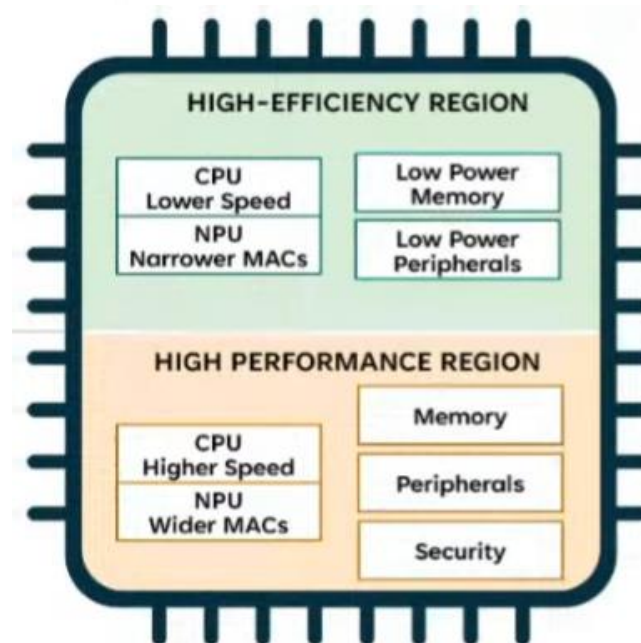
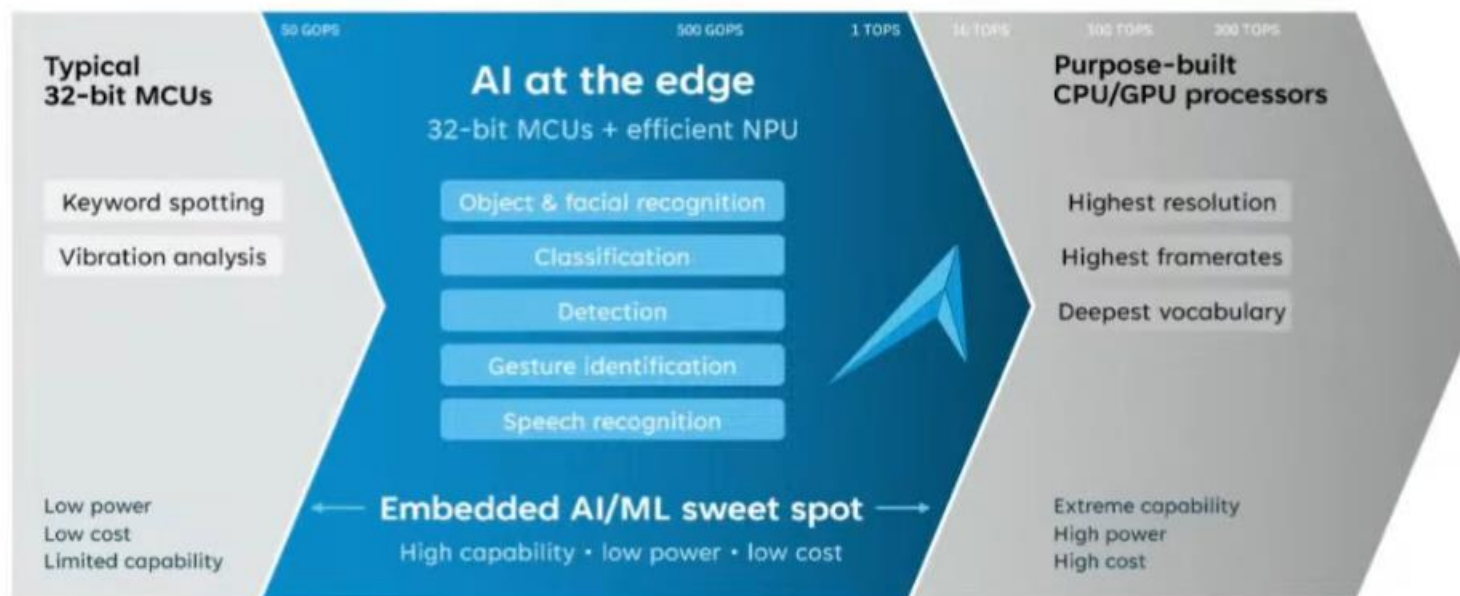
https://youtu.be/_5lvJjrWCv4?t=7554

ALIF/Grovety – 微型文字轉語音應用



Embedded AI/ML sweet spot

AI/ML at the edge. No tradeoffs.



ALIF/Grovety – 微型文字轉語音應用

Four classes of tasks



Extractive Voice Interaction (Answer)

Spoken request →

- an answer from a trusted corpus (voice, img, txt)
- a list of candidates for selection
- a clear “no result” indication

Inspect

Device state →

- answers based on live data and telemetry
- reflects the actual current condition of the device

Safe Voice-Triggered Actions (Act)

Voice command →

- validation within the target domain
- execution via a controlled action path
- or a disambiguation engine if needed

Guide

User interaction →

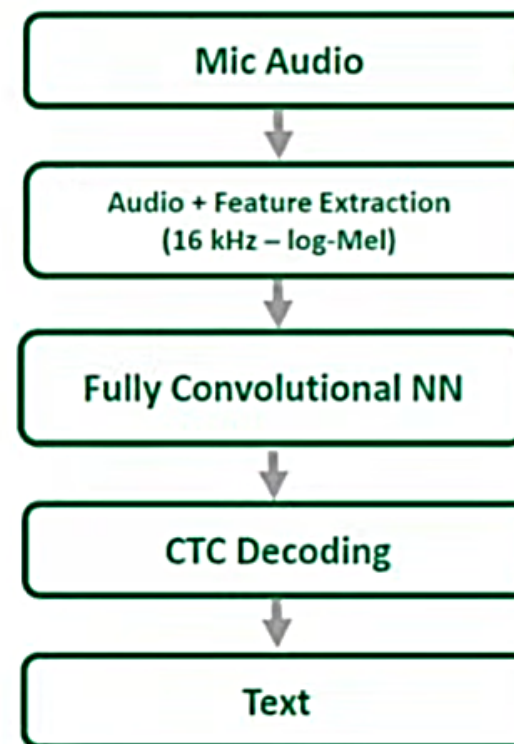
- step-by-step instructions
- stateful, multi-turn guidance

ALIF/Grovety – 微型文字轉語音應用



Components: Speech-to-Text (STT) module

- Based on convolutional architecture
- Fully convolutional model — no RNNs or transformers
- Optimized for deterministic, low-latency execution
- Processes 16 kHz audio via log-Mel spectrogram features
- ~19 million parameters
- 3.9% WER on LibriSpeech test-clean
- Robust to real-world noise (e.g., running water, ventilation), maintaining stable recognition in low-SNR environments



ALIF/Grovety – 微型文字轉語音應用



Components: Deterministic Hybrid NLA&Retrieval Engine

ASR Error Correction

Correcting STT errors
in real time

Hybrid Search

Frequency + fuzzy search, robust to
conversational input

Intent Router

Strict separation of Q&A and device
commands

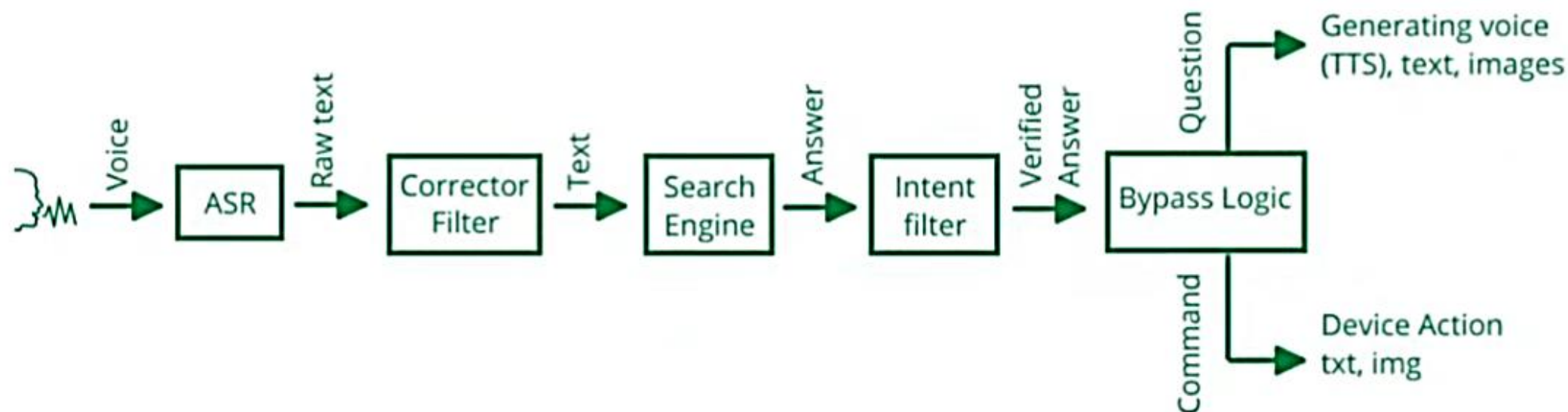
Stateful Context

FSM-based navigation and
disambiguation

ALIF/Grovety – 微型文字轉語音應用



Base Processing Sequence



ALIF/Grovety – 微型文字轉語音應用



Hardware



Alif Ensemble 7

Demo works on Alif Ensemble E7 AI/ML AppKit and Ensemble E8 DevKit.

We use multi-core configuration (High-Performance и High-Efficiency):

- Energy-Efficient Core: Arm® Cortex®-M55, 160 MHz
- Energy-Efficient NPU: Arm Ethos™-U55, 128 MAC/cycle
- Performance Core: Arm® Cortex®-M55, 400 MHz
- Performance NPU: Arm Ethos™-U55, 256 MAC/cycle

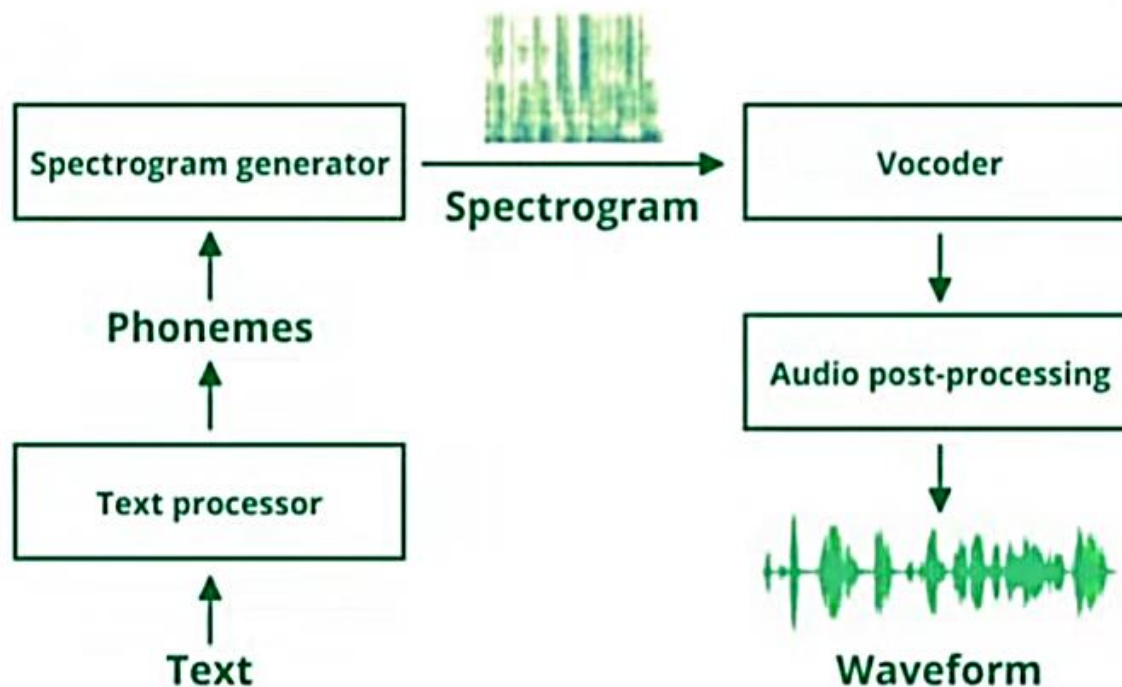
Memory:

- High Endurance MRAM Non-Volatile Memory: Up to 5.5MB
- SRAM: Up to 13.5MB
- Flash: 64MB

ALIF/Grovety – 微型文字轉語音應用



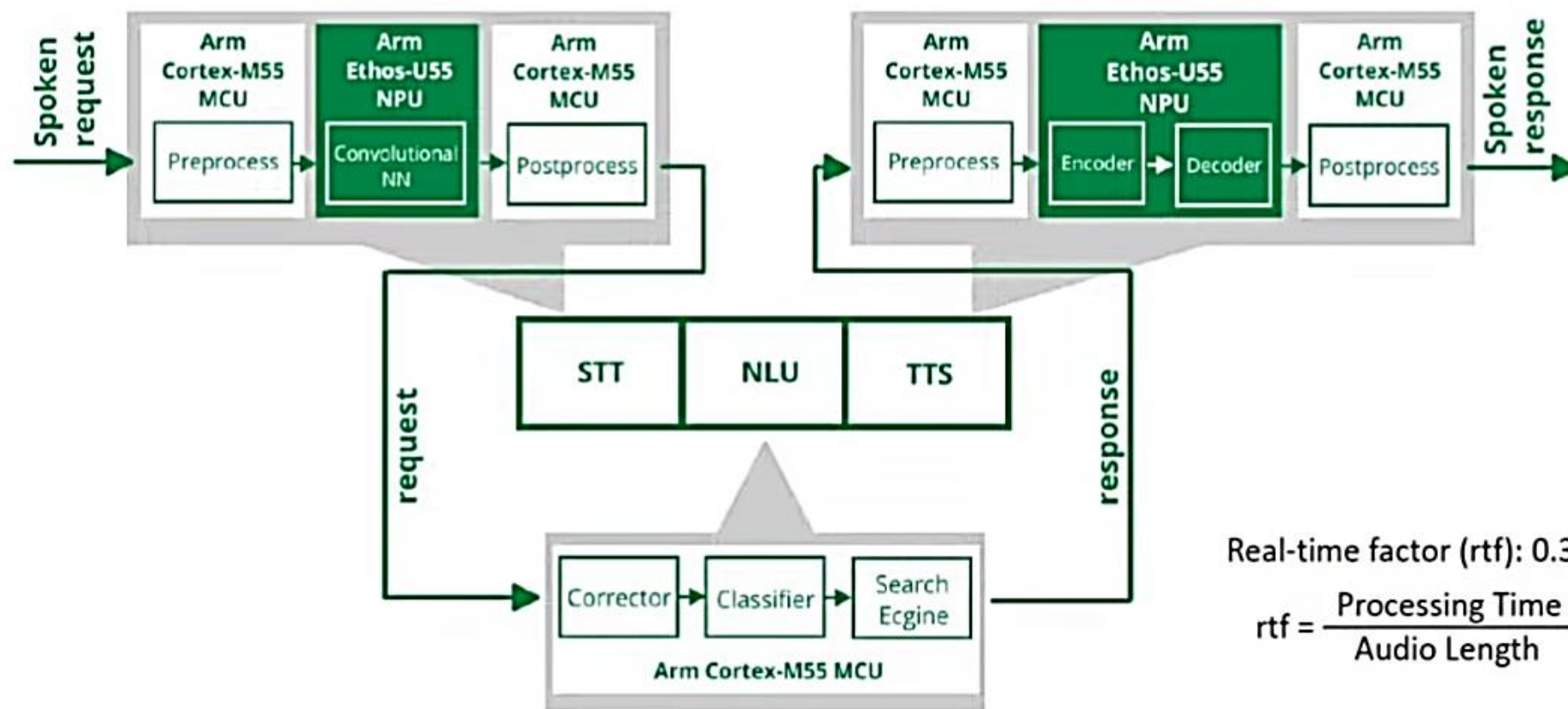
Components: Text-to-Speech (TTS) module



1. Text normalization and phonemization
2. Chunk-based processing for long inputs
3. Neural acoustic model
4. GAN-based neural vocoder
5. Seamless audio stitching

ALIF/Grovety – 微型文字轉語音應用

System architecture



Real-time factor (rtf): 0.3

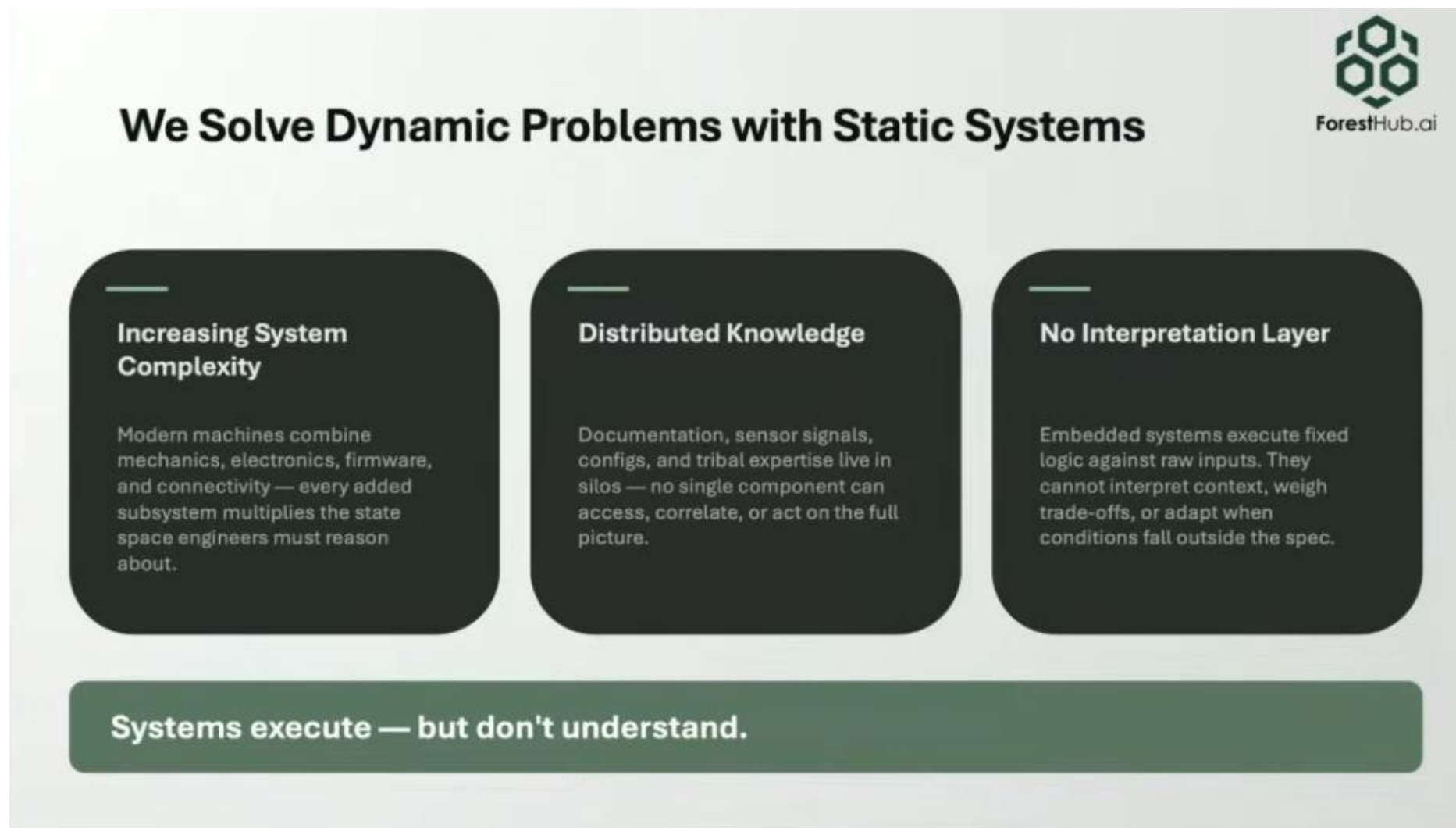
$$rtf = \frac{\text{Processing Time}}{\text{Audio Length}}$$

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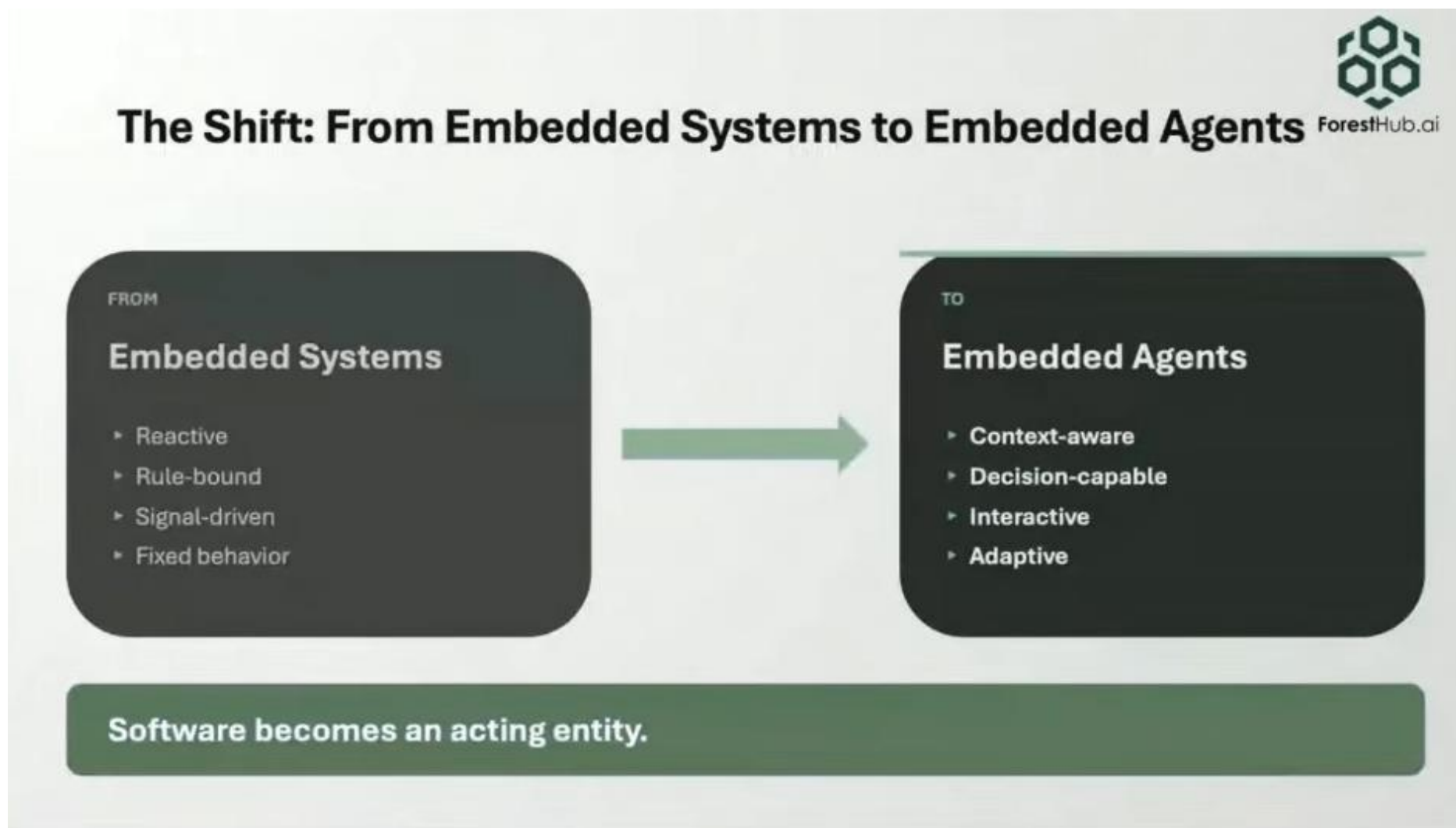


https://youtu.be/_5lvJjrWCv4?t=9067

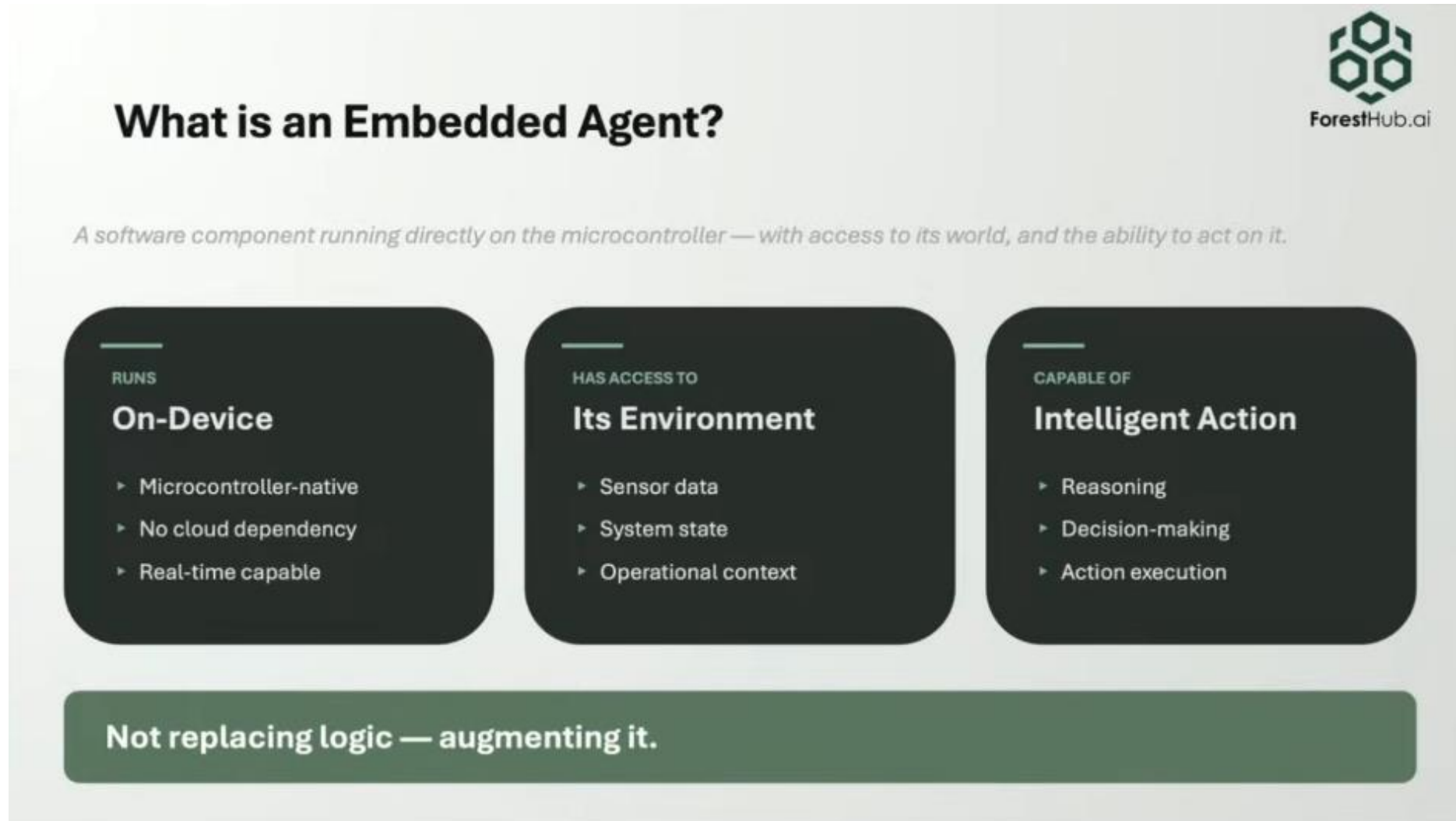
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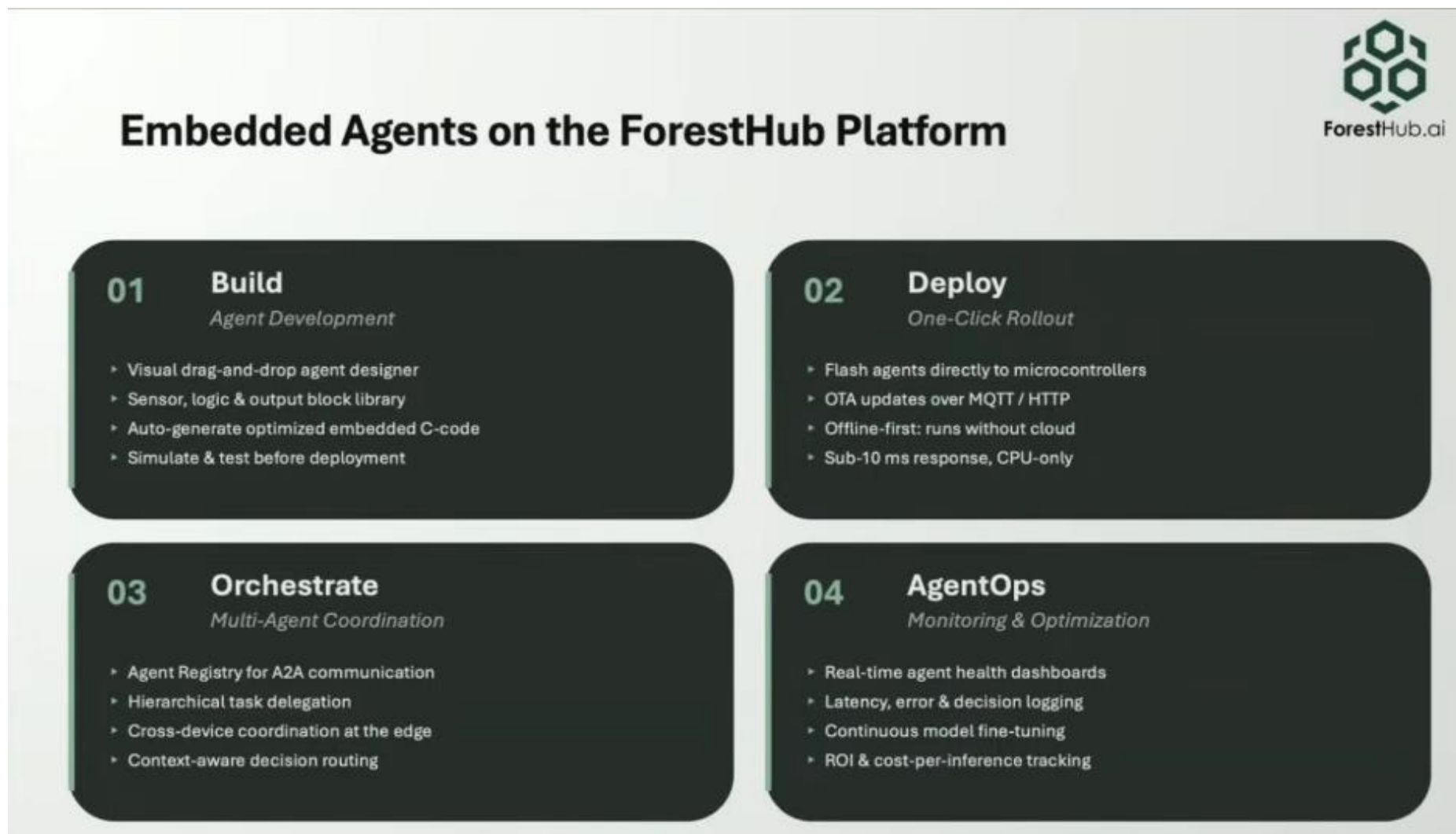
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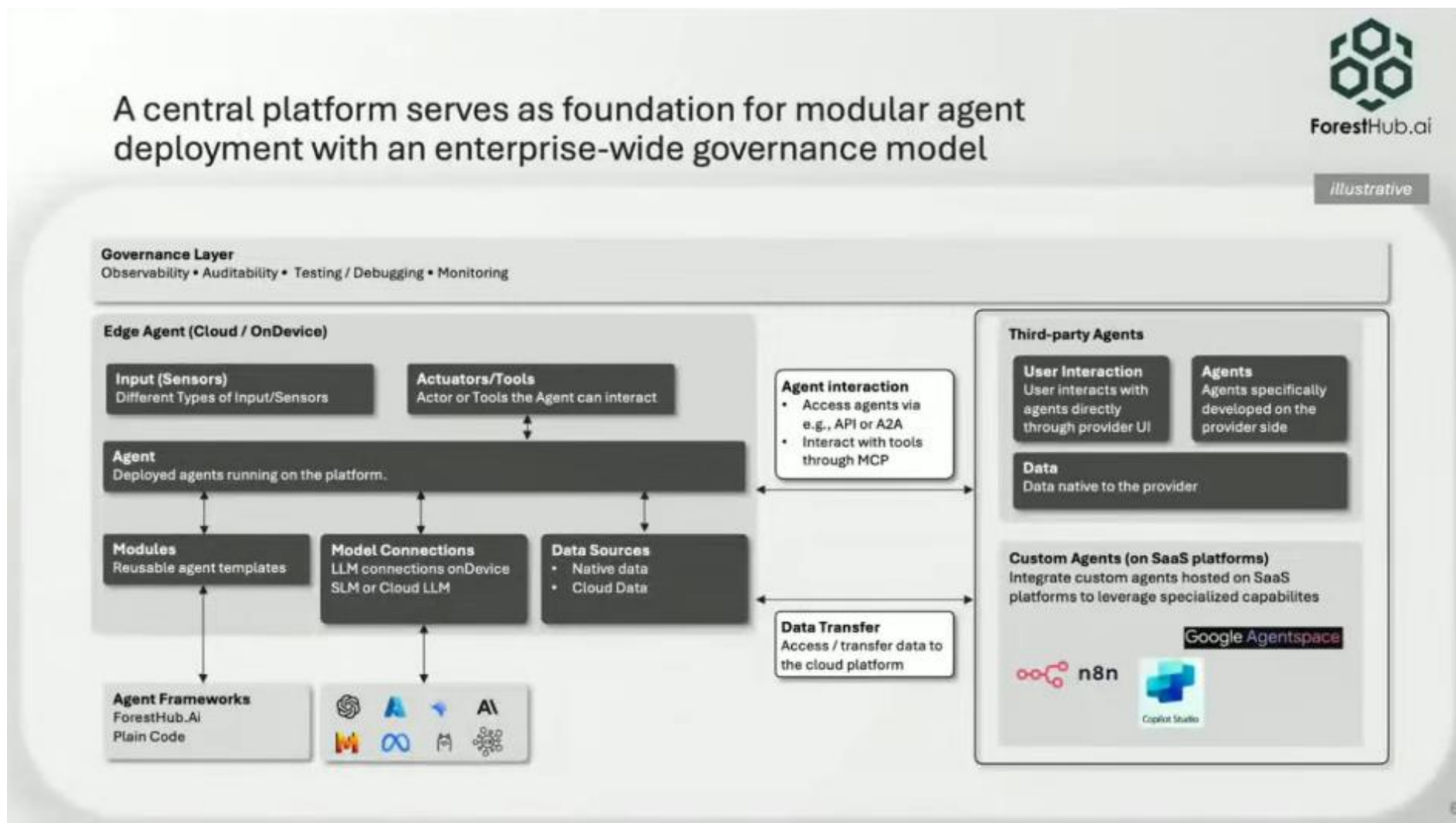
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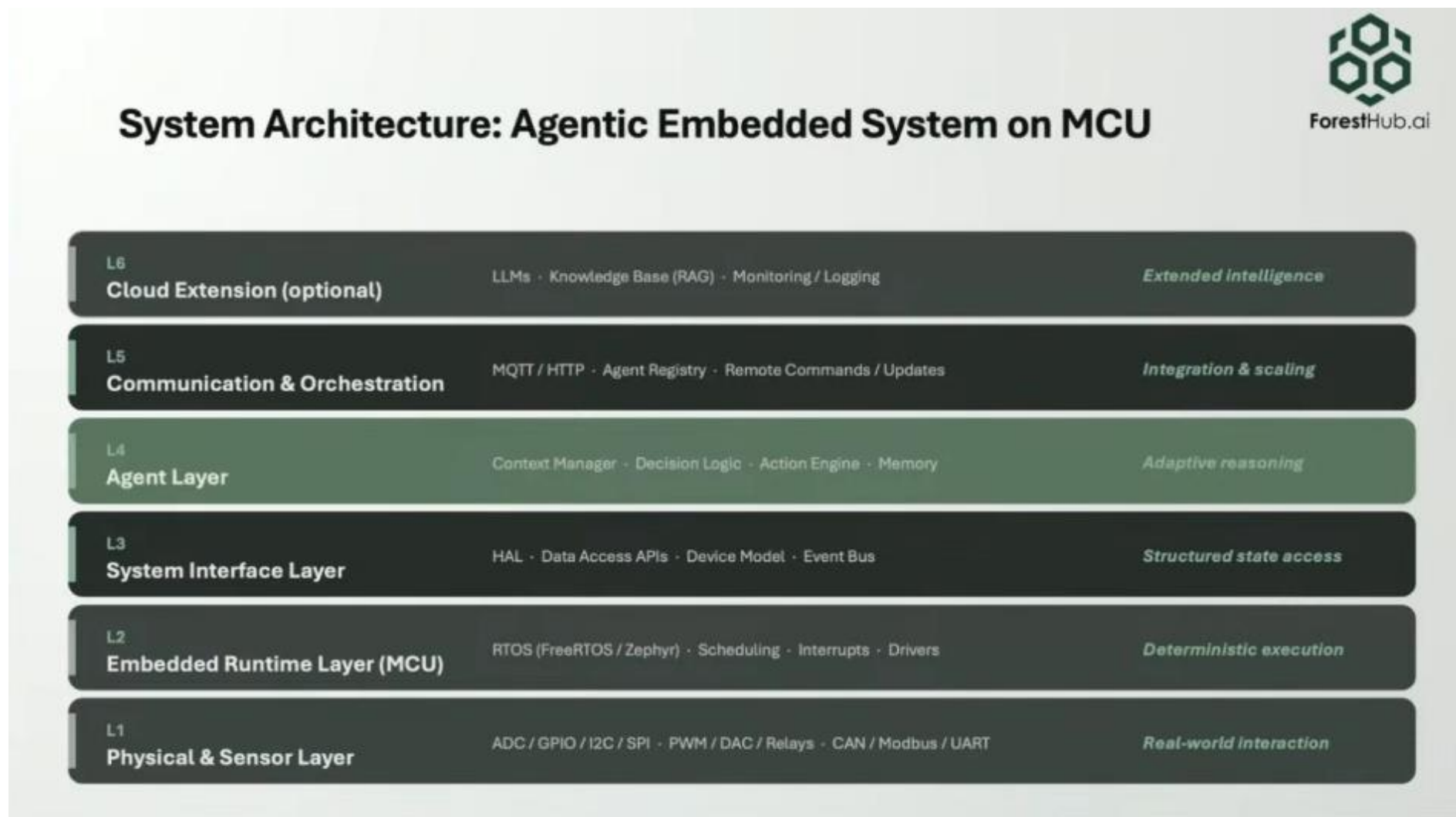
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Closed-Loop Reasoning & Design Principles



DATA FLOW

Closed-Loop Reasoning On-Device



Every loop closes on-device — no network hop required for core reasoning.

KEY DESIGN PRINCIPLES

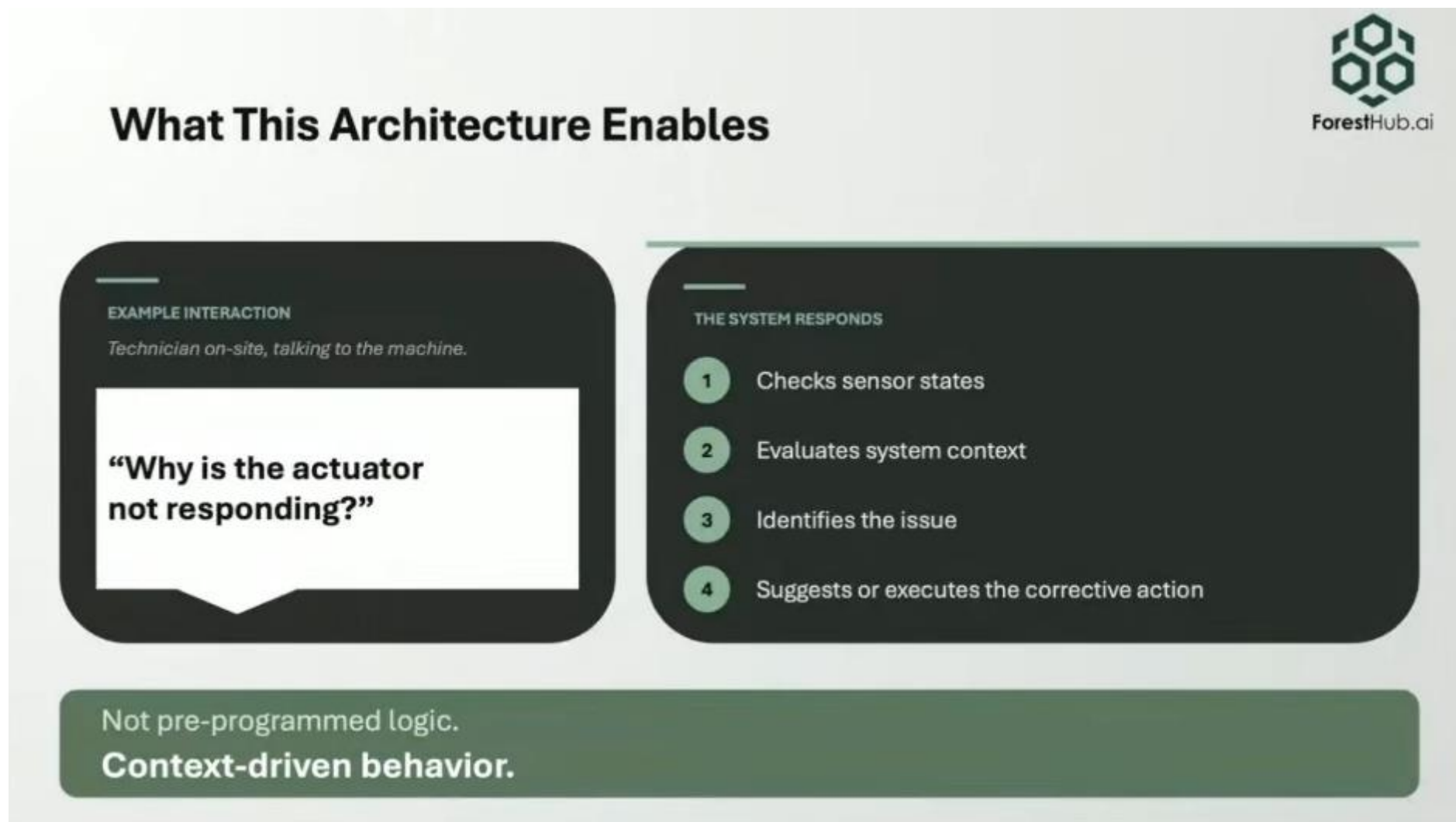
What makes this architecture viable

- **Separation of concerns**
Deterministic control and adaptive reasoning stay in separate layers.
- **On-device first**
Low latency, no cloud dependency for core operation.
- **Hybrid extension**
Cloud is used only when genuinely needed (RAG, LLM fallback).
- **Resource-aware**
Bounded memory and compute — fits a single MCU budget.

Deterministic control remains stable.

Intelligence is added as a separate, composable layer.

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參考文獻

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<https://omnixri.blogspot.com/p/ntust-edge-ai.html>
- Edge AI Foundation, Generative Edge AI: Reasoning, Agentic & Physical Intelligence
Day 1 : <https://www.edgeaifoundation.org/livestreams/generative-edge-ai-reasoning-agentic-physical-intelligence-3>
Day 2 : <https://www.edgeaifoundation.org/livestreams/generative-edge-ai-reasoning-agentic-physical-intelligence-2>
- 許哲豪 , Generative Edge AI: Reasoning, Agentic & Physical Intelligence
<https://hackmd.io/@OmniXRI-Jack/Generative-Edge-AI-Streames-V5>